

Topology

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Preface

Topology is the study of geometric properties and spatial relations that remain invariant under continuous deformations. For anyone stepping into modern mathematics, topology serves as a vital bridge connecting classical analysis, abstract algebra and geometry. These notes are born out of a desire to systematically organize, refine and document my personal journey through this profound and beautiful landscape.

The material presented in this book is structured into three main phases, reflecting a progression from intuitive frameworks to abstract structures and ultimately to algebraic invariants:

- **General and Point-Set Topology (Chapters 1–3, 7):** We begin on the familiar ground of metric spaces, establishing the foundational intuition of open sets and continuity, before lifting these concepts into general topological spaces. Core topological properties—such as compactness, connectivity, separation axioms, and regional properties—are explored thoroughly.
- **Categorical Language and Geometric Constructions (Chapters 4–6, 9, 14):** Uniquely, these notes introduce the preliminaries of Category Theory at an early stage. Reframing products, disjoint unions, pullbacks, and limits through universal properties provides a unifying perspective. This categorical and combinatorial machinery is then applied to study quotient spaces, group actions, simplicial complexes, topological graphs, and the foundational definitions of topological manifolds.
- **An Invitation to Algebraic Topology (Chapters 8, 10–13):** The final arc transitions into algebraic topology, where spaces are investigated via algebraic invariants. Starting with the concept of homotopy and simplicial approximation, we develop the machinery of the fundamental group (Π_1). We explore practical computations using the Seifert–van Kampen Theorem and conclude with an elegant application to Knot Theory and knot groups.

These notes represent a compilation of insights gathered from standard literature, lecture courses, and personal reflections. Mathematics is an ongoing, collaborative pursuit; despite meticulous proofreading, errors, typos, and expository gaps are bound to exist. I warmly welcome any feedback, corrections, or mathematical discussions from fellow readers.

Wang Yuyao

Chapter 1

From Metric to Topology

1.1 Topology induced by Metric

Definition 1.1. Topology

The *topology* induced from (X, d) is the collection of all open sets in (X, d) , denoted as $\mathcal{T}$.

Proposition 1.2. All open balls $B_r(x)$ are open in (X, d) .

Proposition 1.3. All open $U \subset X$ can be written as union of open balls.

Proof. Take $u \in U$. $\exists \epsilon_u > 0$, s.t.: $\{u\} \subset B_{\epsilon_u}(u) \subset U$. Hence,

$$U = \bigcup_{u \in U} \{u\} \subset \bigcup_{u \in U} B_{\epsilon_u}(u) \subset \bigcup_{u \in U} U = U.$$

Then we have: $U = \bigcup_{u \in U} B_{\epsilon_u}(u)$, which is the desired union of open balls. $\square$

Proposition 1.4. Let (X, d) be a metric space, and let $\mathcal{T}$ be the topology induced from (X, d) . Then:

1. Let $\{G_\alpha \mid \alpha \in \mathcal{A}\}$ be a collection of open sets, i.e.: $G_\alpha \in \mathcal{T}$. Then $\bigcup_{\alpha \in \mathcal{A}} G_\alpha \in \mathcal{T}$.
2. Let $G_1, \dots, G_n \in \mathcal{T}$. Then $\bigcap_{i=1}^n G_i \in \mathcal{T}$.

Proof. 1. $\forall x \in \bigcup_{\alpha \in \mathcal{A}} G_\alpha$, $\exists \beta \in \mathcal{A}$, s.t.: $x \in G_\beta$. Since G_β is open, then $\exists \epsilon_x > 0$, s.t.:

$$B_{\epsilon_x} \subset G_\beta \subset \bigcup_{\alpha \in \mathcal{A}} G_\alpha.$$

2. $\forall x \in \bigcap_{i=1}^n G_i$, i.e.: $x \in G_i \forall i = 1, \dots, n$. Then $\exists \epsilon_i > 0$, s.t.: $B_{\epsilon_i}(x) \subset G_i$ for $i = 1, \dots, n$.
Take $\epsilon := \min\{\epsilon_1, \dots, \epsilon_n\}$. Then $B_\epsilon(x) \subset B_{\epsilon_i} \subset G_i$ for all i . Therefore, $B_\epsilon(x) \subset \bigcap_{i=1}^n G_i$. $\square$

Example:

- Denote the topologies induced from $(\mathbb{R}^n, d_2)$ and $(\mathbb{R}^n, d_\infty)$ by $\mathcal{T}_2$ and $\mathcal{T}_\infty$, respectively, where $d_2(x, y) := \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$ and $d_\infty(x, y) := \max_{1 \leq i \leq n} \{x_i, -y_i\}$. Then $\mathcal{T}_2 = \mathcal{T}_\infty$.
- Let $\mathcal{T}$ be a topology induced from $(X, d_{discrete})$, where $d_{discrete}(x, y) := \begin{cases} 0 & , \text{if } x = y \\ 1 & , \text{if } x \neq y \end{cases}$. Then $\mathcal{T} = \text{all subsets of } X$.

Definition 1.5. Closed

A subset $V \subset X$ is closed if $X - V$ is open.

Example: Let $X = \mathbb{R}$ equipped with $d_1(x, y) := |x - y|$.

$$\mathbb{R} - [a, b] = (-\infty, a) \cup (b, \infty) \text{ is open} \Rightarrow [a, b] \text{ is closed.}$$

Proposition 1.6. Let X be a metric space. Then:

1. $\emptyset, X$ are closed in X .
2. If F_α is closed in X , then $\bigcap_{\alpha \in \mathcal{A}} F_\alpha$ is closed in X .
3. If $F_1, \dots, F_k$ is closed, then $\bigcup_{i=1}^k F_i$ is closed in X .

Proof. 1. Since X is open, then $\emptyset = X - X$ is closed in X . Since $\emptyset$ is open, then $X = X - \emptyset$ is closed.

2. Since F_α is closed in X , then $\exists U_\alpha \subset X$ open, s.t.: $F_\alpha = X - U_\alpha$. Then:

$$\bigcap_{\alpha \in \mathcal{A}} F_\alpha = \bigcap_{\alpha \in \mathcal{A}} (X - U_\alpha) = X - \left(\bigcup_{\alpha \in \mathcal{A}} U_\alpha \right)$$

is closed, since $\bigcup_{\alpha \in \mathcal{A}} U_\alpha$ is open.

3. Since F_i is closed in X , then $\exists U_i \subset X$ open, s.t.: $F_i = X - U_i$. Then:

$$\bigcup_{i=1}^k F_i = \bigcup_{i=1}^k (X - U_i) = X - \left(\bigcap_{i=1}^k U_i \right)$$

is closed, since $\bigcap_{i=1}^k U_i$ is open.

□

Definition 1.7. Convergence

Let (X, d) be a metric space. Then $\{x_n\} \rightarrow x$ means $\forall \epsilon > 0, \exists N, \text{ s.t.: } d(x_n, x) < \epsilon \forall n \geq N$.

Proposition 1.8. Let (X, d) be a metric space. Then $\{x_n\} \rightarrow x$ iff $\forall$ open set U containing x , $\exists N, \text{ s.t.: } x_n \in U \forall n \geq N$.

Proof. • $(\Leftarrow)$: $\forall$ given $\epsilon > 0$, take $U := B_\epsilon(x) \ni x$. Then $\exists N, \text{ s.t.: } x_n \in U = B_\epsilon(x) \forall n \geq N$.

- $(\Rightarrow)$: For any open $U \ni x, \exists \epsilon > 0, \text{ s.t.: } B_\epsilon(x) \subset U$. Since $\{x_n\} \rightarrow x$, then $\exists N, \text{ s.t.: } x_n \in B_\epsilon(x) \forall n \geq N$. Hence, $x_n \in B_\epsilon(x) \subset U$.

□

Definition 1.9. Continuity

Let (X, d) and (Y, ρ) be metric spaces. Then $f : X \rightarrow Y$ is continuous at $x_0 \in X$ if $\forall \epsilon > 0, \exists \delta > 0$, s.t.: $d(x, x_0) < \delta \Rightarrow \rho(f(x), f(x_0)) < \epsilon$.

Furthermore, we say f is continuous on X if f is continuous at all points in X .

Lemma 1.10. f is continuous at x iff $\forall \{x_n\} \rightarrow x, \{f(x_n)\} \rightarrow f(x)$.

Proposition 1.11. 1. f is cont at $x \Leftrightarrow \forall$ open $U \ni f(x), \exists \delta > 0$, s.t.: $B_\delta(x) \subset f^{-1}(U)$.

2. f is cont on $X \Leftrightarrow f^{-1}(U)$ is open in X for each open set.

Proof. 1. • $(\Rightarrow)$: For any open U containing $f(x)$, there is a positive ϵ , s.t.: $B_\epsilon(f(x)) \subset U$. Then by the continuity of f , there exists a positive δ , s.t.: $f(B_\delta(x)) \subset B_\epsilon(f(x)) \subset U$, which implies that $B_\delta(x) \subset f^{-1}(U)$.

• $(\Leftarrow)$: For any ϵ , take $\{x_n\} \rightarrow x$. Then $x_n \in B_\delta \subset f^{-1}(B_\epsilon(f(x))) \forall n \geq N$ for some N . Hence, $f(x_n) \in B_\epsilon(f(x))$. Then by lemma 1.9, f is cont at x .

2. • $(\Rightarrow)$: For any x in $f^{-1}(U)$, by proposition 2.11.1, there exists a positive $\delta > 0$, s.t.: $B_\delta(x) \subset f^{-1}(U)$. Hence, $f^{-1}(U)$ is open.

• $(\Leftarrow)$: For any $x \in X$, every open U containing $f(x)$ can imply $f^{-1}(U)$ is open in X . Therefore, there is a positive δ , s.t.: $B_\delta(x) \subset f^{-1}(U)$.

□

1.2 Distance in Metric Space

Definition 1.12. Distance

Let (X, d) be a metric space. Given two non-empty subsets A and B of X , we define by

$$d(A, B) := \inf_{a \in A, b \in B} d(a, b)$$

the *distance between A and B* . Furthermore, if $A = \{a\}$ consists of a single point, then we write

$$d(a, B) := d(\{a\}, B)$$

Example:

- If $A \cap B \neq \emptyset$, then $d(A, B) = 0$.
- But the converse does NOT hold in general: consider $X = \mathbb{R}, A = (0, 1), B = [1, 2]$. Then $d(A, B) = 0$, but $A \cap B = \emptyset$.

Proposition 1.13. *Distance-to-Subspace Proposition*

Let (X, d) be a metric space and let $B \subseteq X$ be a non-empty subset.

1. The map

$$\begin{aligned} X &\rightarrow \mathbb{R}_{\geq 0} \\ x &\mapsto d(x, B) \end{aligned}$$

is continuous.

2. If B is closed, then given $x \in X$, we have

$$d(x, B) \Leftrightarrow x \in B$$

Proposition 1.14. *Disjoint Compact-Closed-Distance Proposition*

Let (X, d) be a metric space and let $K, L \subseteq X$ be two non-empty disjoint subsets. If one of K, L is compact and if the other one is closed in X , then $d(K, L) > 0$.

Proof. WLOG, let K be compact and let L be closed. Consider the map $\Phi : K \rightarrow \mathbb{R}_{\geq 0}$ given by

$$\Phi(x) := d(x, L)$$

which is continuous by the Distance-to-Subspace Proposition.1. Since K is compact, then there exists $x \in K$, s.t.: for any $k \in K$, we have $\Phi(x) \leq \Phi(k)$. Hence,

$$d(K, L) = \inf_{k \in K} d(k, L) = d(x, L)$$

By the closed-ness of L together with the Distance-to-Subspace Proposition.2, $d(x, L) > 0$. □

Proposition 1.15. *Distance-Minimizer Proposition*

Let (X, d) be a metric space and let $K, L \subseteq X$ be two non-empty subsets of X . If one of followings hold:

1. both K and L are compact;
2. $X = \mathbb{R}^n$, K is compact and L is closed in $X = \mathbb{R}^n$,

then there exist $x \in K, y \in L$, s.t.:

$$d(x, y) = d(K, L)$$

Proof. 1. is obvious.

2. Set $s := \max\{\|x\| : x \in K\} + 1 + d(K, L)$ and then set $L' := L \cap \overline{B}_s^n(0)$. Since L is closed, then L' is compact. It's easy to verify that

$$d(K, L') = d(K, L)$$

since $L' \neq \emptyset$ together with the choice of s and the triangle inequality. Therefore, we goes back to 1. □

1.3 The Lebesgue Lemma

Definition 1.16. Diameter

Let (X, d) be a metric space. The *diameter* of a non-empty subset $A \subseteq X$ of X is defined as

$$\text{diam}(A) := \sup_{a, b \in A} d(a, b) \in \mathbb{R}_{\geq 0} \cup \{\infty\}$$

Moreover, we define $\text{diam}(\emptyset) = 0$.

Theorem 1.17. *Lebesgue Lemma*

Let (X, d) be a compact metric space and let $\{U_i\}_{i \in I}$ be an open cover of K . There exists a $\delta > 0$, s.t.: for any $A \subseteq K$ with $\text{diam}(A) < \delta$, there is an $i \in I$ with $A \subseteq U_i$.

Remark: The δ that has the property stated in Lebesgue Lemma is usually called a *Lebesgue number*.

Proof. By the compact-ness of K , we can WLOG let $I = \{1, \dots, n\}$ be a finite set. If $K = U_i$ for some $i \in I$, then every $\delta > 0$ has the desired property. Now we suppose that for any $i \in I$, $U_i \subsetneq K$.

- **Claim 1:** The map $f : K \rightarrow \mathbb{R}$ given by

$$f(x) := \frac{1}{n} \sum_{i=1}^n d(x, K \setminus U_i)$$

is continuous and positive for every $x \in K$.

- *Proof.* By the Distance-to-Subspace Proposition.1, f is continuous. For ever $x \in K$, there exists $i \in I$, s.t.: $x \in U_i$. Since U_i is open, then $K \setminus U_i$ is closed and hence by the Distance-to-Subspace Proposition.2, $d(x, K \setminus U_i) > 0 \Rightarrow f(x) > 0$. □
- **Claim 2:** Let $A \subseteq K$ be a subset of K . If $\text{diam}(A) < \delta$, then $A \subseteq U_i$ for some $i \in I$.
- *Proof.* If $A = \emptyset$, then we're done. Now we suppose that $A \neq \emptyset$. Pick $x \in A$ and then take an $m \in I$ to be such that $d(x, K \setminus U_m) \geq d(x, K \setminus U_i), \forall i \in I$. Since $\text{diam}(A) < \delta$, then $A \subseteq B_\delta(x)$. Note that

$$d(x, K \setminus U_m) \geq \frac{1}{n} \sum_{i=1}^n d(x, K \setminus U_i) = f(x) \geq \delta \tag{*}$$

Hence, $A \subseteq B_\delta(x) \stackrel{(*)}{\subseteq} U_m$. □

Then we're done. □

Corollary 1.18. *Lebesgue Lemma II*

Let $f : [0, 1]^n \rightarrow X$ be a continuous map where X is a topological space and let $\{V_i\}_{i \in I}$ be an open

cover of X . There exists $N \in \mathbb{N}_+$, s.t.: $\forall a_1, \dots, a_n \in \{0, \dots, N-1\}$,

$$f\left(\left[\frac{a_1}{N}, \frac{a_1+1}{N}\right] \times \dots \times \left[\frac{a_n}{N}, \frac{a_n+1}{N}\right]\right) \subseteq V_i$$

for some $i \in I$.

Proof. Note that $[0, 1]^n$ is compact and that $\{f^{-1}(V_i)\}_{i \in I}$ is an open cover of $[0, 1]^n$. Then by the Lebesgue Lemma, there exists $\delta > 0$, s.t.: for any $A \subseteq [0, 1]^n$ with $\text{diam}(A) < \delta$, $A \subseteq f^{-1}(V_i)$ for some $i \in I$. Hence, $f(A) \subseteq V_i$ for some $i \in I$. Take $N := \frac{2\sqrt{n}}{\delta} > \frac{\sqrt{n}}{\delta}$. Since

$$\text{diam}\left(\left[\frac{a_1}{N}, \frac{a_1+1}{N}\right] \times \dots \times \left[\frac{a_n}{N}, \frac{a_n+1}{N}\right]\right) = \frac{\sqrt{n}}{N} = \delta/2 < \delta$$

then we're done. □

1.4 Complete Metric Space

Definition 1.19. Complete

We say that a metric space (X, d) is *complete* if every Cauchy sequence in (X, d) converges. Otherwise, we say it is *incomplete*.

Proposition 1.20. *Every closed subset of a complete space equipped with the same metric of the whole space is also complete.*

Proposition 1.21. *Every compact metric space is complete.*

Proposition 1.22. *Upper Bound-to-Metric Proposition*

Let (M, d) be a metric space.

1. The map $\tilde{d} : M \times M \rightarrow \mathbb{R}_{\geq 0}$ given by

$$\tilde{d}(x, y) := \min\{1, d(x, y)\}$$

defines a metric on M .

2. $\mathcal{T}_{\tilde{d}} = \mathcal{T}_d$.

3. (M, d) is complete $\Leftrightarrow (M, \tilde{d})$ is complete.

Proposition 1.23. *Function Space-Metric Proposition*

Let (M, d) be a metric space and let X be a non-empty topological space. The map $\bar{d} : C(X, M) \times C(X, M) \rightarrow \mathbb{R}_{\geq 0}$ given by

$$\bar{d}(f, g) := \sup_{x \in X} \tilde{d}(f(x), g(x))$$

where $\tilde{d}$ is the metric on M defined in the Upper Bound-to-Metric Proposition, has the following properties:

1. $\tilde{d}$ defines a metric on $C(X, M)$.
2. If X is compact, then the map $\bar{d}' : C(X, M) \times C(X, M) \rightarrow \mathbb{R}_{\geq 0}$ given by

$$\bar{d}'(f, g) := \sup_{x \in X} d(f(x), g(x))$$

also defines a metric on $C(X, M)$, which is equivalent to $\bar{d}$.

3. If (M, d) is complete, then for any Cauchy $(f_n)_{n \in \mathbb{N}_+} \subseteq (C(X, M), \bar{d})$, the map $f : X \rightarrow M$ given by

$$f(x) := \lim_{n \rightarrow \infty} f_n(x)$$

exists, is continuous and in $(C(X, M), \bar{d})$ is a limit of $(f_n)_{n \in \mathbb{N}_+}$. Furthermore, $(C(X, M), \bar{d})$ is complete.

Proof. 3.

- **Claim 1:** For any $x \in X$, $(f_n(x))_{n \in \mathbb{N}_+}$ converges.
- *Proof.* By the def of $\bar{d}$,

$$\tilde{d}(f_n(x), f_m(x)) \leq \bar{d}(f_n, f_m)$$

Since $(f_n)_{n \in \mathbb{N}_+}$ is Cauchy, then so is $(f_n(x))_{n \in \mathbb{N}_+}$ in $(M, \tilde{d})$. By the Upper Bound-to-Metric Proposition.3, $(M, \tilde{d})$ is complete and hence $(f_n(x))_{n \in \mathbb{N}_+}$ converges. $\square$

- **Claim 2:** $\forall \epsilon > 0, \exists N \in \mathbb{N}_+, \forall n \geq N, \forall x \in X$

$$\tilde{d}(f_n(x), f(x)) < \epsilon$$

- *Proof.*

$$\tilde{d}(f_n(x), f(x)) := \tilde{d}(f_n(x), \lim_{m \rightarrow \infty} f_m(x)) \stackrel{\tilde{d} \text{ cont}}{=} \lim_{m \rightarrow \infty} \tilde{d}(f_n(x), f_m(x)) < \epsilon$$

$\square$

- f is continuous.
- *Proof.* For any open $U \subseteq M$ in M , let $x \in f^{-1}(U)$. Since U is open, there exists $\epsilon > 0$, s.t.:

$$B_\epsilon(f(x)) \subseteq U$$

By claim 2 and by the def of $\tilde{d}$, there exists $n \in \mathbb{N}_+$, s.t.: for any $y \in X$, $d(f_n(y), f(y)) < \epsilon/3$. Since f_n is cont, then there exists open $W \ni x$, s.t.: $f_n(W) \subseteq B_{\epsilon/3}(f(x))$. Then for any $y \in W$,

$$d(f(x), f(y)) = d(f(x), f_n(x)) + d(f_n(x), f_n(y)) + d(f_n(y), f(y)) < \epsilon/3 + \epsilon/3 + \epsilon/3 < \epsilon$$

Hence, $W \subseteq B_\epsilon(f(x)) \subseteq U$. $\square$

Then we're done. □

1.5 Space-Filling Curves

Definition 1.24. Space-Filling Paths

Let $m \geq 2$. A surjective continuous map from $[0, 1]$ to $[0, 1]^m$ is called a *space-filling path*.

Example: Hilbert Curve

Theorem 1.25. *Space-Filling Curve Theorem*

For any $m \in \mathbb{N}_+$, there exists a surjective continuous map $f : [0, 1] \rightarrow [0, 1]^m$.

Proof. The case when $m = 1$ is obvious. We do induction on m with the basic case when $m = 2$.

Now we deal with the basic case when $m = 2$. For $n \in \mathbb{N}_+$, we set:

$$P_n := \{0, \frac{1}{4^n}, \frac{2}{4^n}, \dots, \frac{4^n - 1}{4^n}\}, Q_n := \{0, \frac{1}{2^n}, \frac{2}{2^n}, \dots, \frac{2^n - 1}{2^n}\} \times \{0, \frac{1}{2^n}, \frac{2}{2^n}, \dots, \frac{2^n - 1}{2^n}\}$$

Furthermore, for $k \in P_n$, we set:

$$I_k^n := [k, k + \frac{1}{4^n}]$$

and for $(k, l) \in Q_n$, we set:

$$W_{(k,l)}^n := [k, k + \frac{1}{2^n}] \times [l, l + \frac{1}{2^n}]$$

- **Claim 1:** For $n = 0, 1, 2, \dots$, we can find a bijection $\alpha_n : P_n \rightarrow Q_n$ and a continuous map $f_n : [0, 1] \rightarrow [0, 1] \times [0, 1]$ satisfying the following conditions:
 1. $\forall n \in \mathbb{N} \cup \{0\}, \forall k \in P_n$, we have $f_n(I_k^n) \subseteq W_{\alpha_n(k)}^n$.
 2. $\forall n \in \mathbb{N} \cup \{0\}, \forall k \in P_n$, the two endpoints of the interval I_k^n get sent to adjacent vertices of the square $W_{\alpha_n(k)}^n$.
 3. $\forall n \geq s \in \mathbb{N} \cup \{0\}, \forall k \in P_n, l \in P_s$ with $I_k^n \subseteq I_l^s$, we have the inclusion $f_n(I_k^n) \subseteq W_{\alpha_s(l)}^s$.
- *Proof.* We define the maps iteratively.
 - For $n = 0$, $P_0 = \{0\}$ and $Q_0 = \{(0, 0)\}$. Let $\alpha_0 : P_0 \rightarrow Q_0$ be the unique bijection $0 \mapsto (0, 0)$. Furthermore, set

$$f_0 : \begin{matrix} [0, 1] \\ t \mapsto (t, 0) \end{matrix} \rightarrow [0, 1] \times [0, 1]$$

It's clear that the desired properties 1, 2 and 3 are satisfied.

- Suppose that we have constructed bijections $\alpha_0, \dots, \alpha_n$ and continuous maps $f_0, \dots, f_n$ with the desired properties. Given $j \in P_n$. It suffices to define f_{n+1} on $I_j^n = [j, j + \frac{1}{4^n}]$ and α_{n+1} on $j, j + \frac{1}{4^{n+1}}, j + \frac{2}{4^{n+1}}, j + \frac{3}{4^{n+1}} \in P_{n+1}$. Let $(k, l) := \alpha_n(j) \in Q_n$. By 2, we can find a bijection

$$\alpha_{n+1} : \{j, j + \frac{1}{4^{n+1}}, j + \frac{2}{4^{n+1}}, j + \frac{3}{4^{n+1}}\} \mapsto \{(k, l), (k + \frac{1}{2^{n+1}}, l), (k, l + \frac{1}{2^{n+1}}), (k + \frac{1}{2^{n+1}}, l + \frac{1}{2^{n+1}})\}$$

and a map $f_{n+1} : I_j^n \rightarrow W_{(k,l)}^n$ with the following three properties:

- I $f_{n+1} : I_j^n \rightarrow W_{(k,l)}^n$ agrees with the map f_n on the endpoints.
- II Each interval $I_{j+\frac{i}{4^{n+1}}}^{n+1}$ gets sent to the square $W_{\alpha_{n+1}(j+\frac{i}{4^{n+1}})}^{n+1}$.
- III The endpoints of each interval $I_{j+\frac{i}{4^{n+1}}}^{n+1}$ get sent to adjacent vertices of the square $W_{\alpha_{n+1}(j+\frac{i}{4^{n+1}})}^{n+1}$.

□

Now we consider the metric space $(C([0, 1], [0, 1] \times [0, 1]), d)$ of continuous maps from $[0, 1]$ to $[0, 1] \times [0, 1]$, equipped with the metric d defined as

$$d(f, g) := \sup_{t \in [0, 1]} \|f(t) - g(t)\|$$

- **Claim 2:** The sequence $(f_n : [0, 1] \rightarrow [0, 1] \times [0, 1])_{n \in \mathbb{N} \cup \{0\}}$ of maps in claim 1 is a Cauchy sequence in the metric space $(C([0, 1], [0, 1] \times [0, 1]), d)$.
- *Proof.* For any $\epsilon > 0$, pick $N \in \mathbb{N} \cup \{0\}$ with $\frac{1}{2^N} < \frac{\epsilon}{\sqrt{2}}$. For any $m, n \geq N$, any $t \in [0, 1]$,

$$\|f_m(t) - f_n(t)\| \leq \sqrt{2} \cdot \frac{1}{2^n} < \epsilon$$

□

By claim 2 together with the Function Space-Metric Proposition,

$$f : \begin{matrix} [0, 1] \\ t \mapsto f(t) := \lim_{n \rightarrow \infty} f_n(t) \end{matrix} \rightarrow [0, 1] \times [0, 1]$$

is continuous.

- **Claim 3:** $f([0, 1])$ is dense in $[0, 1] \times [0, 1]$.
- *Proof.* For any open $U \subseteq [0, 1] \times [0, 1]$ in $[0, 1] \times [0, 1]$, we need to show that $f([0, 1]) \cap U \neq \emptyset$. Note that $\exists N \in \mathbb{N} \cup \{0\}, (k, l) \in Q_N$, s.t.: $W_{(k,l)}^N \subseteq U$. By the property 1 in claim 1, there exists $x \in [0, 1]$ with $f_N(x) \in W_{(k,l)}^N$. Then by the property 3 in claim 1, for any $n \geq N$, $f_n(x) \in W_{(k,l)}^N$. By the closed-ness of $W_{(k,l)}^N$,

$$f(x) := \lim_{n \rightarrow \infty} f_n(x) \stackrel{\text{closed-ness}}{\in} W_{(k,l)}^N \subseteq U$$

□

- f is surjective.
- *Proof.* By the compact-ness of $[0, 1]$ and the Hausdorff-ness of $[0, 1] \times [0, 1]$, $f([0, 1])$ is closed

in $[0, 1] \times [0, 1]$. By claim 3,

$$f([0, 1]) = \overline{f([0, 1])} = [0, 1] \times [0, 1]$$

□

Then we're done for the basic case when $m = 2$. Now we suppose that there exists a continuous surjective map $\varphi : [0, 1] \rightarrow [0, 1]^m$. Then the proof is immediately done by the following:

$$[0, 1] \xrightarrow{f} [0, 1] \times [0, 1] \xrightarrow{id \times \varphi} [0, 1] \times [0, 1]^m = [0, 1]^{m+1}$$

i.e.: $(id \times \varphi) \circ f$ is continuous and surjective.

□

Proposition 1.26. *Injective Curve-but-Positive Area Proposition*

There exists an injective continuous map $f : [0, 1] \rightarrow [0, 1] \times [0, 1]$, s.t.: $f([0, 1])$ has non-zero 2-dimensional Lebesgue measure.

1.6 Baire Category Theorem

Definition 1.27. Nowhere Dense

Let X be a topological space and let $A \subseteq X$. We say that A is *nowhere dense* in X if there is **NO** open non-empty subset $U \subseteq X$ of X satisfying that $U \cap A$ is dense in U .

Example: The Cantor set is nowhere dense in $[0, 1]$. Moreover, it is uncountable.

Lemma 1.28. *Nowhere Dense-Characterization Lemma*

Let X be a topological space and let $A \subseteq X$ be a subset of X . The followings are equivalent:

1. *A is nowhere dense in X .*
2. *The interior of the closure $\overline{A}$ is empty, i.e.: $\overline{A}^\circ = \emptyset$.*

Proof. A nowhere dense in X

$$\Leftrightarrow \forall \emptyset \neq U \text{ open in } X, U \cap A \text{ is NOT dense in } U$$

$$\Leftrightarrow \forall \emptyset \neq U \text{ open in } X, \exists \emptyset \neq V \text{ open in } U, \text{ s.t.: } A \not\subseteq \overline{V}$$

$$\Leftrightarrow \forall \emptyset \neq U \text{ open in } X, \exists B \subsetneq U \text{ closed with } U \setminus B \cap A = \emptyset, \text{ s.t.: } A \not\subseteq B$$

$$\Leftrightarrow \forall \emptyset \neq U \text{ open in } X, \exists C \subseteq X \text{ closed with } A \subseteq C, \text{ s.t.: } U \not\subseteq C$$

$$\Leftrightarrow \overline{A}^\circ = \emptyset.$$

□

Theorem 1.29. *Baire Category Theorem*

Let X be one of the following:

1. *X is a complete metric space*
2. *X is a regionally compact Hausdorff space.*

Then the union of countably many nowhere dense subsets of X has empty interior.

1.7 Baire Spaces and Completely Metrizable Spaces

Definition 1.30. Baire Spaces

A topological space having the property that the union of countably many nowhere dense subsets of X has empty interior is called a *Baire space*.

Lemma 1.31. Baire Space Characterization Lemma

Let X be a topological space. The followings are equivalent:

1. X is a Baire space.
2. For each countable collection of open dense subsets $\{U_n\}_{n \in \mathbb{N}}$, the intersection $\bigcap_{n \in \mathbb{N}} U_n$ is dense.

Proof. X is a Baire space

$\Leftrightarrow$ for every family $\{W_n\}_{n \in \mathbb{N}}$ of nowhere dense subsets, $(\bigcup_{n \in \mathbb{N}} W_n)^\circ = \emptyset$

NowhereDense-CharacterizationLemma $\Leftrightarrow$ for every family $\{W_n\}_{n \in \mathbb{N}}$ of subsets with $(\overline{W_n})^\circ = \emptyset$, $(\bigcup_{n \in \mathbb{N}} W_n)^\circ = \emptyset$

$\Leftrightarrow$ for every family $\{A_n\}_{n \in \mathbb{N}}$ of closed subsets with $A_n^\circ = \emptyset$, $(\bigcup_{n \in \mathbb{N}} A_n)^\circ = \emptyset$

$\Leftrightarrow$ for every family $\{U_n\}_{n \in \mathbb{N}}$ of dense open subsets, $(\bigcup_{n \in \mathbb{N}} (X \setminus U_n))^\circ = \emptyset$

$\Leftrightarrow$ for every family $\{U_n\}_{n \in \mathbb{N}}$ of dense open subsets, $X \setminus \bigcup_{n \in \mathbb{N}} (X \setminus U_n)$ is dense

$\Leftrightarrow$ for every family $\{U_n\}_{n \in \mathbb{N}}$ of dense open subsets, $\bigcap_{n \in \mathbb{N}} U_n$ is dense. $\square$

Lemma 1.32. If X is a Baire space, then every open $V \subseteq X$ is also a Baire space.

Definition 1.33. Completely Metrizable Space

A topological space $(X, \mathcal{T})$ is called *completely metrizable* if there is a *complete* metric d on X , s.t.: $\mathcal{T} = \mathcal{T}_d$.

Theorem 1.34. Reformulation of the Baire Category Theorem.1

If X is a topological space that is completely metrizable, then X is a Baire space.

Corollary 1.35. $\mathbb{Q}$ is metrizable but NOT completely metrizable.

1.8 Application of the Baire Category Theorem to the Sequences of Maps

Proposition 1.36. Limit Map-Discontinuous Proposition

Let X be a topological space, let (M, d) be a metric space and let $(f_n : X \rightarrow M)_{n \in \mathbb{N}}$ be a sequence of continuous maps which converges pointwisely. If X is a Baire space, then the set of points at which the limit map $f : X \rightarrow M$ is continuous is dense in X .

Proof. Given $N \in \mathbb{N} \cup \{0\}$ and $\epsilon > 0$, we set

$$A_N(\epsilon) := \{x \in X \mid d(f_m(x), f_n(x)) \leq \epsilon, \forall m, n \geq N\}$$

- **Claim 1:** Each $A_N(\epsilon)$ is closed and $\bigcup_{n \in \mathbb{N} \cup \{0\}} A_N(\epsilon) = X$.
- *Proof.* Note that $A_N(\epsilon) = \bigcap_{m, n \geq N} ((d \circ ((f_m, f_n) : X \rightarrow M \times M)) : X \rightarrow \mathbb{R})^{-1}([- \epsilon, \epsilon])$. By the continuity of d , $A_N(\epsilon)$ is closed. The latter statement holds immediately from that f_n conv. pointwisely. $\square$

Given $\epsilon > 0$, set

$$U(\epsilon) := \bigcup_{N \in \mathbb{N} \cup \{0\}} (A_N(\epsilon))^\circ$$

By the Baire Space Characterization Lemma, it suffices to prove the following claim:

- **Claim 2:**
 1. For each $\epsilon > 0$, $U(\epsilon)$ is open and dense in X .
 2. f is continuous at every point in $\bigcap_{m \in \mathbb{N}_+} U(\frac{1}{m})$.
- *Proof.*
 1. It's clear that $U(\epsilon)$'s are open. As for the density of $U(\epsilon)$, for any open $V \subseteq X$ in X , we analyse in the following steps:
 - (a) $V \cap A_N(\epsilon)$ is closed in V .
 - (b) Since X is a Baire space, then by the open-ness of V , V is also a Baire space.
 - (c) By claim 1, $\{V \cap A_N(\epsilon)\}_{N \in \mathbb{N}}$ covers V and hence by (a)(b), none of $V \cap A_N(\epsilon)$ is nowhere dense in V .
 - (d) By the Nowhere Dense-Characterization Lemma, $\overline{V \cap A_N(\epsilon)}^\circ \neq \emptyset$.
 - (e) $(V \cap A_N(\epsilon))^\circ \neq \emptyset$ by (c)(d).

In summary, there exists $N \in \mathbb{N}$, s.t.: $V \cap A_N(\epsilon)$ contains a subset $W \subseteq V$ that is open in V and hence open in X . Therefore, $W \subseteq A_N(\epsilon)^\circ$.

2. For any $x \in \bigcap_{m \in \mathbb{N}_+} U(\frac{1}{m})$, it suffices to show that for any $\epsilon > 0$, there exists open $W \ni x$, s.t.: $d(f(x), f(y)) < \epsilon, \forall y \in W$. We first make a few preparations:
 - (a) Pick $k \in \mathbb{N}_+$ with $\frac{1}{k} < \epsilon/3$.
 - (b) $x \in U(\frac{1}{k}) \Rightarrow x \in A_n(\frac{1}{k})^\circ$.
 - (c) Since f_n is continuous and $A_n(\frac{1}{k})^\circ$ is open, then there exists open $W \ni x$ with $W \subseteq A_n(\frac{1}{k})$, s.t.: $d(f_n(x), f_n(y)) < \epsilon/3$ for any $y \in W$.

For any $z \in W$,

$$d(f(z), f_n(z)) := d(\lim_{m \rightarrow \infty} f_m(z), f_n(z)) \stackrel{d \text{ cont}}{=} \lim_{m \rightarrow \infty} d(f_m(z), f_n(z)) \leq \frac{1}{k} < \frac{\epsilon}{3}$$

which by triangle inequality gives $d(f(x), f(y)) < \epsilon$.

$\square$

$\square$

Chapter 2

Foundations of Topology

2.1 Topological Space

Definition 2.1. Topological Space

A topological space $(X, \mathcal{T})$ consists of a (nonempty) set X , and a family of subsets of X s.t.:

1. $\emptyset, X \in \mathcal{T}$.
2. $U, V \in \mathcal{T}$ implies $U \cap V \in \mathcal{T}$.
3. If $U_\alpha \in \mathcal{T} \forall \alpha \in \mathcal{A}$, then $\bigcup_{\alpha \in \mathcal{A}} U_\alpha \in \mathcal{T}$.

The elements in $\mathcal{T}$ are called open subsets of X . The $\mathcal{T}$ is called a topology on X .

Definition 2.2. Metrizable

$(X, \mathcal{T})$ is metrizable if $\mathcal{T}$ is the family of open subsets in (X, d) .

Example:

- Consider the indiscrete topology $(X, \mathcal{T}_{indis})$, where X contains more than one element and $\mathcal{T}_{indis} := \{\emptyset, X\}$. Then $\mathcal{T}_{indis}$ is a topology on X , but $(X, \mathcal{T}_{indis})$ is NOT metrizable. The reason is:
For any metric d defined on X , let $x, y \in X$ with $x \neq y$. Then take $\epsilon := d(x, y) > 0$. Then $B_{\epsilon/2}(x)$ is open in X , where $x \in B_{\epsilon/2}, y \notin B_{\epsilon/2}$. Hence, $B_{\epsilon/2}$ is neither $\emptyset$ nor X . $\square$
- Consider the cofinite topology $(X, \mathcal{T}_{cofin})$:

$$\mathcal{T}_{cofin} := \{U \mid X - U \text{ is a finite set}\} \cup \{\emptyset\}.$$

Then, $(X, \mathcal{T}_{cofin})$ is NOT a metrizable topology.

Definition 2.3. Topological Equivalence

Two metric spaces are topologically equivalent if they give rise to the same topology.

Example: $(\mathbb{R}^n, d_1)$, $(\mathbb{R}^n, d_2)$, and $(\mathbb{R}^n, d_\infty)$ are topologically equivalent.

Definition 2.4. Closed

Let $(X, \mathcal{T})$ be a topological space. Then $V \subset X$ is closed if $X - V \in \mathcal{T}$.

Proposition 2.5. *Let $(X, \mathcal{T})$ be a topological space. Then:*

1. $\emptyset, X$ are closed in X .
2. V_1, V_2 are closed in X implies $V_1 \cup V_2$ are closed in X .
3. If $\{V_\alpha \mid \alpha \in \mathcal{A}\}$ is a collection of closed subsets in X , then $\bigcap_{\alpha \in \mathcal{A}} V_\alpha$ is closed in X .

Definition 2.6. Convergence

A sequence $\{x_n\}$ of a topological space $(X, \mathcal{T})$ converges to $x \in X$ if $\forall$ open $U \ni x$, $\exists N$, s.t.: $x_n \in U \forall n \geq N$.

Example:

- The topological space induced from the metric space $(\mathbb{R}^n, d_2)$ is called a usual topology on $\mathbb{R}^n$. Then the convergence of sequence in $(\mathbb{R}^n, \mathcal{T}_{usual})$ is the usual convergence in analysis. For $\mathbb{R}^n$ or metric space, the limit of sequence (if exists) is unique. However, this is no longer true if the topology is NOT metrizable.
- Consider $(X, \mathcal{T}_{indis})$. For each $\{x_n\}$ in X , it converges to $x \in X$. For an open $U \ni x$, by the definition of $\mathcal{T}_{indis}$, $U = X$. Therefore, $x_n \in U = X \forall n \geq 1$.
- Consider $(X, \mathcal{T}_{cofin})$, where x is infinite. For each $\{x_n\}$ satisfying $x_m \neq x_n$ for any $m \neq n$. Then $\{x_n\} \rightarrow (\forall)x \in X$.
- Consider $(X, \mathcal{T}_{discrete})$. Then $\{x_n\} \rightarrow x \Leftrightarrow x_n = x \forall n \geq N$.

Remark: The limit of sequence may NOT be unique.

Proposition 2.7. *If $F \subset (X, \mathcal{T})$ is closed, then for any convergent sequence $\{x_n\}$ in F , the limit(s) are also in F .*

Proof. Let $\{x_n\}$ be a sequence in F that converges to x . Suppose that $x \in X - F$. Since F is closed in X , then $X - F$ is open in X . Then $\exists N$, s.t.: $x_n \in X - F \forall n \geq N$, contradiction! $\square$

Remark: The converse may not be true. Furthermore, if $(X, \mathcal{T})$ is metrizable, then the converse holds.

Counterexample(for the rmk): Consider the co-countable topological space $(X = \mathbb{R}, \mathcal{T}_{co-countable})$ where $\mathcal{T}_{co-countable} = \mathcal{T}_{co-co} := \{U \mid X - U \text{ is countable}\} \cup \{\emptyset\}$, and $X = \mathbb{R}$ is uncountable. Then note that $F := [0, 1] \subset X = \mathbb{R}$ is uncountable. Then $\forall \{x_n\} \subset F$, $x_n \rightarrow x \in F$. But $X - F \notin \mathcal{T}_{co-co}$, i.e.: F is NOT closed.

Definition 2.8. Interior, Closure

Let $(X, \mathcal{T})$ be a topological space and $A \subset X$ a subset. Then:

$$A^\circ := \bigcup_{U \subset A, U \text{ is open}} U,$$

$$\bar{A} := \bigcap_{A \subset V, V \text{ is closed}} V.$$

Furthermore, if $\bar{A}$, then we say A is dense in X .

Example:

- For $[a, b)^\circ = (a, b)$, $\overline{[a, b)} = [a, b]$.
- For $X = \mathbb{R}$, $\mathbb{Q}^\circ = \emptyset$ and $\overline{\mathbb{Q}} = \mathbb{R}$.
- Consider $(X, \mathcal{T}_{discrete})$. $S^\circ = S$, $\bar{S} = S$, $\forall S \subset X$.

Proposition 2.9. 1. A° is the largest open set of X contained in A ; $\bar{A}$ is the smallest closed subset of X containing A .

2. If $A \subset B$, then $A^\circ \subset B^\circ$ and $\bar{A} \subset \bar{B}$.

3. A is open in X is equivalent to $A^\circ = A$; A is closed is equivalent to $\bar{A} = A$.

Proposition 2.10.

$$\bar{A} = \{x \in X \mid \forall \text{open } U \ni x, U \cap A \neq \emptyset\}$$

Proof. Define $S := \{x \in X \mid \forall \text{open } U \ni x, U \cap A \neq \emptyset\}$. Then we want to show: $\bar{A} = S$.

- Step 1: Show that S is closed.
 $X - S = \{x \in X \mid \exists \text{open } U_x \ni x, \text{ s.t. : } U_x \cap A = \emptyset\}$. Take $x \in X - S$. Then $\exists U_x$ open, s.t.: $x \in U_x$ and $U_x \cap A = \emptyset$.

Claim: $U_x \subset X - S$: $\forall y \in U_x$ open, there exists an open set $V_y \ni y$, s.t.: $y \in V_y \subseteq U_x$. Hence, by the def of S , $y \in X - S$. Therefore, $U_x \subset X - S$.

Then $X - S = \{x \in X \mid \exists \text{open } U_x \ni x, \text{ s.t. : } U_x \cap A = \emptyset\} = \bigcup_{x \in X - S} \{x\} \subset \bigcup_{x \in X - S} U_x \stackrel{\text{claim}}{\subset} X - S$. Hence, $X - S = \bigcup_{x \in X - S} U_x$. Since U_x is open for all $x \in X - S$, then $X - S = \bigcup_{x \in X - S} U_x$ is open, i.e.: S is closed. Therefore, $S = \bar{S}$.

- Step 2: Show that $\bar{A} \subset S$.
 By definition of S , $A \subset S$. Hence, $\bar{A} \subset \bar{S} \stackrel{\text{step 1}}{=} S$.
- Step 3: Show that $\bar{A} = S$.
 Suppose $\exists z \in S - \bar{A} (\subset X - \bar{A})$. Then there exists an open set $W_z \ni z$, s.t.: $z \in W_z \subseteq X - \bar{A}$. Therefore, $z \notin S$ by the def of S . Contradiction!

□

Definition 2.11. Accumulation Point

Let $A \subset X$ be a subset in a topological space X . We call $x \in X$ are an accumulation point (limit point) of A if $\forall \text{open } U \subset X$ containing x , U satisfies $(U - \{x\}) \cap A \neq \emptyset$. Furthermore, we define the set of accumulation points of A by A' .

Example:

- There are some points in A but not in A' : $A = \{1, 2, \dots\}$, then any point in A is not in A' .
- There are some points in A' but not in A : $A = \{\frac{1}{n} \mid n \geq 1\}$, then $0 \in A'$ but $0 \notin A$.

Proposition 2.12.

$$\bar{A} = A \cup A'$$

Definition 2.13. Sequential Closure

Let A_s be the set of limits of any convergent sequence in A . Then A_s is called the sequential closure of A .

Lemma 2.14.

$$A \subset A_s \subset \bar{A}$$

Proof. • $\forall a \in A, \{a_n = a\} \rightarrow a \in A_s$. Hence, $A \subset A_s$;

- $\forall a \in A_s, \exists \{a_n\} \rightarrow a$, hence, $\forall \text{open } U \ni a, \exists N$, s.t.: $\{a_N, a_{N+1}, \dots\} \subset U \cap A \neq \emptyset$. Then $a \in \bar{A}$. Therefore, $A_s \subset \bar{A}$.

□

Proposition 2.15. Let (X, d) be a metric space. Then $A_s = \bar{A}$.

Proof. Let $a \in \bar{A}$. Then $\exists a_n \in B_\epsilon(a) \cap A$ for all $\epsilon > 0$. Hence, $a_n \rightarrow a$ as $n \rightarrow \infty$, i.e.: $a \in A_s$. Therefore,

$$\bar{A} \subset A_s \subset \bar{A}$$

which implies that $A_s = \bar{A}$.

□

Proposition 2.16. 1. $\overline{X - A} = X - A^\circ$;

2. $(X - B)^\circ = X - \bar{B}$.

Proof. 1. $x \in X - A^\circ \Leftrightarrow x \notin A^\circ \Leftrightarrow \forall \text{open } U \ni x, U \not\subset A \Leftrightarrow \forall \text{open } U \ni x, U \cap (X - A) \neq \emptyset \Leftrightarrow x \in \overline{X - A}$.

2. $x \in (X - B)^\circ \Leftrightarrow \exists \text{open } U \ni x$, s.t.: $U \subset (X - B) \Leftrightarrow U \not\subset B \Leftrightarrow x \notin \bar{B} \Leftrightarrow x \in X - \bar{B}$.

□

Definition 2.17. Boundary

The boundary of A is defined as $\partial A := \bar{A} - A^\circ$.

Proposition 2.18. $\partial A = \bar{A} \cap \overline{X - A}$

Proof. $\partial A := \bar{A} - A^\circ = \bar{A} \cap (X - A^\circ) \stackrel{\text{proposition 2.16.1}}{=} \bar{A} \cap \overline{X - A}$.

□

2.2 Functions on Topological space

Definition 2.19. Continuous Function

Let $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ be a map. Then the function f is continuous, if

$$U \in \mathcal{T}_Y \Rightarrow f^{-1}(U) \in \mathcal{T}_X.$$

Example:

- The identity map $id_X : (X, \mathcal{T}) \rightarrow (X, \mathcal{T})$ defined as $id_X(x) = x$ is continuous.
- The identity map $id : (X, \mathcal{T}_{discrete}) \rightarrow (X, \mathcal{T}_{indiscrete})$ defined as $x \mapsto x$ is continuous, since $id^{-1}(\emptyset) = \emptyset$ and $id^{-1}(X) = X$.
- The identity map $id : (X, \mathcal{T}_{indiscrete}) \rightarrow (X, \mathcal{T}_{discrete})$ defined as $x \mapsto x$ is NOT continuous.

Proposition 2.20. *If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are cont, then $g \circ f$ is cont.*

Proof. $\forall U \in \mathcal{T}_Z, g^{-1}(U) \in \mathcal{T}_Y$. Then $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U)) \in \mathcal{T}_X$. □

Proposition 2.21. *Suppose $f : X \rightarrow Y$ is cont between two topological spaces. Then $\{x_n\} \rightarrow x$ implies $\{f(x_n)\} \rightarrow f(x)$.*

Proof. $\forall$ open $U \ni f(x), f^{-1}(U) \ni x$ is open, then $\exists N > 0$, s.t.: $\{x_N, x_{N+1}, \dots\} \subseteq f^{-1}(U)$, i.e: $\{f(x_N), f(x_{N+1}), \dots\} \subseteq U$. Hence, $\{f(x_n)\}_{n \geq 1} \rightarrow f(x)$. □

Lemma 2.22. *Let X be a topological space and let $\{X_i\}_{i \in I}$ be a family of subsets of X . Then the followings are equivalent:*

1. *For any $U \subseteq X$, U is open in X if $U \cap X_i$ is open in $X_i \forall i$.*
 2. *For any $U \subseteq X$, U is closed in X if $U \cap X_i$ is closed in $X_i \forall i$.*
- Furthermore, if 1 or equivalently 2 holds, then one has*
3. *a map $f : X \rightarrow Y$ (Y topological) is continuous if $f|_{X_i} : X_i \rightarrow Y$ continuous.*

Proof. • $(1 \Rightarrow 2)$: $U \cap X_i$ closed in X_i implies that $X_i \setminus (U \cap X_i) = X_i \setminus U = (X \setminus U) \cap X_i$ is open in X_i . Then by 1, $X \setminus U$ is open in X , which follows that U is closed in X .

- $(2 \Rightarrow 1)$: Similar.
- $(1 \Rightarrow 3)$: For any open V in Y , since

$$(f|_{X_i})^{-1}(V) = f^{-1}(V) \cap X_i$$

is open in X_i , then by 1, $f^{-1}(V)$ is open in X . Hence, f is continuous. □

Definition 2.23. Local Finiteness A family $\{U_i\}_{i \in I}$ of subsets of X is *locally finite* if for any $x \in X$, there exists an open neighborhood $W \ni x$, s.t.:

$$|\{i \in I \mid U_i \cap W \neq \emptyset\}| < \infty$$

Proposition 2.24. *Pasting Proposition*

Let $f : X \rightarrow Y$ be a map between topological spaces.

1. If there is an open cover $\{W_i\}_{i \in I}$ of X , s.t.: $f|_{W_i}$ is continuous, then f is continuous.
2. If there is a locally finite closed cover $\{A_i\}_{i \in I}$ of X , s.t.: $f|_{A_i}$ is continuous, then f is continuous.
3. If X is Hausdorff and if there exists a locally finite cover $\{A_i\}_{i \in I}$ of X , s.t.: A_i is compact and $f|_{A_i}$ is continuous, then f is continuous.

Proof. 1. Note that if $f|_{U_i}$ is continuous, then for any open V in Y , one has $(f|_{U_i})^{-1}(V) = f^{-1}(V) \cap U_i$ is open in U_i . Then by lemma 3.22, 1 immediately holds.

2. Similarly as 1.

3. holds immediately by 2.

□

Definition 2.25. Open and Closed Maps

Let $f : X \rightarrow Y$ be a continuous map between two topological spaces.

- f is called *open* if for any $U \in \mathcal{T}_X$, $f(U) \in \mathcal{T}_Y$.
- f is called *closed* if for any V closed in X , $f(V)$ is closed in Y .

Proposition 2.26. *Let X be topological.*

- U is open in $X \Rightarrow i : U \rightarrow X$, inclusion map, is open.
- A is closed in $X \Rightarrow i : A \rightarrow X$, inclusion map, is closed.

Definition 2.27. Homeomorphism

A homeomorphism between two topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ is a bijection $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ such that f and f^{-1} are both continuous.

Definition 2.28. Proper Maps

A map $f : X \rightarrow Y$ between two topological spaces is called *proper* if the preimage of every compact subset of Y is compact in X .

2.3 Subspace Topology

Definition 2.29. Subspace Topology

Let $A \subset X$ be a nonempty set. The subspace topology of A is defined as:

1. $\mathcal{T}_A := \{U \cap A \mid U \in \mathcal{T}_X\}$.
2. The coarsest topology on A such that the inclusion map $i : (A, \mathcal{T}_A) \rightarrow (X, \mathcal{T}_X)$, $i(x) = x$ is cont; (We say the topology $\mathcal{T}_1$ is coarser than $\mathcal{T}_2$ or $\mathcal{T}_2$ is finer than $\mathcal{T}_1$ if $\mathcal{T}_1 \subset \mathcal{T}_2$.
e.g.: $\mathcal{T}_{discrete}$ is the finest topology and $\mathcal{T}_{indiscrete}$ is the coarsest topology.)
3. $\mathcal{T}_A$ is the (unique) topology such that for all $(Y, \mathcal{T}_Y)$,

$$f : (Y, \mathcal{T}_Y) \rightarrow (X, \mathcal{T}_X) \text{ is cont} \Leftrightarrow i \circ f : (Y, \mathcal{T}_Y) \rightarrow (X, \mathcal{T}_X) \text{ is cont.}$$

Proposition 2.30. *The definition 1 and 2 of subspace topology are equivalent.*

Proof. For all $S \subseteq X$, $i^{-1}(S) = S \cap A \Leftrightarrow \mathcal{T}_A = \{U \cap A \mid U \in \mathcal{T}_X\}$. □

Proposition 2.31. *The definition 2 and 3 of subspace topology are equivalent.*

Proof. • (2 $\Rightarrow$ 3):

- ($\Rightarrow$): For any $U \in \mathcal{T}_X$, $(i \circ f)^{-1}(U) = f^{-1}(i^{-1}(U)) = f^{-1}(U \cap A) \in \mathcal{T}_Y$. Therefore, f is cont.
- ($\Leftarrow$): For any $U' \in \mathcal{T}_A$, $\exists U'' \in \mathcal{T}_X$, s.t.: $U' = U'' \cap A$. Hence,

$$f^{-1}(U') = f^{-1}(U'' \cap A) = f^{-1}(i^{-1}(U'')) = (i \circ f)^{-1}(U'') \in \mathcal{T}_Y.$$

Therefore, $i \circ f$ is cont.

- (3 $\Rightarrow$ 2): Take $Y = A$ and $f = id_A$. Since id_A is cont, then we're done. □

Proposition 2.32. 1. *The definition 1 of subspace topology defines a topology of A .*
2. *Closed sets of A under subspace topology are of the form $V \cap A$, where V is closed in X .*

Proposition 2.33. *Suppose $(A, \mathcal{T}_A) \subseteq (X, \mathcal{T}_X)$ is a subspace topology and $B \subseteq A \subseteq X$. Then:*

1. $\bar{B}^A = \bar{B}^X \cap A$;
2. $(B^\circ)^X \subseteq (B^\circ)^A$.

Proof. 1. By proposition 2.26.2, $\bar{B}^X \cap A$ is closed in A . Hence, $\bar{B}^A \subseteq \bar{B}^X \cap A$, since $B = B \cap A \subseteq \bar{B}^X \cap A$.

On the other hand, since $\bar{B}^A$ is closed in A , then $\bar{B}^A = V \cap A$ for some closed V . Then $B \subseteq \bar{B}^A \subseteq V$. Hence, $\bar{B}^X \subseteq \bar{V}^X = V$. Therefore, $\bar{B}^X \cap A \subseteq V \cap A = \bar{B}^A$.

Then we have $\bar{B}^A = \bar{B}^X \cap A$.

2. For any $x \in (B^\circ)^X$, $\exists U_x \in \mathcal{T}_X$, s.t.: $x \in U_x \subseteq B$. Then $U_x \cap A \in \mathcal{T}_A$ and $x \in U_x \cap A \subseteq B$. Hence, $x \in (B^\circ)^A$. Therefore, $(B^\circ)^X \subseteq (B^\circ)^A$. □

2.4 Basis of Topology

Definition 2.34. Basis

Let $(X, \mathcal{T})$ be a topological space. A family of subsets $\mathcal{B}$ in X is a basis of $\mathcal{T}$ if

- $\mathcal{B} \subseteq \mathcal{T}$, i.e.: everything in $\mathcal{B}$ is open;
- Every $U \in \mathcal{T}$ can be written as union of elements in $\mathcal{B}$, i.e.: $\forall U \in \mathcal{T}, \exists \{B_i\} \subseteq \mathcal{B}$, s.t.: $U = \bigcup_i B_i$.

Example:

- $\mathcal{B} = \mathcal{T}$ is a basis of $\mathcal{T}$.
- For $X = \mathbb{R}^n$, $\mathcal{B} := \{B_r(\mathbf{x}) \mid \mathbf{x} \in \mathbb{Q}^n, r \in \mathbb{Q} \cap (0, \infty)\}$ is a basis of the usual topology of X . For any interval $(a, b) \subseteq \mathbb{R}^1$, $(a, b) = \bigcup_{i \in I} (p_i, q_i)$ for $p_i, q_i \in \mathbb{Q}$. Therefore, $\mathcal{B}$ is countable.

Definition 2.35. Second-countability

If $(X, \mathcal{T})$ is a countable basis, e.g.: $\mathbb{R}^n$, then we say $(X, \mathcal{T})$ is a second-countable space.

Proposition 2.36. *Let $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ be topological spaces, and let $\mathcal{B}$ be a basis of $\mathcal{T}_Y$. Then:*

$$f : X \rightarrow Y \text{ is cont} \Leftrightarrow f^{-1}(B) \text{ is open in } X \quad \forall B \in \mathcal{B}.$$

In other words, to check whether f is cont, it suffices to check $f^{-1}(B)$ is open for all $B \in \mathcal{B}$.

Proof. • $(\Rightarrow)$: Since $\mathcal{B} \subseteq \mathcal{T}_Y$, then by the definition of continuity, $f^{-1}(B)$ is open for all $B \in \mathcal{B}$.
 • $(\Leftarrow)$: Let $U \in \mathcal{T}_Y$. Then $U = \bigcup_{i \in I} B_i$, where $B_i \in \mathcal{B}$. Hence, $f^{-1}(U) = f^{-1}(\bigcup_{i \in I} B_i) = \bigcup_{i \in I} f^{-1}(B_i)$ is open, since $f^{-1}(B_i)$ are open. □

Corollary 2.37. *Let $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ be a bijection. Suppose there is a basis $\mathcal{B}_X$ of $\mathcal{T}_X$ such that $\{f(B) \mid B \in \mathcal{B}_X\}$ forms a basis of $\mathcal{T}_Y$. Then $X \cong Y$ are homeomorphic.*

Proof. Suppose $W \in \mathcal{T}_Y$. Then by hypothesis, $W = \bigcup_{i \in I} f(B_i)$, $B_i \in \mathcal{B}_X$, which can imply that $f^{-1}(W) = \bigcup_{i \in I} B_i \in \mathcal{T}_X$. Hence, f is cont.

On the other hand, similarly, f^{-1} is cont. □

Proposition 2.38. *Let X be a set, and let $\mathcal{B}$ be a collection of subsets satisfying:*

1. X is a union of sets in $\mathcal{B}$, i.e.: every $x \in X$ lies in some $B_x \in \mathcal{B}$.
2. The intersection $B_1 \cap B_2 \quad \forall B_1, B_2 \in \mathcal{B}$ is a union of sets in $\mathcal{B}$, i.e.: $\forall B_1, B_2 \in \mathcal{B}, \forall x \in B_1 \cap B_2, \exists B'_x \in \mathcal{B}$, s.t.: $x \in B'_x \subseteq B_1 \cap B_2$.

Then the collection of subsets $\mathcal{T}_{\mathcal{B}}$, formed by taking any union of sets in $\mathcal{B}$, is a topology, and $\mathcal{B}$ is a basis of $\mathcal{T}_{\mathcal{B}}$.

Proof. Firstly, for $x \in X$, $B_x \in \mathcal{B}$, by the hypothesis 1, $X = \bigcup_{x \in X} B_x \in \mathcal{T}_{\mathcal{B}}$.

Now we show that $\mathcal{T}_{\mathcal{B}}$ is closed under intersection:

Let $T_1, T_2 \in \mathcal{T}_{\mathcal{B}}$ and let $x \in T_1 \cap T_2$. Then T_i can be written as a union of subsets in $\mathcal{B}$. Hence, $x \in B_1 \subseteq T_1$ for some $B_1 \in \mathcal{B}$ and $x \in B_2 \subseteq T_2$ for some $B_2 \in \mathcal{B}$. Then $x \in B_1 \cap B_2$, and then $x \in B_x \subseteq B_1 \cap B_2$ for some $B_x \in \mathcal{B}$. Therefore, $\bigcup_{x \in B_1 \cap B_2} \{x\} \subseteq \bigcup_{x \in B_1 \cap B_2} B_x \subseteq B_1 \cap B_2$. On the other hand, it's clear that $\bigcup_{x \in B_1 \cap B_2} B_x \supseteq B_1 \cap B_2$. Therefore, $\bigcup_{x \in B_1 \cap B_2} B_x = B_1 \cap B_2 \in \mathcal{T}_{\mathcal{B}}$.

Finally, the property that $\mathcal{T}_{\mathcal{B}}$ is closed under union operations can be checked directly, which completes the proof. $\square$

2.5 Product Space and Disjoint Union

Definition 2.39. Product Topology

Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be two topological spaces. Consider the family of subsets in $X \times Y$:

$$\mathcal{B}_{X \times Y} := \{U \times V \mid U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}.$$

It forms a basis of a topology on $X \times Y$. The induced topology from $\mathcal{B}_{X \times Y}$ is called product topology.

Proposition 2.40. $\mathcal{B}_{X \times Y}$ does form a basis of a topology of $X \times Y$.

Proof. We apply proposition 2.32 to check whether $\mathcal{B}_{X \times Y}$ forms a basis:

1. For any $(x, y) \in X \times Y$, we imply $x \in X$ and $y \in Y$. Note that $X \in \mathcal{T}_X$ and $Y \in \mathcal{T}_Y$, we imply $(x, y) \in X \times Y \in \mathcal{B}_{X \times Y}$.
2. Suppose $U_1 \times V_1, U_2 \times V_2 \in \mathcal{B}_{X \times Y}$. Then $(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2)$, where $(U_1 \cap U_2) \in \mathcal{T}_X, (V_1 \cap V_2) \in \mathcal{T}_Y$. Hence, $(U_1 \times V_1) \cap (U_2 \times V_2) \in \mathcal{B}_{X \times Y}$.

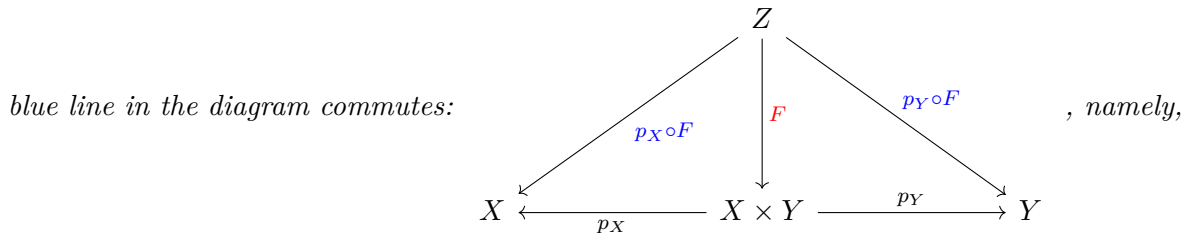
$\square$

Proposition 2.41. Let $X \times Y$ be endowed with product topology. Then the projection mappings defined as:

$$p_X : X \times Y \rightarrow X \text{ with } p_X(x, y) := x \text{ and } p_Y : X \times Y \rightarrow Y \text{ with } p_Y(x, y) := y$$

are cont. Moreover,

1. The product topology is the coarsest topology on $X \times Y$ s.t.: p_X and p_Y are both cont.
2. Let Z be a topological space. Then the product topology is the unique topology that the red and



the mapping $F : Z \rightarrow X \times Y$ is cont iff both $p_X \circ F : Z \rightarrow X$ and $p_Y \circ F : Z \rightarrow Y$ are cont.

Proof. For any open U , $p_X^{-1}(U) = U \times Y \in \mathcal{B}_{X \times Y} \subseteq \mathcal{T}_{X \times Y}$, i.e.: $p_X^{-1}(U)$ is open. Similar for p_Y . Hence, p_X and p_Y are cont. Moreover,

1. It suffices to show that any topology $\mathcal{T}$ meeting the condition in 2 must contain $\mathcal{T}_{product}$. Then For any $U \in \mathcal{T}_X$ and $V \in \mathcal{T}_Y$, $p_X^{-1}(U) = U \times Y \in \mathcal{T}$ and $p_Y^{-1}(V) = X \times V \in \mathcal{T}$. Hence, $(U \times Y) \cap (X \times V) = U \times V \in \mathcal{T}$, which implies $\mathcal{B}_{X \times Y} \subseteq \mathcal{T}$. Since $\mathcal{T}$ is closed for union operation on subsets, we imply $\mathcal{T}_{product} \subseteq \mathcal{T}$.

2. • **Step 1:** Show that $\mathcal{T}_{product}$ satisfies the commutative diagram.

- ($\Rightarrow$): F, p_X, p_Y are cont, which implies that $p_X \circ F$ and $p_Y \circ F$ are cont.
- ($\Leftarrow$): For any $U \in \mathcal{T}_X$, $V \in \mathcal{T}_Y$, $F^{-1}(U \times V) = (p_X \circ F)^{-1}(U) \cap (p_Y \circ F)^{-1}(V)$. Since $p_X \circ F$ and $p_Y \circ F$ are cont, then $(p_X \circ F)^{-1}(U)$ and $(p_Y \circ F)^{-1}(V)$ are open.

- **Step 2:** Show that uniqueness of $\mathcal{T}_{product}$.

Let $\mathcal{T}$ be another topology $X \times Y$ satisfying the commutative diagram:

- Take $Z = (X \times Y, \mathcal{T})$ and consider the identity mapping $F = id : Z \rightarrow Z$, which is cont. Hence, $p_X \circ F$ and $p_Y \circ F$ are cont, i.e.: p_X, p_Y are cont. By 1, $\mathcal{T}_{product} \subseteq \mathcal{T}$.
- Take $Z = (X \times Y, \mathcal{T}_{product})$ and consider the identity mapping $F = id : Z \rightarrow Z$. Note that $p_X \circ F = p_X$ and $p_Y \circ F = p_Y$, which are cont. Hence, $F : (X \times Y, \mathcal{T}_{product}) \rightarrow (X \times Y, \mathcal{T})$ is a cont identity mapping, which implies $U = id^{-1}(U) \in \mathcal{T}_{product}$ for all $U \in \mathcal{T}$. Therefore, $\mathcal{T} \subseteq \mathcal{T}_{product}$.

Hence, $\mathcal{T} = \mathcal{T}_{product}$.

□

Definition 2.42. Disjoint Union

Let $X \times Y$ be two topological spaces. Then the disjoint union of X and Y is:

$$X \sqcup Y := (X \times \{0\}) \cup (Y \times \{1\})$$

where $\mathcal{T}_{X \sqcup Y}$ consists of all $U \subseteq X \sqcup Y$ satisfying:

- $U \cap (X \times \{0\})$ is open in $X \times \{0\}$;
- $U \cap (Y \times \{1\})$ is open in $Y \times \{1\}$,

is a topology.

In other words, S is open in $X \sqcup Y$ iff S can be expressed as $S = (U \times \{0\}) \cup (V \times \{1\})$, where $U \subseteq X$ is open and $V \subseteq Y$ is open.

Chapter 3

Properties of Topological Spaces

3.1 Hausdorff Property

Definition 3.1. First Separation Axiom, T_1

A topological space $(X, \mathcal{T}_X)$ is said to satisfy the first separation axiom (or to be a T_1 space) if for every pair of distinct points $x, y \in X$, there exists an open set $U \subseteq X$, s.t.: $x \in U$ and $y \notin U$.

Proposition 3.2. A topological space X satisfies the first separation axiom (T_1) iff every singleton set $\{x\}$ is closed in X .

Proof. • $(\Rightarrow)$: Fix $x \in X$. For any $y \in X - \{x\}$, there is an open $U_y \ni y$ such that $x \notin U_y$.
Then:

$$X - \{x\} = \bigcup_{y \neq x} \{y\} \subseteq \bigcup_{y \neq x} U_y \subseteq X - \{x\}.$$

Hence, $X - \{x\} = \bigcup_{y \neq x} U_y$, where U_y is open. Therefore, $\{x\}$ is closed.

- $(\Leftarrow)$: For any $x, y \in X$ with $x \neq y$, set $U := X - \{y\}$, which is open, containing x , and not containing y . Hence, X is a T_1 space. □

Definition 3.3. Hausdorff Space

A topological space X is called Hausdorff (or said to satisfy the second separation axiom, T_2) if for every pair of distinct points $x, y \in X$, there exists open neighborhoods U of x and V of y such that $U \cap V = \emptyset$.

Example (A topological space that is a T_1 space may not necessarily be second separable):

Consider $\mathcal{T}_{co-finite}$. Then X is a T_1 space but NOT Hausdorff if X is infinite:

Suppose on the contrary that for given $x \neq y$, there are two open sets U, V such that $x \in U$ and $y \in V$, and let $U \cap V = \emptyset$, which implies $X = X - (U \cap V) = (X - U) \cup (X - V)$. Then the union of two finite sets equals X which is infinite. Contradiction. □

Proposition 3.4. If the topological space $(X, \mathcal{T})$ is Hausdorff, then all sequences $\{x_n\}$ in X has at most one limit.

Proof. Suppose on the contrary that $\{x_n\} \rightarrow a$ and $\{x_n\} \rightarrow b$ with $a \neq b$. Then by Hausdorffness, there are two open $U, V \in \mathcal{T}$ such that $U \cap V = \emptyset$ and $a \in U, b \in V$. Then $\exists N, M > 0$, s.t.: $\{x_N, x_{N+1}, \dots\} \subseteq U$ and $\{x_M, x_{M+1}, \dots\} \subseteq V$. Take $K := \max\{M, N\} + 1$. Then $x_K \in U \cap V = \emptyset$, contradiction. $\square$

Proposition 3.5. *Let X, Y be two Hausdorff spaces. Then the product space $X \times Y$, equipped with the product topology, is Hausdorff.*

Proof. For any $(x_1, y_1) \neq (x_2, y_2) \in X \times Y$, either $x_1 \neq x_2$ or $y_1 \neq y_2$. WLOG, let $x_1 \neq x_2$. By the Hausdorffness, there are two sets $U, V \subseteq X$, s.t.: $x_1 \in U, x_2 \in V$ and $U \cap V = \emptyset$. Then $(x_1, y_1) \in U \times Y, (x_2, y_2) \in V \times Y$, and $(U \times Y) \cap (V \times Y) = \emptyset$. $\square$

Definition 3.6. Topological Embedding

A continuous injective map $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ is called a topological embedding if $f : X \rightarrow f(X)$ is a homeomorphism onto its image, where $f(X)$ is equipped with the subspace topology from Y .

Remark: Equivalently, f is an embedding if it is cont, injective, and for every open set $U \subseteq X$, there exists an open set $V \subseteq Y$ such that $U = f^{-1}(V) \cap X$.

Proposition 3.7. *Let $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ be a topological embedding. If Y is Hausdorff, then X is Hausdorff.*

Proof. f is embedding implies that $f : X \rightarrow f(X)$ is homeomorphism onto its image, where $f(X)$ is equipped with the subspace topology from Y . Take distinct points $a, b \in X$. Then $f(a) \neq f(b)$ in Y .

By the Hausdorffness of Y , there are two open sets $U, V \subseteq Y$, s.t.: $f(a) \in U, f(b) \in V$, and $U \cap V = \emptyset$.

Hence, $U \cap f(X)$ and $V \cap f(X)$ are the open neighborhoods of $f(a)$ and $f(b)$ in $f(X)$, respectively, with $(U \cap f(X)) \cap (V \cap f(X)) = \emptyset$.

Since $f : X \rightarrow f(X)$ is a homeomorphism, then $f^{-1}(U \cap f(X))$ and $f^{-1}(V \cap f(X))$ are two open neighborhoods of a and b , respectively, with $f^{-1}(U \cap f(X)) \cap f^{-1}(V \cap f(X)) = \emptyset$. $\square$

Remark: If f is only cont and injective, the conclusion may fail. For example, let X be any non-Hausdorff space and let $f : X \rightarrow \{*\}$ be the constant injective map into a one-point Hausdorff space. Then f is cont and injective, but X is NOT Hausdorff.

Corollary 3.8. *Let $f : X \rightarrow Y$ be a homeomorphism. Then X is Hausdorff iff Y is Hausdorff. In particular, Hausdorffness is a topological property, i.e.: a property preserved under homeomorphism.*

3.2 Connectivity

Definition 3.9. Connected Space

A topological space $(X, \mathcal{T})$ is said to be disconnected if there exist non-empty open sets $U, V \in \mathcal{T}$

such that $U \cap V = \emptyset, U \cup V = X$.

If no such pair exists, then X is called connected.

Proposition 3.10. *For a topological space $(X, \mathcal{T})$, the following are equivalent:*

1. X is connected.
2. The only subsets of X which are both open and closed are $\emptyset$ and X .
3. Every cont fct $f : X \rightarrow \{0, 1\}$ (where $\{0, 1\}$ is equipped with the discrete topology) is constant.

Proof. • (1 $\Rightarrow$ 2): Suppose $U \subseteq X$ is both open and closed. Then U and $X - U$ are disjoint open sets with $U \cup (X - U) = X$. By connectivity of X , either $U = \emptyset$ or $X - U = \emptyset$. Hence, $U = \emptyset$ or $U = X$.

• (2 $\Rightarrow$ 3): Let $f : X \rightarrow \{0, 1\}$ be cont. Define $U := f^{-1}(\{0\})$ and $V := f^{-1}(\{1\})$. Then U, V are open disjoint, and $U \cup V = X$. By 2, either $(U, V) = (X, \emptyset)$ or $(U, V) = (\emptyset, X)$. Thus, f is constant.

• (3 $\Rightarrow$ 2): Suppose $U \subseteq X$ is both open and closed. Define

$$f(x) := \begin{cases} 0, & \text{if } x \in U \\ 1, & \text{otherwise.} \end{cases}$$

This map f is cont. Then by 3, f is constant. Hence, $U = \emptyset$ or $U = X$.

• (2 $\Rightarrow$ 1): Suppose on the contrary that X is disconnected, i.e.: there are disjoint non-empty open sets $U, V \subseteq X$ with $U \cup V = X$. Then $U = X - V$ is both open and closed, contradicting 2.

□

Corollary 3.11. *The interval $[a, b] \subseteq \mathbb{R}$ is connected.*

Proof. Suppose on the contrary that there exists a cont fct $f : [a, b] \rightarrow \{0, 1\}$ that takes 2 values. Construct the mapping $\tilde{f} : [a, b] \xrightarrow{f} \{0, 1\} \xrightarrow{i} \mathbb{R}$ with $\tilde{f} = i \circ f$.

Note that $\{0, 1\} \subseteq \mathbb{R}$ denotes the subspace topology, we imply the inclusion mapping $i : \{0, 1\} \rightarrow \mathbb{R}$ with $s \mapsto s$ is cont. Then the composition of cont mappings is cont, i.e.: $\tilde{f}$ is cont. Since f can take two values, there are $p, q \in [a, b]$, s.t.: $\tilde{f}(p) = (i \circ f)(p) = 0$ and $\tilde{f}(q) = (i \circ f)(q) = 1$. Then by IVT, there is $r \in [a, b]$ such that $\tilde{f}(r) = (i \circ f)(r) = 1/2$. Hence, $f(r) = 1/2$, contradiction. □

Definition 3.12. Connected Subset

A non-empty subset $S \subseteq X$ is connected if $(S, \mathcal{T}_S)$, where $\mathcal{T}_S$ denotes the subspace topology on S , is a connected topological space.

Equivalently, $S \subseteq X$ is connected if, whenever $U, V \subseteq X$ are open sets such that $S \subseteq U \cup V$, $(U \cap V) \cap S = \emptyset$, then it follows that either $U \cap S = \emptyset$ or $V \cap S = \emptyset$.

Proposition 3.13. *If $f : X \rightarrow Y$ is cont and $A \subseteq X$ is connected, then $f(A)$ is connected. In other words, the cont image of a connected set is connected.*

Proof. Suppose $U, V \subseteq Y$ are open s.t.: $f(A) \subseteq U \cup V$ and $(U \cap V) \cap f(A) = \emptyset$. Then $A \subseteq f^{-1}(U) \cup f^{-1}(V)$ and $(f^{-1}(U) \cap A) \cap (f^{-1}(V) \cap A) = \emptyset$. By the connectivity of A , either $f^{-1}(U) \cap A = \emptyset$ or $f^{-1}(V) \cap A = \emptyset$. Therefore, either $f(A) \cap U = \emptyset$ or $f(A) \cap V = \emptyset$. $\square$

Proposition 3.14. *Let $B, A_i \ i \in I$ be connected subsets of X , s.t.: $B \cap A_i \neq \emptyset$ for all $i \in I$. Then $B \cup (\bigcup_{i \in I} A_i)$ is connected.*

Proof. Consider a cont fct $f : B \cup (\bigcup_{i \in I} A_i) \rightarrow \{0, 1\}$. Let $f_i := f|_{A_i} : A_i \xrightarrow{i} B \cup (\bigcup_{i \in I} A_i) \xrightarrow{f} \{0, 1\}$ and $f_B := f|_B : B \xrightarrow{i} B \cup (\bigcup_{i \in I} A_i) \xrightarrow{f} \{0, 1\}$. Then by the connectivity of B, A_i, f_i, f_B are constant fcts.

Suppose $f_B \equiv 1$. Then $f|_{B \cap A_i} \equiv 1$ for all $i \in I$. Hence, $f_i \equiv 1$ for all $i \in I$. Therefore, $f \equiv 1$ is a constant fct. $\square$

Corollary 3.15. *If $\{A_i\}_{i \in I}$ is a family of connected subsets of X s.t.: for all $i, j \in I$,*

$$A_i \cap A_j \neq \emptyset,$$

then the union $\bigcup_{i \in I} A_i$ is connected.

Proposition 3.16. *If X and Y are connected spaces, then the product space $X \times Y$, equipped with the product topology, is connected.*

Proof. Fix $y_0 \in Y$. The subset $B := X \times \{y_0\} \subseteq X \times Y$ is homeomorphic to X , hence connected. For any $x \in X$, the subset $C_x := \{x\} \times Y \subseteq X \times Y$ is homeomorphic to Y , hence connected. Moreover, for any $x \in X$, $B \cap C_x = \{(x, y_0)\} \neq \emptyset$. Therefore, by proposition 3.14, the union $B \cup (\bigcup_{x \in X} C_x) = X \times Y$ is connected. $\square$

Definition 3.17. Path-Connected

Let $(X, \mathcal{T})$ be a topological space.

- **(Path):** A path between two points $x, y \in X$ is a cont map $p : [0, 1] \rightarrow X$ with $p(0) = x$ and $p(1) = y$.
- **(Path-connected Space):** X is said to be path-connected if any two points in X can be joined by a path.
- **(Path-connected Subset):** A subset $A \subseteq X$ is path-connected if it is path-connected with respect to the subspace topology. Equivalently, A is path-connected for every pair of points $x, y \in A$, there is a cont map $p : [0, 1] \rightarrow X$ with $p(0) = x$ and $p(1) = y$.

Proposition 3.18. *All path-connected spaces are connected.*

Proof. Fix $x_0 \in X$. For any $y \in X$, by path-connected-ness, there is a cont map

$$p_y : [0, 1] \rightarrow X \text{ with } p_y(0) = x_0 \text{ and } p_y(1) = y.$$

Define $C_y := p_y([0, 1]) \subseteq X$. By proposition 3.13, each C_y is connected. Moreover, for any $y_1, y_2 \in X$, $\{x_0\} \subseteq C_{y_1} \cap C_{y_2} \neq \emptyset$. Hence, by corollary 3.15, $X = \bigcup_{y \in X} C_y$ is connected. $\square$

Proposition 3.19. *If $A \subseteq B \subseteq \bar{A}$ and A is connected, then B is connected.*

Proof. Suppose on the contrary that X is not connected. Then there are two disjoint non-empty open sets $U, V \subseteq X$, s.t.: $U \cup V = X$ (therefore, $(*)$: U and V are closed in X).

Then $A = A \cap X = A \cap (U \cup V) = (A \cap U) \cup (A \cap V)$. Hence, by the connectivity of A , WLOG, $A \subseteq U$. Then $X = \bar{A} \cap X \subseteq \bar{U} \cap X \stackrel{(*)}{=} U$, contradiction. $\square$

Example (Topologist's comb):

The converse of proposition 3.18 is false.

Consider the subset $X \subseteq \mathbb{R}^2$ defined by

$$X = ([0, 1] \times \{0\}) \cup \left(\bigcup_{n \in \mathbb{N}_+} \left\{ \frac{1}{n} \right\} \times [0, 1] \right) \cup \{(0, 1)\}$$

called Topologist's comb, which is connected but NOT path-connected.

- X is NOT path-connected:

Suppose that there is a path $p : [0, 1] \rightarrow X$ with $p(0) = (0, 1)$ and $p(1) = (x, y) \in X - \{(0, 1)\}$.

Define $A := \{t \in [0, 1] \mid p(t) = (0, 1)\}$.

- A is closed, since it is the preimage of the closed singleton $\{(0, 1)\}$.
- A is open: fix $t_0 \in A$. By the continuity of p , there exists $\delta > 0$ such that $\|p(t) - (0, 1)\| < \frac{1}{2}$, $t \in [0, 1] \cap (t_0 - \delta, t_0 + \delta)$. Hence, $p(t)$ cannot lie on the x-axis. Therefore, by the def of X , x-coordinate of $p(t)$ must belong to $\{0\} \cup \{1/n \mid n \geq 1\}$.
Let $I := [0, 1] \cap (t_0 - \delta, t_0 + \delta)$. Consider the cont fct $f := \pi_x \circ p : I \rightarrow \mathbb{R}$ where $\pi_x(a, b) = a$ is the x-coordinate. Since I is connected, then $f(I)$ is connected. But $f(I) \subseteq \{0\} \cup \{1/n \mid n \geq 1\}$, whose only connected subsets are singletons. Since $f(t_0) = 0$, then we deduce $f(I) = \{0\}$. Hence, $p(t) = (0, 1)$ for all $t \in I$, i.e.: $I \subseteq A$, which means A is open.

Therefore, A is non-empty, closed, and open in the connected interval $[0, 1]$, so $A = [0, 1]$. Hence, any such path is constant, and no path connects $(0, 1)$ to another point of X .

- X is connected. The proof is omitted here, but it relies on showing that any separation would disconnect the horizontal interval from the vertical combs, which is impossible.

Definition 3.20. Concatenation

Let X be a topological space and let $\alpha, \beta : [0, 1] \rightarrow X$ be two paths with $\alpha(1) = \beta(0)$. Then define the *concatenation* of α and β , denoted as $\alpha * \beta$, by $\alpha * \beta : [0, 1] \rightarrow X$ with

$$\alpha * \beta(t) := \begin{cases} \alpha(2t) & , \text{if } t \in [0, 1/2] \\ \beta(1 - 2t) & , \text{if } t \in [1/2, 1] \end{cases}$$

Definition 3.21. Path-Equivalence

Let $x, y \in X$, topological space. We say x and y are path-equivalent if there exists a path $\gamma : [0, 1] \rightarrow X$, s.t.:

$$\gamma(0) = x, \gamma(1) = y$$

Proposition 3.22. *For every topological X , path-equivalence defines an equivalence relation.*

Definition 3.23. Path-components

Let X be a topological space. We call the path-equivalence classes of points the *path-components* of X , denoted as $\Pi_0(X)$.

In particular, if X is non-empty, consisting of a simple path-component, then we write $\Pi_0(X) = 0$.

Proposition 3.24. Path-Component Proposition

Let X be a topological space.

1. X is the disjoint union of its path-components.
2. Each path-component of X is path-connected.
3. Each path-component C of X is a maximal path-connected subsets, i.e.: if $C \subsetneq D$, then D is NOT path-connected.
4. Let $A \subseteq X$ be a subset of X .
 - (a) If no point in A is path-equivalent to a point in $X \setminus A$, then A is the union of path-components of X .
 - (b) If A is path-connected, then A is contained in a path-component of X .
 - (c) If A is path-connected and if no point in A is path-component to a point in $X \setminus A$, then A is a path-component of X .

Proof. It's trivial for 1,2,3,4(b),4(c). As for 4(a), suppose on the contrary that there are path-components $\{C_i\}_{i \in I}$, s.t.:

$$\bigcup_{i \neq j} C_i \subsetneq A \subsetneq \bigcup_{i \in I} C_i$$

for some $j \in I$. Consider $a \in A \setminus \bigcup_{i \neq j} C_i \subseteq C_j$ and $b \in C_j \setminus A$. Note that $a, b \in C_j$, contradicting the assumption of 4(a). $\square$

Definition 3.25. Connectable

Let x and y be two points in a topological space X . We say x and y are connectable if there exists a connected subset of X containing both x and y .

Proposition 3.26. *For every topological X , the notion of being connectable gives an equivalence relation.*

Definition 3.27. Component

Let X be topological. We call the equivalence classes of connectable points on X the *component* of X .

Proposition 3.28. *Component Proposition*

Let X be a topological space.

1. As a set X is the disjoint union of its components.
2. Each component of X is connected.
3. Each component C of X is a maximal connected subset of X , i.e.: if $C \subsetneq D$, then D is NOT connected.
4. Let $A \subseteq X$ be a subset of X .
 - (a) If C is a component of X , then $A \cap C$ is a component of A .
 - (b) If A is both closed and open in X , then A is the union of components of X .
 - (c) If A is both closed and open in X and if A is connected, then A is a component of X .
5. Each component of X is the union of path-components of X .

Proof. It's trivial for 1,2,3,4(a),4(c),5. As for 4(b), let C be a component of X . Note that $A \cap C$ is both closed and open in C . By (2), the connectivity of C , $A \cap C = \emptyset$ or $A \cap C = C$. $\square$

Proposition 3.29. *Let X be a topological space.*

1. Every component of X is closed subset of X .
2. If X has only finitely many components, then each component is also an open subset of X .

Proof. 1. Let C be a component of X . If $C = X$, then we're done, since $\emptyset \in \mathcal{T}_X$. Otherwise, take $P \in X \setminus C$. Then by the component proposition 3, $C \cup \{P\}$ is NOT connected. Therefore, by def there exist open (in $C \cup \{P\}$) nonempty disjoint U, V , s.t.:

$$C \cup \{P\} = U \sqcup V$$

WLOG, let $C \cap U \neq \emptyset$. Since C is connected and $C = (U \cap C) \sqcup (V \cap C)$, then $V \cap C = \emptyset$. Hence, $V = \{P\}$, open in $C \cup \{P\}$. Therefore, by the subspace topology, there exists an open subset W in X , s.t.:

$$W \cap (C \cup \{P\}) = V = \{P\}$$

Hence, $W \cap C = \emptyset$ and $P \in W$, which follows that $X \setminus C$ is open $\Leftrightarrow C$ is closed.

2. Let C be a component of X . By assumption,

$$X \setminus C = \bigcup_{i=1}^n C_i$$

where C_i 's are components of X . Since the union is finite and each C_i is closed by 1, then $X \setminus C$ is closed and hence C is open.

□

Proposition 3.30. *Let $f : X \rightarrow Y$ be a continuous map between two topological spaces.*

1. *Let C be a (path-)component of Y . The preimage $f^{-1}(C)$ is the union of (path-)components of X .*
2. *Let B be a (path-)component of X . There exists a (path-)component C of Y , s.t.: $f(B) \subseteq C$.*

3.3 Compactness

Definition 3.31. Cover

Let X be a topological space. An open cover of X is a collection of open sets U_i , $i \in I$, s.t.: $X = \bigcup_{i \in I} U_i$. Moreover, a subcover of X is a subcollection $\{U_j \mid j \in J \subseteq I\}$ s.t.: $X = \bigcup_{j \in J} U_j$.

Definition 3.32. Compact

Let X be a topological space. We say X is compact if $\forall$ open cover of X , there is always a subcover with finitely many open sets.

Remark: A compact set can be regarded as a point.

Example:

- Heine-Borel Theorem:

$$X \subseteq \mathbb{R}^n \text{ cpt} \Leftrightarrow X \text{ closed and bounded.}$$

- Let $K \subseteq \mathbb{R}^n$ be cpt and let $\mathcal{C}(K) := \{f : K \rightarrow \mathbb{R}\}$. Define a metric:

$$d_\infty(f, g) := \sup_{x \in K} \{|f(x) - g(x)|\}.$$

Then we have Ascoli-Aizela Theorem:

$$A \subseteq \mathcal{C}(K) \text{ cpt} \Leftrightarrow A \text{ closed, bounded, and equicontinuous.}$$

Remark:

$A \subseteq X$ is cpt if the following two equivalent conditions hold:

- A is cpt using subspace topology;
- For any $\{U_i\}_{i \in I}$ open in X , s.t.: $A \subseteq \bigcup_{i \in I} U_i$, there is a finite subcover $A \subseteq \bigcup_{j=1}^n U_{i_j}$.

Proposition 3.33. *Let $f : X \rightarrow Y$ be a cont fct between two topological spaces. If $A \subseteq X$ is cpt, then $f(A)$ is cpt.*

Proof. Suppose $f(A) \subseteq \bigcup_{i \in I} U_i$ open cover in Y . Then $A = f^{-1}(\bigcup_{i \in I} U_i) = \bigcup_{i \in I} f^{-1}(U_i)$. By the cptness of A , $A \subseteq \bigcup_{j=1}^n f^{-1}(U_{i_j})$. Hence, $f(A) \subseteq \bigcup_{j=1}^n U_{i_j}$. $\square$

Corollary 3.34. *Let K be cpt and let $f : K \rightarrow \mathbb{R}$ be cont. Then $f(K)$ is closed and bounded in $\mathbb{R}$. In particular, there are $m, M \in \mathbb{R}$, s.t.: $m \leq f(x) \leq M$ for all $x \in K$.*

Lemma 3.35. *Let X be Hausdorff and let $K \subseteq X$ be cpt. Then for any $x \in X - K$, there are two open sets $U \ni x$ and $V \supseteq K$, s.t.: $U \cap V = \emptyset$.*

Proof. For any $k \in K$, clearly $x \neq k$, so we have $U_k \ni x$ and $V_k \ni k$ such that $U_k \cap V_k = \emptyset$, by the Hausdorff-ness of X . Therefore, $\{V_k \mid k \in K\}$ is an open cover of X , i.e.: $\bigcup_{i=1}^n V_{k_i} \supseteq K$. Now take $U := \bigcap_{i=1}^n U_{k_i}$ and $V := \bigcup_{i=1}^n V_{k_i}$. Then we're done. $\square$

Corollary 3.36. *Let X be Hausdorff and let $K \subseteq X$ be cpt. Then K is closed.*

Proof. For any $x \in X - K$, by lemma 4.25, there exists an open set $U_x \ni x$ s.t.: $U_x \subseteq X - K$. Then $X - K$ is open. $\square$

Proposition 3.37. *If X, Y are cpt, then $X \times Y$ is cpt using the product topology.*

Proof. Suppose $A = \{W_i\}_{i \in I}$ is an open cover of $X \times Y$. Then $X \times Y = \bigcup_{i \in I} W_i$. We can write:

$$X \times Y = \bigcup_{i \in I} W_i = \bigcup_{i \in I} \bigcup_{j \in J} U_{ij} \times V_{ij}$$

where $U_{ij} \in \mathcal{T}_X$ and $V_{ij} \in \mathcal{T}_Y$.

For each $y \in Y$, $X \times \{y\} \subseteq \bigcup_{i,j} U_{ij} \times V_{ij} =: \bigcup_k U_k \times V_k$ where $\{k = (i, j) \mid i \in I, j \in J\}$. Since $X \times \{y\}$ is cpt, then $X \times \{y\} \subseteq \bigcup_{s \in F_y} U_s \times V_s$ where $|F_y| < \infty$. Then:

$$\begin{aligned} y \in \bigcap_{s \in F_y} V_s &=: V_y \Rightarrow Y = \bigcup_{y \in Y} V_y \\ &\stackrel{Y \text{ cpt}}{\Rightarrow} Y = \bigcup_{l=1}^m V_{y_l} \end{aligned}$$

Hence,

$$X \times Y = \bigcup_{l=1}^m X \times V_{y_l} \subseteq \bigcup_{l=1}^m \bigcup_{s \in F_{y_l}} U_s \times V_s \subseteq \bigcup_{l=1}^m \bigcup_{s \in F_{y_l}} W_s.$$

$\square$

Definition 3.38. Product Topology for infinite product

Let $\{(X_i, \mathcal{T}_i)\}_{i \in I}$ topological spaces. Then the product topology of $\prod_{i \in I} X_i := \{(x_i)_{i \in I} \mid x_i \in X_i\}$ has a basis of the form:

$$\{\prod_{i \in I} U_i \mid U_i \in \mathcal{T}_i \text{ for all } i \in I \text{ and } U_i = X_i \text{ for all but finitely many } i \in I\}.$$

Indeed, the product topology defined above is the coarsest topology satisfying the property that

$$p_i : \prod_{j \in I} X_j \rightarrow X_i \text{ with } p_i(x_j) := x_i \text{ is cont for all } i \in I.$$

Lemma 3.39. *Let $\mathcal{B}$ be a collection of open subsets of $X \times Y \times Z$ with Y compact. Suppose that there exists a point $x_0 \in X$, s.t.: for any open neighborhood of x_0 U , $U \times Y \times Z$ is NOT finitely covered by $\mathcal{B}$. Then $\exists y_0 \in Y$, s.t.: $\forall$ open $U \ni x_0, V \ni y_0$, $U \times V \times Z$ is NOT finitely covered by $\mathcal{B}$.*

Proof. Suppose on the contrary that for any $y \in Y$, there exists an open $U_y \ni x_0, V_y \ni y$, s.t.:

$$U_y \times V_y \times Z \subseteq \bigcup_{j=1}^{m_y} W_{y,j}$$

where $W_{y,j} \in \mathcal{B}$.

Since $\{V_y \mid y \in Y\}$ is an open cover of $\text{cpt } Y$, then there is an open finite subcover of Y $\{V_{y_i}\}_{i=1}^n$. Then set $U := \bigcap_{i=1}^n U_{y_i}$. Then:

$$U \times Y \times Z \subseteq \bigcup_{i=1}^n (U_{y_i} \times V_{y_i} \times Z) \subseteq \bigcup_{i=1}^n \bigcup_{j=1}^{m_{y_i}} W_{y_i,j}$$

contradicting the hypothesis of lemma. □

Theorem 3.40. *Tychonoff's Theorem*

Let $\{X_i \mid i \in I\}$ be cpt topological spaces. Then $\prod_{i \in I} X_i$ is cpt in product topology.

Remark: We'll only prove the case when I is countable. For the case when I is uncountable, we need axiom of choice (well-ordering).

Proof. Let $\mathcal{B}$ be a collection of open sets of $X = \prod_{i=1}^{\infty} X_i$, s.t.: $\mathcal{B}$ does NOT cover X finitely.

- **Claim 1:** there exists a point $x_1 \in X_1$, s.t.: for any $U_1 \ni x_1$ open in $\mathcal{B}$, $\mathcal{B}$ does NOT cover $U_1 \times X_2 \times X_3 \times \cdots$ finitely.

Proof of claim 1: Suppose on the contrary that for any $x \in X_1$, there exists $U_x \ni x$ open, s.t.:

$$U_x \times X_2 \times X_3 \times \cdots \subseteq \bigcup_{i=1}^{m_x} W_{x,i}.$$

Since $\{U_x\}$ covers X_1 , then we have $\{U_{x_j}\}_{j=1}^p$ covers X_1 . Hence,

$$X_1 \times X_2 \times \cdots \subseteq \bigcup_{j=1}^p U_{x_j} \times X_2 \times X_3 \times \cdots \subseteq \bigcup_{j=1}^p \bigcup_{i=1}^{m_{x_j}} W_{x_j,i}.$$

Therefore, $\mathcal{B}$ covers $X_1 \times X_2 \times X_3 \times \cdots$ finitely, contradiction!

Then by claim 1 and lemma 4.29, if we regard U_1 in claim 1 as X in lemma 3.28, X_2 as Y in lemma 4.29, and $X_3 \times X_4 \times \cdots$ as Z in lemma 3.28, then there exists a point $x_2 \in X_2$, s.t.: for any open $U_1, \ni x_1, U_2, \ni x_2$, $\mathcal{B}$ does NOT cover $U_1 \times U_2 \times X_3 \times X_4 \times \cdots$ finitely.

Repeat this procedure. Then we get (*): $\exists \mathbf{x} = (x_1, \cdots, x_n, \cdots) \in \prod_{i=1}^{\infty} X_i$, s.t.: for any $n \in \mathbb{N}_+$, $\forall U_i \ni x_i, (i = 1, \cdots, n)$, $U_1 \times \cdots \times U_n \times X_{n+1} \times \cdots$ can NOT be covered by $\mathcal{B}$ finitely.

- **Claim 2:** $\mathbf{x} \notin \bigcup_{W \in \mathcal{B}} W$.

Proof of claim 2: Suppose on the contrary that $\mathbf{x} \in W_b$ for some $W_b \in \mathcal{B}$. Then $\mathbf{x} \in W_b \subseteq \bigcup(\Pi_{i=1}^{\infty} U_i)$, where $(\Pi_{i=1}^{\infty} U_i)$ is in the basis of product topology of X . Hence,

$$\mathbf{x} \in \Pi_{i=1}^{\infty} U_i \Rightarrow \mathbf{x} \in U_1 \times \cdots \times U_N \times X_{N+1} \times \cdots \subseteq W_b$$

contradicting (*).

Then by claim 2, we implies that X is NOT contained in $\bigcup_{W \in \mathcal{B}} W$. □

Lemma 3.41. *Any closed subset of a compact set is compact.*

Proof. Suppose X is cpt, V is closed in X , and $\{U_i\}_{i \in I}$ is an open cover of V . Then $\{U_i\}_{i \in I} \cup (X - V)$ is an open cover of X . Hence, by the cpt-ness of X , $\{U_{i_k}\}_{k=1}^m \cup (X - V)$ covers X . Therefore, $\{U_{i_k}\}_{k=1}^m$ covers V . □

Theorem 3.42. *Suppose X is compact, Y is Hausdorff, and $f : X \rightarrow Y$ is cont and bijective. Then f^{-1} is cont, and moreover, f is a homeomorphism.*

Proof. Take a closed set $V \subseteq X$, and by lemma 3.30, V is cpt. Then by the continuity of f , $f(V)$ is cpt. Since Y is Hausdorff, then $f(V)$ is closed. Therefore, $(f^{-1})^{-1}(V) = f(V)$ is closed whenever V is closed, i.e.: f^{-1} is cont. □

Example: Consider $f : S^1 \times S^1 \rightarrow \mathbb{R}^3$ with $f(\theta, \phi) := \{(R+r \cos \theta) \cos \phi, (R+r \cos \theta) \sin \phi, r \sin \phi\}$, where $S^1 := \{(\cos \theta, \sin \theta) \mid \theta \in \mathbb{R}\}$. Then $f(S^1 \times S^1)$ forms a torus, which can be denoted by T^2 . It's easy to check that f is cont, $S^1 \times S^1$ is cpt (i.e.: closed and bdd in $\mathbb{R}^2$), and $f(S^1 \times S^1)$ is Hausdorff. Then by theorem 3.30, $S^1 \times S^1 \cong f(S^1 \times S^1) = T^2$ are homeomorphic.

Definition 3.43. Cantor Set

Let $C_0 := [0, 1]$. Then we define $\{C_i\}_{i=1}^{\infty}$ iteratively as follows:

$$C_i := \left\{ \frac{1}{3}x \mid x \in C_{i-1} \right\} \cup \left\{ \frac{2}{3} + \frac{1}{3}x \mid x \in C_{i-1} \right\} = \frac{1}{3}C_{i-1} \cup \left(\frac{2}{3} + \frac{1}{3}C_{i-1} \right)$$

Then we define *Cantor Set* by

$$C := \bigcap_{i=0}^{\infty} C_i$$

Proposition 3.44. *The Cantor Set is compact in $\mathbb{R}$.*

3.4 Normal Topological Space

Definition 3.45. Normal Topological Space

A topological space X is called *normal* if any two disjoint closed subsets A, B of X can be separated by open neighborhoods, i.e.: there exists two open sets U, V , s.t.: $A \subseteq U, B \subseteq V$ and $U \cap V = \emptyset$.

Proposition 3.46. *Every compact Hausdorff space is normal.*

Proposition 3.47. *Metric Spaces-Normal Proposition*

Let (X, d) be a metric space. Then $(X, \mathcal{T}_d)$ is a normal topological space.

Proof. Given two disjoint closed subsets A, B of B , if one of A or B is empty, then the desired open neighborhoods are $\emptyset$ and X . Otherwise, A, B are both non-empty. Note that $\forall x \in X$,

$$d(x, A) + d(x, B) > 0$$

So we can consider a well-defined map $f : X \rightarrow \mathbb{R}$ given by

$$f(x) := \frac{d(x, A)}{d(x, A) + d(x, B)}$$

which is continuous by the Distance-to-Subspace Proposition.1. It's clear that $f|_A \equiv 0, f|_B \equiv 1$. Then

$$A \subseteq U := f^{-1}((-\infty, \frac{1}{4})), B \subseteq V := f^{-1}((\frac{3}{4}, \infty))$$

are the two desired open neighborhoods. □

Proposition 3.48. *Normal-Open-Closed-Neighborhood Proposition*

Let X be a topological space and let $A \subseteq X$ be a closed subset and $U \subseteq X$ be an open subset with $A \subseteq U$. If X is normal, then there exists an open $V \subseteq X$ in X , s.t.:

$$A \subseteq V \subseteq \overline{V} \subseteq U$$

Proof. Since $A \subseteq U$, then $A \cap (X \setminus U) = \emptyset$, where $A, X \setminus U$ are closed. By the normality of X , there are disjoint open neighborhoods V of A , W of $X \setminus U$. It suffices to show that $\overline{V} \subseteq U$. Note that $V \subseteq X \setminus W$ with $X \setminus W$ closed. Hence,

$$\overline{V} \subseteq \overline{X \setminus W} = X \setminus W \subseteq U$$

□

Proposition 3.49. *Closed Disjoint Embedding Proposition*

Let Y be a normal space.

1. *Let $A_1, \dots, A_k \subseteq Y$ be distinct subsets. If A_i 's are closed in Y , then the natural map $A_1 \sqcup \dots \sqcup A_k \rightarrow A_1 \cup \dots \cup A_k$ is a homeomorphism.*
2. *Let $X_1, \dots, X_k$ be compact spaces and let $\varphi_i : X_i \rightarrow Y$ be embeddings with disjoint images. If Y is Hausdorff, then the map*

$$\bigsqcup_{i=1}^k \varphi_i : \bigsqcup_{i=1}^k X_i \rightarrow Y$$

is an embedding.

- Proof.* 1. Given $j \in \{1, \dots, k\}$, note that $\bigcup_{i \neq j} A_i$ is closed and that it is disjoint with A_j closed in Y . Then by the normality of Y , there exists an open neighborhood of A_j that doesn't intersect any of the other A_i . Hence, by the Disjoint Embedding Proposition, we're done.
2. By the compact-ness of X_i and the Hausdorff-ness of Y , the images of embeddings $\varphi_i(X_i)$'s are closed in Y . Then by 1, we're done.

□

3.5 Urysohn's Lemma

Lemma 3.50. *Urysohn's Lemma*

Let X be a topological space. The followings are equivalent:

1. For any two disjoint closed subsets A, B , any $r < s \in \mathbb{R}$, there exists a continuous map $f : X \rightarrow [r, s]$ with $f|_A \equiv r, f|_B \equiv s$.
2. X is normal.

Proof. • (1 $\Rightarrow$ 2): For any disjoint closed A, B in X , by 1, $f^{-1}((-\infty, \frac{r+s}{2}))$ and $f^{-1}((\frac{r+s}{2}, \infty))$ are the desired disjoint open neighborhoods of A respectively B .

- (2 $\Rightarrow$ 1): Given two disjoint closed A, B subsets of X , it suffices to show that there is a continuous map $f : X \rightarrow [0, 1]$, s.t.: $f|_A \equiv 0, f|_B \equiv 1$.

– **Claim 1:** Set $P := [0, 1] \cap \mathbb{Q}$. There exists a family $\{U_s\}_{s \in P}$ of open subsets of X with the following properties:

1. $A \subseteq U_0$
2. $\forall p \in P$ with $p < 1$, we have $U_p \subseteq X \setminus B$
3. $\forall s, t \in P$ with $s < t$, we have $\overline{U_s} \subseteq U_t$.

– *Proof. of claim 1:* Since P is countable, then there is a bijective $\varphi : \mathbb{N} \cup \{0\} \rightarrow P$. WLOG, let $\varphi(0) = 0$ and $\varphi(1) = 1$. Set $U_1 := X \setminus B$, which is open by the closed-ness of B . Since X is normal and since A is closed, then by the Normal-Open-Closed-Neighborhood Proposition, there is an open neighborhood U_1 of A , s.t.: $\overline{U_0} \subseteq U_1$.

Now we suppose that we're already constructed $U_{\varphi(0)} = U_0, U_{\varphi(1)} = U_1, U_{\varphi(2)}, \dots, U_{\varphi(n)}$ with the required properties. Then set $P_n := \{\varphi(0), \dots, \varphi(n)\}$ and $y := \varphi(n+1)$. By the injectivity of φ , since $\varphi(0) = 0$ and $\varphi(1) = 1$, then $y \in (0, 1)$. Since P_n is finite, then $\exists x, z \in P_n$, s.t.:

$$p \leq x < y, \forall p \in P_n \text{ and } y < z \leq q, \forall q \in P_n$$

Then by the property 3, $\overline{U_x} \subseteq U_z$. Hence, by the Normal-Open-Closed-Neighborhood Proposition, there is an open U_y , s.t.:

$$\overline{U_x} \subseteq U_y \subseteq \overline{U_y} \subseteq U_z$$

Therefore, we're done. $\square$

$\forall p \in (-\infty, 0) \cap \mathbb{Q}$, we set $U_p := \emptyset$ and $\forall p \in (1, \infty) \cap \mathbb{Q}$, we set $U_p := X$. Consider $f : X \rightarrow \mathbb{R}$ given by

$$f(x) := \inf\{p \in \mathbb{Q} \mid x \in U_p\}$$

By the properties 1,2, $f|_A \equiv 0$, $f|_B \equiv 1$ and $f(X) \subseteq [0, 1]$. Then it remains to prove the continuity of f .

– **Claim 2:** Let $r \in \mathbb{Q}$.

$$1. x \in \overline{U}_r \Rightarrow f(x) \leq r, \text{ i.e.: } f(x) > r \Rightarrow x \notin \overline{U}_r.$$

$$2. x \notin U_r \Rightarrow f(x) \geq r, \text{ i.e.: } f(x) < r \Rightarrow x \in U_r.$$

– *Proof. of claim 2:* It's trivial for 1,2 in claim 2 if $r < 0$ or $r > 1$. Now we suppose that $r \in P = [0, 1] \cap \mathbb{Q}$. Then by the property 3 in claim 1,

$$x \in \overline{U}_r \xrightarrow{3} x \in U_s, \forall s \in P, \text{ with } s > r \xrightarrow{P \text{ dense in } [0,1]} f(x) \leq r$$

and similarly for 2. $\square$

– **Claim 3:** Given any $r \in \mathbb{Q}$, we have

$$f^{-1}([0, r]) = \bigcap_{n \in \mathbb{N}_+} \overline{U}_{r+\frac{1}{n}}, f^{-1}([0, r)) = \bigcup_{n \in \mathbb{N}_+} U_{r-\frac{1}{n}}$$

– *Proof. of claim 3:*

$$\begin{aligned} * f^{-1}([0, r]) &= f^{-1}(\bigcap_{n \in \mathbb{N}_+} [0, r + \frac{1}{n})) = \bigcap_{n \in \mathbb{N}_+} f^{-1}([0, r + \frac{1}{n})) \stackrel{\text{claim.2.2}}{\subseteq} \bigcap_{n \in \mathbb{N}_+} U_{r+\frac{1}{n}} \subseteq \\ &\bigcap_{n \in \mathbb{N}_+} \overline{U}_{r+\frac{1}{n}} \stackrel{\text{claim.2.1}}{\subseteq} \bigcap_{n \in \mathbb{N}_+} f^{-1}([0, r + \frac{1}{n}]) = f^{-1}(\bigcap_{n \in \mathbb{N}_+} [0, r + \frac{1}{n}]) = f^{-1}([0, r]). \end{aligned}$$

$$\begin{aligned} * f^{-1}([0, r)) &= f^{-1}(\bigcup_{n \in \mathbb{N}_+} [0, r - \frac{1}{n})) = \bigcup_{n \in \mathbb{N}_+} f^{-1}([0, r - \frac{1}{n})) \stackrel{\text{claim.2.2}}{\subseteq} \bigcup_{n \in \mathbb{N}_+} U_{r-\frac{1}{n}} \subseteq \\ &\bigcup_{n \in \mathbb{N}_+} \overline{U}_{r-\frac{1}{n}} \stackrel{\text{claim.2.1}}{\subseteq} \bigcup_{n \in \mathbb{N}_+} f^{-1}([0, r - \frac{1}{n}]) = f^{-1}(\bigcup_{n \in \mathbb{N}_+} [0, r - \frac{1}{n}]) = f^{-1}([0, r)). \end{aligned}$$

$\square$

– **Claim 4:** f is continuous.

– *Proof. of claim 4:* $\forall c < d \in \mathbb{Q}$,

$$f^{-1}((c, d)) = f^{-1}([0, d)) \setminus f^{-1}([0, c]) \stackrel{\text{claim.3}}{=} \bigcup_{n \in \mathbb{N}_+} U_{d-\frac{1}{n}} \setminus \bigcap_{n \in \mathbb{N}_+} \overline{U}_{c+\frac{1}{n}}$$

is open in X . $\square$

$\square$

3.6 Tietze Extension Theorem

Theorem 3.51. *Tietze Extension Theorem*

Let X be a topological space, let $A \subseteq X$ be a closed subset of X and let $f : A \rightarrow \mathbb{R}$ be a continuous map. If X is normal, then $\exists$ a continuous map $g : X \rightarrow \mathbb{R}$ with $g|_A = f$. If A is non-empty, then one can find such g , s.t.:

$$\inf g(X) = \inf f(A) \in \mathbb{R} \cup \{-\infty\}, \sup g(X) = \sup f(A) \in \mathbb{R} \cup \{\infty\}$$

Theorem 3.52. *Tietze Extension Theorem II*

Let X be a topological space and let $A \subseteq X$ be a closed subset of X . Suppose that for $i = 1, \dots, n$, we're given a closed interval $I_i \subseteq \mathbb{R}$ and suppose that we're given a continuous map $f : A \rightarrow I_1 \times \dots \times I_n \subseteq \mathbb{R}^n$. If X is normal, then there is a map $g : X \rightarrow I_1 \times \dots \times I_n$ with $g|_A = f$.

3.7 Local and Regional Properties

Definition 3.53. Let P be a property of topological spaces and let X be a topological space.

- We say X is *regionally* P if for any $x \in X$, any $U \ni x$, there exists $V \subseteq X$ with property P , s.t.:

$$x \in V \subseteq U$$

- We say X is *locally* P if for any $x \in X$, any $U \ni x$, there exists **open** $V \subseteq X$ with property P , s.t.:

$$x \in V \subseteq U$$

Remark: Locally property $\Rightarrow$ regionally property.

Proposition 3.54. *Basis-with-property Proposition*

Let P be a property of topological spaces and let X be a topological space. If X is locally P , then the open subsets of X that have P form a basis for the topology of X .

Proposition 3.55. *Locally Path-Connected Proposition*

Let X be a topological space.

1. If X is locally path-connected, then the following statements hold:

(a) X is connected $\Leftrightarrow X$ is path-connected.

(b) The components of X are precisely the path-components of X .

2. X is locally path-connected, then it is also locally connected.

3. If X is locally connected, then every component of X is both closed and open.

Proof. 1. (a) ($\Leftarrow$): Obviously. ($\Rightarrow$) is a special case of (b).

- (b) • **Claim 1:** Each path-component of X is open.

Proof. Let P be a path-component of X . Pick $x \in X$. For any $y \in P$, since X is path-connected, then we can take a path $\gamma : [0, 1] \rightarrow X$ given by $\gamma(0) = x$ and $\gamma(1) = y$. Since X is locally path-connected, then there exists an open path-connected U , s.t.: $y \in U \subseteq P$. Hence, $\gamma([0, 1]) \cup U$ is path-connected. Therefore, by the path-component proposition 4(b), $\gamma([0, 1]) \cup U \subseteq P$. Then we have:

$$y \in U \subseteq \gamma([0, 1]) \cup U \subseteq P$$

with U open which follows that P is open. $\square$

- **Claim 2:** Each path-component of X is closed.

Proof. Let P be a path-component of X . By the path-component proposition 1, $X \setminus P = \bigsqcup_i P_i$, where P_i 's are the components of X . By claim 1, $X \setminus P$ is open and hence P is closed. $\square$

Let C be a component of X . By the component proposition 2,5, C is connected and there exists a family of path-components $\{P_i\}_{i \in I}$, s.t.:

$$C = \bigsqcup_{i \in I} P_i$$

By claim 1 and claim 2, each P_i is both open and closed. Since C is connected, then I consists of a single element, i.e.: C is path-component of X .

2. Obviously.
3. Let C be a component of X . Then by proposition 4.29.1, C is closed. For any $x \in C$, since X is locally connected, then there exists an open connected U , s.t.: $x \in U \subseteq C$. Therefore, C is open. $\square$

Proposition 3.56. *Let X be a Hausdorff topological space.*

1. *If X is compact, then X is also regionally compact.*
2. *If $\forall x \in X$, there exists a compact neighborhood of x , then X is regionally compact.*

Proof. 1. Let $x \in X$ and let U be a neighborhood of x .

Claim: There exists an open neighborhood A of x with $\overline{A} \subseteq U$.

Proof. Since X is Hausdorff, then $\{x\}$ is closed in X . By proposition 4.46, X is normal. Hence, we can separate the closed sets $\{x\}$ and $X \setminus U^\circ$ by open disjoint neighborhoods A and B , i.e.: $A \cap B = \emptyset$, $x \in A$ and $X \setminus U^\circ \subseteq B \Rightarrow X \setminus B \subseteq U^\circ \subseteq U$. Hence,

$$x \in A \subseteq X \setminus B \subseteq U$$

Since B is open, then $X \setminus B$ is closed and hence

$$x \in A \subseteq \overline{A} \subseteq X \setminus B \subseteq U$$

□

Since $\overline{A}$ is closed and X is compact, then $\overline{A}$ is compact.

2. For any $x \in X$, any $U \ni x$, by hypothesis, there exists a compact $K \ni x$. By 1, K is regionally compact. Hence, there exists a compact L , s.t.:

$$x \in L \subseteq U \cap K \subseteq U$$

□

Definition 3.57. Precompact

Let X be a topological space. A subset $A \subseteq X$ is called *precompact* if $\overline{A}$ is compact.

Proposition 3.58. *Regionally Compact–Locally Precompact Proposition*

Let X be a Hausdorff space that is regionally compact. Then for any $x \in X$, any open $U \ni x$, there exists an open $V \ni x$, s.t.: $\overline{V}$ is compact and s.t.: $x \in V \subseteq \overline{V} \subseteq U$, i.e.: X is locally precompact.

Definition 3.59. Let P be a property of maps between two topological spaces and let $f : X \rightarrow Y$ be a map between two topological spaces.

- We say f is *locally P* at $x \in X$ if there exist an open $U \ni x$ and an open $V \ni f(x)$, s.t.: $f : U \rightarrow V$ has the property P .
- We say f is *locally P* if it is locally P at any $x \in X$.

Proposition 3.60. *If a map $f : X \rightarrow Y$ between two topological spaces is bijective and a local homeomorphism, then it is a homeomorphism.*

Proposition 3.61. *Let $f : X \rightarrow Y$ be a continuous map. If Y is Hausdorff and regionally compact, then*

1. f proper $\Rightarrow f$ closed.
2. f injective and proper $\Leftrightarrow$ closed embedding.

Proof. 1. Let $A \subseteq X$ be a closed subset of X . For any $y \in Y \setminus f(A)$, since Y is regionally compact and Hausdorff, then by proposition 4.52, there exists an open $U \ni y$, s.t.: $y \in U \subseteq \overline{U}$ with $\overline{U}$ compact. Since f is proper, then $f^{-1}(\overline{U})$ is compact and hence so is $f^{-1}(\overline{U}) \cap A$. Therefore, $f(A) \cap \overline{U} = f(f^{-1}(\overline{U}) \cap A)$ is compact. Since Y is Hausdorff, then $f(A) \cap \overline{U}$ is closed in Y . Set $V := U \setminus (f(A) \cap \overline{U}) = U \setminus f(A)$ is an open neighborhood of $y \in Y \setminus f(A)$ that is contained in $Y \setminus f(A)$. Therefore, $Y \setminus f(A)$ is open, i.e.: $f(A)$ is closed.

2. • ($\Rightarrow$): By a, we already then get f is closed. Furthermore, f is closed $\Rightarrow f(X)$ is closed in $Y \Rightarrow f(X)$ is compact in $Y \Rightarrow X$ is compact. Then by theorem 4.42, f is a homeomorphism between $X \cong f(X)$.
- ($\Leftarrow$): It suffices to show that f is proper. Let $K \subseteq Y$ be a compact subset of Y . Since f is closed, then $f(X)$ is closed $\Rightarrow f(X) \cap K$ is compact. Since f is embedding, then $f^{-1}(f(X) \cap K) = X \cap f^{-1}(K) = f^{-1}(K)$ is compact.

□

Proposition 3.62. *Closed-Embedding-to- $\mathbb{R}^n$ Proposition*

Let X be a topological space, $n \in \mathbb{N}_+$ and let $f : X \rightarrow \mathbb{R}^n$ be an injective continuous map. Suppose that X is the union of compact subsets $\{K_i\}_{i \in I}$, s.t.: for any $r > 0$, there are finitely many $i_1, \dots, i_l \in I$, s.t.:

$$f(K_j) \cap \overline{B}_r^n \neq \emptyset$$

Then f is a closed embedding.

Proof.

Lemma 3.63. *Proper Criterion*

Let $f : X \rightarrow Y$ be a continuous map between two topological spaces with Y Hausdorff.

Let $\{L_j\}_{j \in J}$ be a family of compact subsets of Y satisfying that each compact subset of Y is contained in some L_j . If each $f^{-1}(L_j)$ is compact, then f is proper.

2. Let X be covered by compact subsets $\{K_i\}_{i \in I}$ and let $L \subseteq Y$ be compact in Y . If there are finitely many $i \in I$, s.t.: $f(K_i) \subseteq L$, then $f^{-1}(L)$ is compact in X .

Proof. 1. For any compact L in Y , there exists $j \in J$, s.t.: $L \subseteq L_j$. Hence,

$$f^{-1}(L) = f^{-1}(L \cap L_j) = f^{-1}(L) \cap f^{-1}(L_j)$$

is compact in X since $f^{-1}(L)$ is closed and $f^{-1}(L_j)$ is compact.

2.

$$f^{-1}(L) = f^{-1}(L) \cap \bigcup_{i \in I, f(K_i) \subseteq L} K_i$$

is compact since $f^{-1}(L)$ is closed and $\bigcup_{i \in I, f(K_i) \subseteq L} K_i$ is compact.

□

Since $\mathbb{R}^n$ is Hausdorff, then by lemma above (proper criterion)2, for any $r > 0$, $f^{-1}(\overline{B}_r^n)$ is compact in X . Hence, by lemma above (proper criterion)1, f is proper. Since $\mathbb{R}^n$ is Hausdorff and regionally compact, then proposition 4.55, f is injective and proper $\Leftrightarrow f$ is a closed embedding. □

Chapter 4

Preliminaries of Category Theory

4.1 Category

Definition 4.1. Category

A category $\mathcal{C}$ consists of the followings:

- *Objects*: a class $Ob(\mathcal{C})$, the collection of all elements in $\mathcal{C}$.
- *Morphisms*: for any $X, Y \in Ob(\mathcal{C})$, there exists a set $Mor_{\mathcal{C}}(X, Y)$ consisting of maps from X to Y with the following properties:
 - If $f \in Mor_{\mathcal{C}}(X, Y)$, $g \in Mor_{\mathcal{C}}(Y, Z)$, then

$$g \circ f \in Mor_{\mathcal{C}}(X, Z)$$

- For any $f \in Mor_{\mathcal{C}}(W, X)$, $g \in Mor_{\mathcal{C}}(X, Y)$, $h \in Mor_{\mathcal{C}}(Y, Z)$,

$$(h \circ g) \circ f = h \circ (g \circ f)$$

- For any $X \in Ob(\mathcal{C})$, there exists $id_X \in Mor_{\mathcal{C}}(X, X)$, s.t.:

$$id_X \circ f = f \quad \forall W \in Ob(\mathcal{C}), \forall f \in Mor_{\mathcal{C}}(W, X)$$

$$g \circ id_X = g \quad \forall Y \in Ob(\mathcal{C}), \forall g \in Mor_{\mathcal{C}}(X, Y)$$

Proposition 4.2. Let $\mathcal{C}$ be a category. For any $X \in Ob(\mathcal{C})$, the identity morphism of $Mor_{\mathcal{C}}(X, X)$ exists and is unique.

Proof. The existence of id_X is from the def of category. As for the uniqueness, suppose that id_X, id'_X are both the identity morphism of $Mor_{\mathcal{C}}(X, X)$. Then

$$id_X = id_X \circ id'_X = id'_X$$

□

Definition 4.3. Examples of Categories

- The category $\mathcal{Set}$ of sets:
 - $Ob(\mathcal{Set})$ consists of all sets
 - $Mor_{\mathcal{Set}}(X, Y)$ consists of all maps from X to Y .
- The category $\mathcal{Top}$ of topological spaces:
 - $Ob(\mathcal{Top})$ consists of all topological spaces
 - $Mor_{\mathcal{Top}}(X, Y)$ consists of all continuous maps from X to Y .
- The category $\mathcal{Vec}_{\mathbb{F}}$ of vector spaces over $\mathbb{F}$:
 - $Ob(\mathcal{Vec}_{\mathbb{F}})$ consists of all vector spaces over $\mathbb{F}$
 - $Mor_{\mathcal{Vec}_{\mathbb{F}}}(X, Y)$ consists of all linear transformation from X to Y .
- The category $\mathcal{Group}$ of groups:
 - $Ob(\mathcal{Group})$ consists of all groups
 - $Mor_{\mathcal{Group}}(X, Y)$ consists of all group homomorphisms from X to Y .

Similarly for defining the category $\mathcal{AbGroup}$ of abelian groups, the category $\mathcal{Ring}$ of rings and the category $\mathcal{CRing}$ of commutative rings.

- A *pair of sets* is a pair (X, A) where X is a set and A is a subset of X . Then we define the category $\mathcal{PairSet}$ of pairs of sets:
 - $Ob(\mathcal{PairSet})$ consists of all pairs of sets
 - $Mor_{\mathcal{PairSet}}((X, A), (Y, B))$ consists of all maps f from X to Y with $f(A) \subseteq B$.
- Let $(P, \leq)$ be a *partially ordered set*. Fix a set $\{*\}$ consisting of a single element. We define the order category $\mathcal{Order}_{(P, \leq)}$ to be the category given by
 - $Ob(\mathcal{Order}_{(P, \leq)}) := P$
 - $Mor_{\mathcal{Order}_{(P, \leq)}}(x, y) := \begin{cases} \{*\} & , \text{if } x \leq y \\ \emptyset & , \text{otherwise} \end{cases}$

Definition 4.4. Opposite Category

Let $\mathcal{C}$ be a category. The *opposite category* $\mathcal{C}^{op}$ of $\mathcal{C}$ is defined as

- $Ob(\mathcal{C}^{op}) := Ob(\mathcal{C})$
- $Mor_{\mathcal{C}^{op}}(X, Y) := Mor_{\mathcal{C}}(Y, X)$

with the composition given by $f \circ_{\mathcal{C}^{op}} g := g \circ_{\mathcal{C}} f$.

Definition 4.5. Subcategory

Let $\mathcal{C}$ be a category. A *subcategory* is a category $\mathcal{D}$ satisfying that

- $Ob(\mathcal{D}) \subseteq Ob(\mathcal{C})$
- for any $X, Y \in Ob(\mathcal{D})$, $Mor_{\mathcal{D}}(X, Y) \subseteq Mor_{\mathcal{C}}(X, Y)$.

Furthermore, we say that $\mathcal{D}$ is a *full subcategory* if $Mor_{\mathcal{D}}(X, Y) = Mor_{\mathcal{C}}(X, Y)$ for all objects X, Y .

Example:

- $AbGroup$ is a full subcategory of $Group$.
- Let $\mathcal{C}$ be given by $Ob(\mathcal{C}) = Ob(Group)$ and $Mor_{\mathcal{C}}(X, Y)$ consists of all group isomorphisms from X to Y . Then $\mathcal{C}$ is a subcategory of $Group$.

4.2 Isomorphisms

Definition 4.6. Let $\mathcal{C}$ be a category and let $X, Y \in Ob(\mathcal{C})$.

- Let $f \in Mor_{\mathcal{C}}(X, Y), g \in Mor_{\mathcal{C}}(Y, X)$
 - g is said to be a *left inverse* of f if $g \circ f = id_X$.
 - g is said to be a *right inverse* of f if $f \circ g = id_Y$.
 - g is said to be a *inverse* of f if g is both a left and a right inverse of f .
- If $f \in Mor_{\mathcal{C}}(X, Y)$ admits an inverse, then we say f is *invertible* or that f is an *isomorphism*.
- We denote by $Mor_{\mathcal{C}}(X, Y)_{inv}$ the set of all invertible morphisms.
- $X, Y \in Ob(\mathcal{C})$ are called *isomorphic* if $Mor_{\mathcal{C}}(X, Y)_{inv} \neq \emptyset$.
- For any $X \in Ob(\mathcal{C})$, we define the *isomorphism type* of X as the class of all objects that are isomorphic to X .

Example:

- In $Vec_{\mathbb{F}}$, the invertible morphisms are precisely the isomorphisms which are the same as in linear algebra.
- In Set , the isomorphisms are precisely the bijections.
- In Top , the isomorphisms are precisely the homeomorphisms.
- Consider a category $\mathcal{C}$ given by
 - $Ob(\mathcal{C}) :=$ the set with unique element $*$
 - $Mor_{\mathcal{C}}(*, *) := G$

where G denotes an fixed arbitrary group, with the component

$$g \circ f := g \cdot f$$

where $\cdot$ denotes the multiplication in G . By the def of group, every morphism is an isomorphism.

Definition 4.7. Groupoids

Categories in which every morphism is an isomorphism are called *groupoids*.

Proposition 4.8. *Let $\mathcal{C}$ be a category, let $X, Y \in \text{Ob}(\mathcal{C})$ and let $f \in \text{Mor}_{\mathcal{C}}(X, Y)$.*

1. *If f admits a left inverse $\bar{g}$ and a right inverse g , then they agree, i.e.: $g = \bar{g}$. In particular, f is invertible.*
2. *If f admits an inverse, then the inverse is unique.*
3. *$\text{Mor}_{\mathcal{C}}(X, X)_{\text{inv}}$ forms a group under the composition of morphisms.*

Proof. 1. $g = \text{id}_X \circ g = (\bar{g} \circ f) \circ g = \bar{g} \circ (f \circ g) = \bar{g} \circ \text{id}_Y = \bar{g}$.

2. Let g, g' be inverses of f . Then g is a left inverse of f and g' is a right inverse of f . Hence, by 1, $g = g'$.

□

Definition 4.9. We denote the inverse of f as f^{-1} .

Proposition 4.10. *Let $\mathcal{C}$ be a category.*

1. *Let $f \in \text{Mor}_{\mathcal{C}}(X, Y), g \in \text{Mor}_{\mathcal{C}}(Y, Z)$. If two morphisms out of $f, g, g \circ f$ are invertible, then so is the third and one has $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.*
2. *"Being isomorphic" gives an equivalence relation in the class of objects of $\mathcal{C}$.*

4.3 Monomorphisms and Epimorphisms

Definition 4.11. Let $\mathcal{C}$ be a category.

- A morphism $g \in \text{Mor}_{\mathcal{C}}(X, Y)$ is called a *monomorphism* if $f_1, f_2 \in \text{Mor}_{\mathcal{C}}(W, X)$ with $g \circ f_1 = g \circ f_2$ can imply that $f_1 = f_2$.
- A morphism $g \in \text{Mor}_{\mathcal{C}}(X, Y)$ is called a *epimorphism* if $h_1, h_2 \in \text{Mor}_{\mathcal{C}}(Y, Z)$ with $h_1 \circ g = h_2 \circ g$ can imply that $h_1 = h_2$.

Proposition 4.12. *Monomorphism Proposition*

Let $\mathcal{C}$ be one of the following categories: 1.Group, 2.AbGroup, 3.Vec $_{\mathbb{C}\mathbb{F}}$, 4.Ring. If $\varphi : G \rightarrow H$ is a monomorphism in $\mathcal{C}$, then φ , viewed as a map between sets, is injective.

Proof. • **Step 1:** $\mathcal{C}$ is the first three categories. Consider

$$K := \ker \varphi := \{g \in G \mid \varphi(g) = e\}$$

where e denotes the identity of group or the zero element in the vector space. Note that $K \in \text{Ob}(\mathcal{C})$. Now consider the trivial homomorphism $\alpha : K \rightarrow G$ and the inclusion map

$i : K \rightarrow G$. Then $\varphi \circ \alpha = \varphi \circ i$. Since φ is monomorphism, then $\alpha = i$. Hence, $K = \{e\} \Leftrightarrow \varphi$ is injective.

- **Step 2:** $\mathcal{C} = \mathcal{R}ing$. Let $a_1, a_2 \in G$ be such that $\varphi(a_1) = \varphi(a_2)$. Consider the polynomial ring $\mathbb{Z}[x]$ and the two homomorphisms $f_1, f_2 : \mathbb{Z}[x] \rightarrow G$ given by $f_1(x) := a_1$ and $f_2(x) := a_2$. Then

$$(\varphi \circ f_1)(x) = (\varphi \circ f_2)(x)$$

and hence $\varphi \circ f_1 = \varphi \circ f_2$. Since φ is monomorphism, then $f_1 = f_2$. Therefore, $a_1 = a_2$. $\square$

Proposition 4.13. *Epimorphism Proposition*

Let $\mathcal{C}$ be one of the following categories: 1. $\mathcal{G}roup$, 2. $\mathcal{A}b\mathcal{G}roup$, 3. $\mathcal{V}ec_{\mathbb{F}}$. If $\varphi : G \rightarrow H$ is a epimorphism in $\mathcal{C}$, then φ , viewed as a map between sets, is surjective.

Proof. The case when $\mathcal{C} = \mathcal{A}b\mathcal{G}roup$ and $\mathcal{V}ec_{\mathbb{F}}$ are similarly as in the *step 1* of monomorphism proposition. As for $\mathcal{C} = \mathcal{G}roup$, consider

$$H/\varphi(G) := \{h\varphi(G) \mid h \in H\}$$

Let $\{*\}$ be a set of single element which is NOT an element in $H/\varphi(G)$. Set

$$Y := \{*\} \cup H/\varphi(G)$$

and denote by $Bij(Y)$ the group of bijections: $Y \rightarrow Y$. Consider $\alpha : H \rightarrow Bij(Y)$ given by

$$\alpha(h)(y) := \begin{cases} (h \cdot k)\varphi(G) & , \text{if } y = k\varphi(G) \in H/\varphi(G) \\ * & , \text{if } y = * \end{cases}$$

It's easy to show that α is a homomorphism. Let $\sigma \in Bij(Y)$ be the bijection that swaps $e\varphi(G) \in Y$ with $*$ and that keeps all other elements of Y fixed, i.e.:

$$\sigma(y) := \begin{cases} * & , \text{if } y = e\varphi(G) \\ e\varphi(G) & , \text{if } y = * \\ y & \text{otherwise} \end{cases}$$

Define $\beta : H \rightarrow Bij(Y)$ by $\beta(h) := \sigma \circ \alpha(h) \circ \sigma^{-1}$.

- **Claim:** $\alpha \circ \varphi = \beta \circ \varphi : G \rightarrow Bij(Y)$
- *Proof.* For any $g \in G$, $(\alpha \circ \varphi)(g) = \alpha(\varphi(g)) \in Bij(Y)$. Since $\alpha(\varphi(g))(*) := *$ and $\alpha(\varphi(g))(e\varphi(G)) := e\varphi(G)$, then by the def of σ , $\alpha(\varphi(g)) \neq \sigma \Leftrightarrow \alpha(\varphi(g))$ and σ commute. Hence

$$\alpha(\varphi(g)) = \sigma^{-1} \circ (\sigma \circ \alpha(\varphi(g))) = \sigma^{-1} \circ \alpha(\varphi(g)) \circ \sigma := \beta(\varphi(g))$$

Therefore, $\alpha \circ \varphi = \beta \circ \varphi$ □

Since φ is an epimorphism, then $\alpha = \beta : H \rightarrow \text{Bij}(Y)$. For any $h \in H$, $\alpha(h) = \beta(h) := \sigma^{-1} \circ \alpha(h) \circ \sigma \Rightarrow \alpha(h)$ and σ commute $\Leftrightarrow \alpha(h)(e\varphi(G)) = e\varphi(G)$. Hence,

$$h\varphi(G) = e\varphi(G) \Rightarrow h \in \varphi(G) \Rightarrow H \subseteq \varphi(G) (\subseteq H) \Rightarrow \varphi(G) = H$$

□

Example: Note that in the epimorphism proposition doesn't say anything about *Ring*, so here's the counterexample:

the inclusion map $i : \mathbb{Z} \rightarrow \mathbb{Q}$ is an epimorphism but $|\mathbb{Z}| < |\mathbb{Q}| \Leftrightarrow$ it can NOT be surjective.

4.4 Initial and Terminal Objects

Definition 4.14. Let $\mathcal{C}$ be a category.

- An object $X \in \text{Ob}(\mathcal{C})$ is said to be *initial* if for any $Y \in \text{Ob}(\mathcal{C})$, there uniquely exists a morphism from X to Y .
- An object $X \in \text{Ob}(\mathcal{C})$ is said to be *terminal* if for any $Y \in \text{Ob}(\mathcal{C})$, there uniquely exists a morphism from Y to X .

Example:

- In *Set* and *Top*, $\emptyset$ is initial and $\{*\}$ is terminal.
- In *Group* and *AbGroup*, $\{*\}$ is both initial and terminal.
- The initial object of a category $\mathcal{C}$ is a terminal object of the opposite category $\mathcal{C}^{op}$.

Proposition 4.15. Let $\mathcal{C}$ be a category. If an initial object exists, then it is unique up to isomorphism. Same as terminal object.

4.5 Functors

4.5.1 Covariant Functors

Definition 4.16. Covariant Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be two categories. A *covariant functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of a map $F : \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{D})$ and $\forall X, Y \in \text{Ob}(\mathcal{C})$ of a map $F : \text{Mor}_{\mathcal{C}}(X, Y) \rightarrow \text{Mor}_{\mathcal{D}}(F(X), F(Y))$, satisfying the following axioms:

- For any $X \in \text{Ob}(\mathcal{C})$,

$$F(id_X) = id_{F(X)}$$

- For any $\varphi \in \text{Mor}_{\mathcal{C}}(X, Y)$, $\psi \in \text{Mor}_{\mathcal{C}}(Y, Z)$

$$F(\psi \circ \varphi) = F(\psi) \circ F(\varphi) \in \text{Mor}_{\mathcal{D}}(F(X), F(Z))$$

Remark (Convention): If the covariant functor F is understood from the context, then given a morphisms $\varphi \in \text{Mor}_{\mathcal{C}}(X, Y)$, we often write φ_* instead of $F(\varphi)$. Then the axioms become:

$$(id_X)_* = id_{F(X)} \quad \& \quad (\psi \circ \varphi)_* = \psi_* \circ \varphi_*$$

Proposition 4.17. Fixed Domain-Covariant Functor Proposition

Let $\mathcal{D}$ be a category, let Set be the category of sets and let $W \in \text{Ob}(\mathcal{D})$. The map $F : \mathcal{D} \rightarrow \text{Set}$ given by the map

$$\begin{aligned} \text{Ob}(\mathcal{D}) &\rightarrow \text{Ob}(\text{Set}) \\ X &\mapsto \text{Mor}_{\mathcal{D}}(W, X) \end{aligned}$$

together with the map

$$\begin{aligned} \text{Mor}_{\mathcal{D}}(X, Y) &\rightarrow \text{Mor}_{\text{Set}}(\text{Mor}_{\mathcal{D}}(W, X), \text{Mor}_{\mathcal{D}}(W, Y)) \\ \varphi &\mapsto \left(\text{Mor}_{\mathcal{D}}(W, X) \rightarrow \text{Mor}_{\mathcal{D}}(W, Y) \right) \\ &\quad g \mapsto \varphi \circ g \end{aligned}$$

is a covariant functor from the category $\mathcal{D}$ to the category Set .

4.5.2 Contravariant Functors

Definition 4.18. Contravariant Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be two categories. A *contravariant functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of a map $F : \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{D})$ and $\forall X, Y \in \text{Ob}(\mathcal{C})$ of a map $F : \text{Mor}_{\mathcal{C}}(X, Y) \rightarrow \text{Mor}_{\mathcal{D}}(F(Y), F(X))$, satisfying the following axioms:

- For any $X \in \text{Ob}(\mathcal{C})$,

$$F(id_X) = id_{F(X)}$$

- For any $\varphi \in \text{Mor}_{\mathcal{C}}(X, Y)$, $\psi \in \text{Mor}_{\mathcal{C}}(Y, Z)$

$$F(\psi \circ \varphi) = F(\varphi) \circ F(\psi) \in \text{Mor}_{\mathcal{D}}(F(Z), F(X))$$

Remark (Convention): If the contravariant functor F is understood from the context, then given a morphisms $\varphi \in \text{Mor}_{\mathcal{C}}(X, Y)$, we often write φ^* instead of $F(\varphi)$. Then the axioms become:

$$(id_X)^* = id_{F(X)} \quad \& \quad (\psi \circ \varphi)^* = \varphi^* \circ \psi^*$$

Proposition 4.19. Fixed Target-Contravariant Functor Proposition

Let $\mathcal{D}$ be a category, let Set be the category of sets and let $W \in \text{Ob}(\mathcal{D})$. The map $F : \mathcal{D} \rightarrow \text{Set}$ given by the map

$$\begin{aligned} \text{Ob}(\mathcal{D}) &\rightarrow \text{Ob}(\text{Set}) \\ X &\mapsto \text{Mor}_{\mathcal{D}}(X, W) \end{aligned}$$

together with the map

$$\begin{aligned} \text{Mor}_{\mathcal{D}}(X, Y) &\rightarrow \text{Mor}_{\text{Set}}(\text{Mor}_{\mathcal{D}}(Y, W), \text{Mor}_{\mathcal{D}}(X, W)) \\ \varphi &\mapsto \left(\text{Mor}_{\mathcal{D}}(Y, W) \rightarrow \text{Mor}_{\mathcal{D}}(X, W) \right) \\ &\quad g \mapsto g \circ \varphi \end{aligned}$$

is a contravariant functor from the category $\mathcal{D}$ to the category Set .

Proposition 4.20. *Functor-Opposite Category Proposition*

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. If

$$\begin{aligned} F : \quad &\mathcal{C} \rightarrow \mathcal{D} \\ &X \mapsto F(X) \\ (X \xrightarrow{\varphi} Y) &\mapsto (F(X) \xrightarrow{F(\varphi)} F(Y)) \end{aligned}$$

is a contravariant functor between two categories, then

$$\begin{aligned} F : \quad &\mathcal{C} \rightarrow \mathcal{D}^{\text{op}} \\ &X \mapsto F(X) \\ (X \xrightarrow{\varphi} Y) &\mapsto (F(Y) \xrightarrow{F(\varphi)} F(X)) \end{aligned}$$

is a covariant functor. The same statement also with the roles of "covariant" and "contravariant" swapped.

Proposition 4.21. *Functor Composition Proposition*

Let $\mathcal{C}, \mathcal{D}$ and $\mathcal{E}$ be three categories.

1. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{E}$ be covariant functors. The map

$$\begin{aligned} \text{Ob}(\mathcal{C}) &\rightarrow \text{Ob}(\mathcal{E}) \\ X &\mapsto G(F(X)) \end{aligned}$$

and the maps

$$\begin{aligned} \text{Mor}_{\mathcal{C}}(X, Y) &\rightarrow \text{Mor}_{\mathcal{E}}(G(F(X)), G(F(Y))) \\ \varphi &\mapsto G(F(\varphi)) \end{aligned}$$

define a covariant functor from $\mathcal{C}$ to $\mathcal{E}$, denoted by $G \circ F$.

2. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{E}$ be contravariant functors. The map

$$\begin{aligned} \text{Ob}(\mathcal{C}) &\rightarrow \text{Ob}(\mathcal{E}) \\ X &\mapsto G(F(X)) \end{aligned}$$

and the maps

$$\begin{aligned} \text{Mor}_{\mathcal{C}}(X, Y) &\rightarrow \text{Mor}_{\mathcal{E}}(G(F(X)), G(F(Y))) \\ \varphi &\mapsto G(F(\varphi)) \end{aligned}$$

define a covariant functor from $\mathcal{C}$ to $\mathcal{E}$, denoted by $G \circ F$.

3. Similarly, one can define the composition of a covariant functor with a contravariant functor and one obtains a contravariant functor.

Proposition 4.22. *Every functor (regardless of whether it is covariant or contravariant) sends invertible morphisms to invertible morphisms.*

4.6 Natural Transformations

Definition 4.23. Let $\mathcal{C}$ and $\mathcal{D}$ be two categories and let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be two covariant functors.

- A *natural transformation* between the functors F and G assigns to each $X \in Ob(\mathcal{C})$ a morphism $\Phi_X : F(X) \rightarrow G(X)$ in $\mathcal{D}$, s.t.: $\forall \mu \in Mor_{\mathcal{C}}(X, Y)$, the following diagram of morphisms in $\mathcal{D}$ commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{F(\mu)} & F(Y) \\ \downarrow \Phi_X & & \downarrow \Phi_Y \\ G(X) & \xrightarrow{G(\mu)} & G(Y) \end{array}$$

- If $\forall X \in Ob(\mathcal{C})$, the morphism $\Phi_X : F(X) \rightarrow G(X)$ is an isomorphism, then we refer to the natural transformation as a *natural isomorphism*. In this case, given $X \in Ob(\mathcal{C})$, we often say that $F(X)$ and $G(X)$ are *naturally isomorphic*.

The same way one can also define the notion of a natural transformation and a natural isomorphism between two contravariant functors.

Example:

- Consider $\mathcal{Vec}_{\mathbb{F}}$. Given $V \in \mathcal{Vec}_{\mathbb{F}}$, we write

$$V^* := Hom_{\mathbb{F}}(V, \mathbb{F})$$

and given $f : V \rightarrow W$ where $W \in \mathcal{Vec}_{\mathbb{F}}$, we denote by $f^* : W^* \rightarrow V^*$ the induced map. We consider two covariant functors from $\mathcal{Vec}_{\mathbb{F}}$ to $\mathcal{Vec}_{\mathbb{F}}$:

$$id : (f : V \rightarrow W) \mapsto (f : V \rightarrow W), F : (f : V \rightarrow W) \mapsto ((f^*)^* : V^{**} \rightarrow W^{**})$$

For each V , we define $\Phi_V : V \rightarrow V^{**}$ by

$$\Phi_V : v \mapsto \left(\begin{array}{c} V \rightarrow V^{**} \\ V^* \rightarrow \mathbb{F} \\ (\varphi : V \rightarrow \mathbb{F}) \mapsto \varphi(v) \end{array} \right)$$

These maps (Φ_V) define a natural transformation from id to F . Furthermore, if we restrict $\mathcal{Vec}_{\mathbb{F}}$ to $\mathcal{Vec}^{\dim < \infty}_{\mathbb{F}}$, then the above Φ_V 's show that any finite-dimensional $\mathbb{F}$ -vector space V is naturally isomorphic to its double dual V^{**} .

- Let $\mathbb{F}$ be a field. Denote by $\mathcal{Epi}$ the following category:
 - $Ob(\mathcal{Epi})$ consists of all epimorphisms $\varphi : V \rightarrow W$ between two vector spaces over $\mathbb{F}$
 - $Mor_{\mathcal{Epi}}(\varphi : V \rightarrow W, \varphi' : V' \rightarrow W')$ consists of all pairs of $\mathbb{F}$ -linear maps $(\alpha : V \rightarrow V', \beta :$

$W \rightarrow W'$) satisfying that the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ \downarrow \alpha & & \downarrow \beta \\ V' & \xrightarrow{\varphi'} & W' \end{array}$$

The composition of (α, β) and (α', β') is defined as $(\alpha, \beta) \circ (\alpha', \beta') := (\alpha' \circ \alpha, \beta' \circ \beta)$ provided $\alpha' \circ \alpha$ and $\beta' \circ \beta$ are defined.

The following maps define covariant functors from $\mathcal{Epi}$ to $\mathcal{Vec}_{\mathbb{F}}$:

$$F : \begin{array}{c} \mathcal{Epi} \rightarrow \mathcal{Vec}_{\mathbb{F}} \\ (\varphi: V \rightarrow W) \mapsto V / \ker \varphi \\ (\alpha, \beta) \mapsto \left(\begin{array}{c} V / \ker \varphi \rightarrow V' / \ker \varphi' \\ [\mathbf{v}] \mapsto [\alpha(\mathbf{v})] \end{array} \right) \end{array}, G : \begin{array}{c} \mathcal{Epi} \rightarrow \mathcal{Vec}_{\mathbb{F}} \\ (\varphi: V \rightarrow W) \mapsto W \\ (\alpha, \beta) \mapsto \beta \end{array}$$

It's easy to verify that the maps $\begin{array}{c} V / \ker \varphi \rightarrow W \\ [\mathbf{v}] \mapsto \varphi(\mathbf{v}) \end{array}$ are well-defined and form a natural isomorphism from F to G .

- Consider the covariant functors

$$F : \begin{array}{c} \mathcal{Ring} \rightarrow \mathcal{Ring} \\ R \mapsto \mathbb{Z} \\ (f: R \rightarrow S) \mapsto (id_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}) \end{array}, id : \begin{array}{c} \mathcal{Ring} \rightarrow \mathcal{Ring} \\ R \mapsto R \\ (f: R \rightarrow S) \mapsto (f: R \rightarrow S) \end{array}$$

For each commutative unital ring R , consider the ring homomorphism:

$$\Phi_R : \begin{array}{c} \mathbb{Z} \rightarrow R \\ n \mapsto n \cdot 1_R \end{array}$$

For unital ring homomorphism $f : R \rightarrow S$, i.e.: $f(1_R) = 1_S$, the following diagram commutes:

$$\begin{array}{ccc} F(R) = \mathbb{Z} & \xrightarrow{F_*(f)=id} & \mathbb{Z} = F(S) \\ \downarrow \Phi_R & & \downarrow \Phi_S \\ id(R) = R & \xrightarrow{id_*(f)=f} & S = id(S) \end{array}$$

Hence, Φ_R 's define a natural transformation from F to id . Therefore, given a commutative unital ring, we refer to this ring homomorphism $\mathbb{Z} \rightarrow R$ as the natural ring homomorphism.

4.7 Equivalence of Categories

Definition 4.24. Let $\mathcal{C}$ and $\mathcal{D}$ be two categories.

- An *equivalence* of the categories $\mathcal{C}$ and $\mathcal{D}$ is a tuple (F, G, μ, ν) where $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ are two covariant functors and $\mu : G \circ F \rightarrow id_{\mathcal{C}}$ and $\nu : F \circ G \rightarrow id_{\mathcal{D}}$ are natural isomorphisms.
- A *weak equivalence* of the categories $\mathcal{C}$ and $\mathcal{D}$ is a covariant functor $F : \mathcal{C} \rightarrow \mathcal{D}$ that admits a covariant functor $G : \mathcal{D} \rightarrow \mathcal{C}$ and two natural isomorphisms $\mu : G \circ F \rightarrow id_{\mathcal{C}}$ and $\nu : F \circ G \rightarrow id_{\mathcal{D}}$, s.t.: (F, G, μ, ν) is an equivalence of categories.
- We say that $\mathcal{C}$ and $\mathcal{D}$ are *equivalent categories* if there exists an equivalence of categories $\mathcal{C}$ and $\mathcal{D}$.

Remark: Given a weak equivalence F of $\mathcal{C}$ and $\mathcal{D}$, the corresponding G, μ, ν may NOT be uniquely defined.

Proposition 4.25. *Weak Equivalence of Categories Proposition*

Let $\mathcal{C}$ and $\mathcal{D}$ be two categories and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a covariant functor. The following statements are equivalent:

- $F : \mathcal{C} \rightarrow \mathcal{D}$ is a weak equivalence of categories $\mathcal{C}$ and $\mathcal{D}$.
- The following three statements hold:
 - F is full, i.e.: $\forall C_1, C_2 \in Ob(\mathcal{C})$, the map $Mor_{\mathcal{C}}(C_1, C_2) \rightarrow Mor_{\mathcal{D}}(F(C_1), F(C_2))$ is surjective.
 - F is faithful, i.e.: $\forall C_1, C_2 \in Ob(\mathcal{C})$, the map $Mor_{\mathcal{C}}(C_1, C_2) \rightarrow Mor_{\mathcal{D}}(F(C_1), F(C_2))$ is injective.
 - F is essentially surjective (or dense), i.e.: each $D \in \mathcal{D}$ is isomorphic to an object of the form $F(C)$ for some object $C \in Ob(\mathcal{C})$.

4.8 Product and Coproduct

Definition 4.26. Product

Let $\mathcal{C}$ be a category and let $\{X_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$. A *product* of $\{X_i\}_{i \in I}$ in the category $\mathcal{C}$ is an object $P \in Ob(\mathcal{C})$ together with a family of morphisms $\{p_i : P \rightarrow X_i\}_{i \in I}$, s.t.: the following universal property is satisfied: given an object $Z \in Ob(\mathcal{C})$ and a family of morphisms $\{f_i : Z \rightarrow X_i\}_{i \in I}$, there exists a unique morphism $\Phi : Z \rightarrow P$, s.t.: $\forall i \in I$, $f_i = p_i \circ \Phi$, i.e.: the following diagram commutes:

$$\begin{array}{ccc}
 Z & \xrightarrow{\exists! \Phi} & P \\
 & \searrow f_i & \downarrow p_i \\
 & & X_i
 \end{array}$$

Proposition 4.27. Let $\mathcal{C}$ be a category and let $\{X_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$. If we have products $(P, \{p_i : P \rightarrow X_i\}_{i \in I})$ and $(P', \{p'_i : P' \rightarrow X_i\}_{i \in I})$, then there exist unique isomorphisms

$\Phi : P' \rightarrow P$ and $\Phi' : P \rightarrow P'$ which are inverses of one another and such that $\forall i \in I$, the following diagram commutes:

$$\begin{array}{ccc} P' & \xrightleftharpoons[\Phi']{\Phi} & P \\ & \searrow p'_i & \swarrow p_i \\ & X_i & \end{array}$$

Remark (Notation): Let $\mathcal{C}$ be a category and let $\{X_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$ such that a product exists.

- We usually denote the product by $\prod_{i \in I} X_i$ and we refer to the morphisms $\prod_{i \in I} X_i \rightarrow X_i$ as *natural projections*. Furthermore, given morphisms $f_i : Z \rightarrow X_i$, we denote the corresponding morphism $Z \rightarrow \prod_{i \in I} X_i$ by $\prod_{i \in I} f_i$.
- If $I = \{1, \dots, k\}$, then we often denote the product by $X_1 \times \dots \times X_k$. Furthermore, given morphisms $f_i : Z \rightarrow X_i$, $i = 1, \dots, k$, we denote the corresponding morphism $Z \rightarrow X_1 \times \dots \times X_k$ by $(f_1, \dots, f_k)$.

Proposition 4.28. *Product-of-Sets-and-Groups Proposition*

1. Let $\mathbf{Set}$ be the category of sets and let $\{X_i\}_{i \in I}$ be a family of sets. The product in $\mathbf{Set}$ is precisely the product:

$$\prod_{i \in I} X_i := \{f : I \rightarrow \bigcup_{i \in I} X_i \mid f(i) \in X_i, \forall i \in I\}$$

together with maps $\{f \mapsto f(i)\}_{i \in I}$.

2. Let $\mathbf{Group}$ be the category of groups and let $\{X_i\}_{i \in I}$ be a family of groups. If $I \neq \emptyset$, then the product in $\mathbf{Set}$ is precisely the product:

$$\prod_{i \in I} X_i := \{f : I \rightarrow \bigcup_{i \in I} X_i \mid f(i) \in X_i, \forall i \in I\}$$

together with group homomorphisms $\{f \mapsto f(i)\}_{i \in I}$.

3. The same approach as in 2 also applies to $\mathbf{AbGroup}$ and $\mathbf{Vec}_{\mathbb{F}}$.

Proof. 1. Let $Z \in \mathbf{Ob}(\mathbf{Set})$ and let $\{f_i : Z \rightarrow X_i\}_{i \in I}$ be a family of maps between sets. Define

$$\Phi : z \mapsto \begin{pmatrix} Z \rightarrow \prod_{i \in I} X_i \\ I \rightarrow \bigcup_{i \in I} X_i \\ i \mapsto f_i(z) \end{pmatrix}$$

Then it follows immediately from the def that $\forall i \in I$, $f_i = p_i \circ \Phi$. For the uniqueness of Φ , let $\Psi : Z \rightarrow \prod_{i \in I} X_i$ be a continuous map, s.t.: $\forall i \in I$, $f_i = p_i \circ \Psi$. Then for any $z \in Z$, any

$$i \in I,$$

$$\Phi(z)(i) = p_i(\Phi(z)) = f_i(z) = p_i(\Psi(z))$$

□

Example: Let $(P, \leq)$ be a partially ordered set. Consider the corresponding order category $\mathcal{O}rder_{(P, \leq)}$. Given a family $\{X_i\}_{i \in I}$ of objects in $\mathcal{O}rder_{(P, \leq)}$, i.e.: given a family $\{X_i\}_{i \in I}$ of elements of P , one can show that

$$\Pi_{i \in I} X_i := \begin{cases} \inf(\{X_i\}_{i \in I}) \\ \emptyset \end{cases}, \text{ if no such infimum exist}$$

is the product.

Definition 4.29. Coproducts

Let $\mathcal{C}$ be a category and let $\{X_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$. A *coproduct* of $\{X_i\}_{i \in I}$ in the category $\mathcal{C}$ is an object $C \in \text{Ob}(\mathcal{C})$ together with a family of morphisms $\{\iota_i : X_i \rightarrow C\}_{i \in I}$ satisfying the following property: given an object $Z \in \text{Ob}(\mathcal{C})$ and a family of morphisms $\{f_i : X_i \rightarrow Z\}_{i \in I}$, there exists a unique morphism $\Phi : C \rightarrow Z$, s.t.: $f_i = \Phi \circ \iota_i$ for any i , s.t.: the following diagram commutes:

$$\begin{array}{ccc} X_i & & \\ \downarrow \iota_i & \searrow f_i & \\ C & \xrightarrow{\exists! \Phi} & Z \end{array}$$

Remark: Note that by def, coproducts in $\mathcal{C}$ are products in the opposite category $\mathcal{C}^{op}$.

Proposition 4.30. Let $\mathcal{C}$ be a category and let $\{X_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$. Given two coproducts $(C, \{\iota_i : X_i \rightarrow C\}_{i \in I})$ and $(C', \{\iota'_i : X_i \rightarrow C'\}_{i \in I})$, there exist unique isomorphisms $\Phi : C \rightarrow C'$ and $\Phi' : C' \rightarrow C$ which are inverses of one another and such that $\forall i \in I$, the following diagram commutes:

$$\begin{array}{ccc} & X_i & \\ & \downarrow \iota'_i & \\ C & \xrightarrow{\Phi} & C' \\ & \xleftarrow{\Phi'} & \end{array}$$

Remark (Notation): Let $\mathcal{C}$ be a category and let $\{X_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$, s.t.: a coproduct exists. Often we denote the coproduct by

$$\bigsqcup_{i \in I} X_i$$

and we refer to the morphisms $X_i \rightarrow \bigsqcup_{i \in I} X_i$ as *natural injections*.

Proposition 4.31. *Coproduct-of-Sets-and-Abelian Groups Proposition*

1. Let $\mathbf{Set}$ be the category of sets and let $\{X_i\}_{i \in I}$ be a family of sets. The coproduct in $\mathbf{Set}$ is the disjoint union

$$\bigsqcup_{i \in I} X_i := \bigcup_{i \in I} (X_i \times \{i\})$$

together with the natural injection of X_i .

2. Let $\mathbf{AbGroup}$ be the category of abelian groups and let $\{X_i\}_{i \in I}$ be a family of groups. The coproduct in $\mathbf{AbGroup}$ is given by

$$\bigsqcup_{i \in I} X_i = \bigoplus_{i \in I} X_i := \{(g_i)_{i \in I} \in \prod_{i \in I} X_i \mid \text{all but finitely many } g_i \text{ are trivial}\}$$

with the homomorphisms

$$x \mapsto \begin{pmatrix} X_i \rightarrow \bigoplus_{i \in I} X_i \\ I \rightarrow \bigcup_{i \in I} X_i \\ j \mapsto \begin{cases} x_i & , \text{if } i = j \\ e_{X_j} & , \text{if } i \neq j \end{cases} \end{pmatrix}$$

3. The same approach as in 2 also applies to $\mathbf{Vec}_{\mathbb{F}}$.

4.9 Limits and Colimits

Definition 4.32. Diagram

Let $(P, \leq)$ be a partially ordered set and let $\mathcal{C}$ be a category. A *diagram* in $\mathcal{C}$ over $(P, \leq)$ is a covariant functor from $\mathcal{O}rder_{(P, \leq)}$ to $\mathcal{C}$.

Definition 4.33. Limits

Let $(P, \leq)$ be a partially ordered set, let $\mathcal{C}$ be a category and let F be a diagram in $\mathcal{C}$ over $(P, \leq)$. A *limit* of F consists of an object $L \in \mathcal{O}b(\mathcal{C})$ together with morphisms $f_x : L \rightarrow F(x)$, $\forall x \in P$ satisfying the following two conditions:

- $\forall x \leq y, f_y = F(x \rightarrow y) \circ f_x : L \rightarrow F(y)$.
- The following universal property holds: if $Z \in \mathcal{O}b(\mathcal{C})$ and if we are given morphisms $g_x : Z \rightarrow F(x)$, $\forall x \in P$, s.t.: 1 is satisfied, then $\exists!$ morphism $\Phi : Z \rightarrow L$, s.t.: $g_x = f_x \circ \Phi \quad \forall x \in P$.

Proposition 4.34. *Uniqueness of Limit*

Let $(P, \leq)$ be a partially ordered set, let $\mathcal{C}$ be a category and let F be a diagram in $\mathcal{C}$ over $(P, \leq)$. Suppose that we're given two limits $(L, \{f_x : L \rightarrow F(x)\}_{x \in P})$ and $(L', \{f'_x : L' \rightarrow F(x)\}_{x \in P})$. Then there exist uniquely determined isomorphisms $\Phi : L' \rightarrow L$ and $\Phi' : L \rightarrow L'$ which are inverses of

one another and such that $\forall x \in P$, the following diagram commutes:

$$\begin{array}{ccc} L' & \begin{array}{c} \xrightarrow{\Phi} \\ \xleftarrow{\Phi'} \end{array} & L \\ & \searrow f'_x & \swarrow f_x \\ & F(x) & \end{array}$$

Remark (Notation): If a limit exists, then often we denote it by $\lim F$.

Definition 4.35. Pullback

Consider $P = \{0, x, y\}$ with the partial order $\leq$ that is given by the trivial relations $0 \leq 0, x \leq x, y \leq y$ and the nontrivial relations $x \leq 0, y \leq 0$. If $(P, \leq)$ gives rise to the diagram:

$$\begin{array}{ccc} & x & \\ & \downarrow & \\ y & \longrightarrow & 0 \end{array} \Rightarrow \begin{array}{ccc} & X & \\ & \downarrow \varphi_X & \\ Y & \xrightarrow{\varphi_Y} & A \end{array}$$

then the corresponding limit is called a *pullback*:

$$\begin{array}{ccc} \text{pullback } L & \xrightarrow{f_X} & X \\ f_Y \downarrow & \lrcorner & \downarrow \varphi_X \\ Y & \xrightarrow{\varphi_Y} & A \end{array}$$

Here the symbol " $\lrcorner$ " indicates that L is the pullback of the " $\lrcorner$ -shaped" diagram.

Example: In *Group*,

$$\begin{array}{ccc} G & \xrightarrow{\ker \varphi} & G \\ \downarrow \varphi & \Rightarrow & \downarrow \lrcorner \downarrow \varphi \\ \{e\} & \longrightarrow & H \end{array}$$

Proposition 4.36. Pullback Proposition

Let $\mathcal{C}$ be a category.

1. Suppose that $\mathcal{C}$ admits a terminal object $*$. Then the pullback of the diagram:

$$\begin{array}{ccc} X \times Y & \xrightarrow{p_X} & X \\ p_Y \downarrow & \lrcorner & \downarrow \varphi_X \\ Y & \xrightarrow{\varphi_Y} & * \end{array}$$

is given by the product $X \times Y$.

2. Let

$$\begin{array}{ccc} L & \xrightarrow{f_X} & X \\ f_Y \downarrow & \lrcorner & \downarrow \varphi_X \\ Y & \xrightarrow{\varphi_Y} & A \end{array}$$

be a pullback.

(a) If φ_X is an isomorphism, then so is the "opposite" morphism f_Y .

(b) If φ_X is a monomorphism, then so is the "opposite" morphism f_Y .

3. Given pullbacks

$$\begin{array}{ccc} L & \xrightarrow{f_X} & X \\ f_Y \downarrow & \lrcorner & \downarrow \varphi_X \\ Y & \xrightarrow{\varphi_Y} & A \end{array} \quad \text{and} \quad \begin{array}{ccc} X & \xrightarrow{\theta} & Z \\ \varphi_X \downarrow & \lrcorner & \downarrow \psi_Z \\ A & \xrightarrow{\psi_A} & B \end{array}$$

the "composition"

$$\begin{array}{ccc} L & \xrightarrow{\theta \circ f_X} & Z \\ f_Y \downarrow & \lrcorner & \downarrow \psi_Z \\ Y & \xrightarrow{\psi_A \circ \varphi_Y} & B \end{array}$$

is also a pullback.

Proof. 1. (b) Let $\alpha : Z \rightarrow L$ and $\beta : Z \rightarrow L$ be two morphisms such that $f_Y \circ \alpha = f_Y \circ \beta$. Hence,

$$\varphi_X \circ f_X \circ \alpha = \varphi_Y \circ f_Y \circ \alpha = \varphi_Y \circ f_Y \circ \beta = \varphi_X \circ f_X \circ \beta$$

Since φ_X is monomorphism, then $f_X \circ \alpha = f_X \circ \beta$. By the universal property of limit, $\exists! \Phi : Z \rightarrow L$, s.t.: $f_Y \circ \Phi = f_Y \circ \alpha = f_Y \circ \beta$ and s.t.: $f_X \circ \Phi = f_X \circ \alpha = f_X \circ \beta$. Hence, $\Phi = \alpha = \beta$. □

Definition 4.37. Colimits

Let $(P, \leq)$ be a partially ordered set, let $\mathcal{C}$ be a category and let F be a diagram in $\mathcal{C}$ over $(P, \leq)$. A *colimit* of F consists of an object $C \in \text{Ob}(\mathcal{C})$ together with morphisms $f_x : F(x) \rightarrow C$, $\forall x \in P$ satisfying the following two conditions:

- $\forall x \leq y$, $f_x = f_y \circ F(x \rightarrow y) : F(x) \rightarrow C$.
- The following universal property holds: if $Z \in \text{Ob}(\mathcal{C})$ and if we are given morphisms $g_x : F(x) \rightarrow Z$, $\forall x \in P$, s.t.: 1 is satisfied, then $\exists!$ morphism $\Phi : C \rightarrow Z$, s.t.: $g_x = f_x \circ \Phi \circ f_x \quad \forall x \in P$.

Proposition 4.38. Uniqueness of Colimit

Let $(P, \leq)$ be a partially ordered set, let $\mathcal{C}$ be a category and let F be a diagram in $\mathcal{C}$ over $(P, \leq)$. Suppose that we're given two colimits $(C, \{f_x : C \rightarrow F(x)\}_{x \in P})$ and $(C', \{f'_x : C' \rightarrow F(x)\}_{x \in P})$. Then there exist uniquely determined isomorphisms $\Phi : C \rightarrow C'$ and $\Phi' : C' \rightarrow C$ which are inverses

of one another and such that $\forall x \in P$, the following diagram commutes:

$$\begin{array}{ccc} F(x) & & \\ f_x \downarrow & \searrow f'_x & \\ C & \xrightleftharpoons[\Phi']{\Phi} & C' \end{array}$$

Remark: If a colimit exists, then often we denote it by $\text{colim } F$.

Definition 4.39. Pushout

Consider $P = \{0, x, y\}$ with the partial order $\leq$ that is given by the trivial relations $0 \leq 0, x \leq x, y \leq y$ and the nontrivial relations $0 \leq x, 0 \leq y$. If $(P, \leq)$ gives rise to the diagram:

$$\begin{array}{ccc} 0 & \longrightarrow & x \\ \downarrow & & \Rightarrow \varphi_Y \downarrow \\ y & & Y \end{array} \quad \begin{array}{ccc} A & \xrightarrow{\varphi_X} & X \\ & & \downarrow f_X \\ Y & \xrightarrow{f_Y} & \text{pushout } C \end{array}$$

then the corresponding colimit is called a *pushout*:

$$\begin{array}{ccc} A & \xrightarrow{\varphi_X} & X \\ \varphi_Y \downarrow & \lrcorner & \downarrow f_X \\ Y & \xrightarrow{f_Y} & \text{pushout } C \end{array}$$

Here the symbol " $\lrcorner$ " indicates that C is the pushout of the " $\lrcorner$ -shaped" diagram.

Example: Let G be a group and let $A \triangleleft G$ be a normal group of G . Consider the following pushout diagram in *Group*:

$$\begin{array}{ccc} A & \longrightarrow & \{e\} \\ \downarrow & & \downarrow \\ G & & G/A \end{array} \quad \Rightarrow \quad \begin{array}{ccc} A & \longrightarrow & \{e\} \\ \downarrow & \lrcorner & \downarrow \\ G & \longrightarrow & G/A \end{array}$$

Proposition 4.40. *Pushout Proposition*

Let $\mathcal{C}$ be a category.

1. Suppose that $\mathcal{C}$ admits a initial object $*$. Then the pushout of the diagram:

$$\begin{array}{ccc} * & \xrightarrow{\varphi_X} & X \\ \varphi_Y \downarrow & \lrcorner & \downarrow \iota_X \\ Y & \xrightarrow{\iota_Y} & X \sqcup Y \end{array}$$

is given by the coproduct $X \sqcup Y$.

2. Let

$$\begin{array}{ccc} A & \xrightarrow{\varphi_X} & X \\ \varphi_Y \downarrow & & \downarrow f_X \\ Y & \xrightarrow{f_Y} & C \end{array}$$

be a pushout.

(a) If φ_X is an isomorphism, then so is the "opposite" morphism f_Y .

(b) If φ_X is a monomorphism, then so is the "opposite" morphism f_Y .

3. Given pushouts

$$\begin{array}{ccc} A & \xrightarrow{\varphi_X} & X \\ \varphi_Y \downarrow & & \downarrow f_X \\ Y & \xrightarrow{f_Y} & C \end{array} \quad \text{and} \quad \begin{array}{ccc} X & \xrightarrow{\theta} & Z \\ f_X \downarrow & & \downarrow g_Z \\ L & \xrightarrow{g_L} & W \end{array}$$

the "composition"

$$\begin{array}{ccc} A & \xrightarrow{\theta \circ \varphi_X} & Z \\ \varphi_Y \downarrow & & \downarrow g_Z \\ Y & \xrightarrow{g_L \circ f_Y} & W \end{array}$$

is also a pushout.

Chapter 5

Product and Disjoint Unions of Topological Spaces

Recall: the defs of product topological and disjoint unions are introduced in the previous chapter.

5.1 Product of Topological Spaces

5.1.1 Properties of Product Topological Spaces

Proposition 5.1. *Product Metric*

1. Let $(X_1, d_1), \dots, (X_k, d_k)$ be metric spaces. We define the product metric d on the product $X_1 \times \dots \times X_k$ as follows:

$$d((x_1, \dots, x_k), (y_1, \dots, y_k)) := \sum_{i=1}^k d_i(x_i, y_i)$$

The product topology on $X_1 \times \dots \times X_k$ agrees with the metric topology coming from the product metric.

2. Let $n_1, \dots, n_m \in \mathbb{N} \cup \{0\}$. The map $\varphi : \mathbb{R}^{n_1 + \dots + n_m} \rightarrow \mathbb{R}^{n_1} \times \dots \times \mathbb{R}^{n_m}$ given by

$$\varphi(x_{1,1}, \dots, x_{1,n_1}, \dots, x_{m,1}, \dots, x_{m,n_m}) := ((x_{1,1}, \dots, x_{1,n_1}), \dots, (x_{m,1}, \dots, x_{m,n_m}))$$

is a homeomorphism. The analogous statement also holds if we replace " $\mathbb{R}$ " by " $\mathbb{C}$ ".

Proof. 1. We can only prove it with the case $k = 2$ and then do induction on k . It's easy to verify that the product metric d gives a metric on $X_1 \times X_2$. Denote the product topology as $\mathcal{T}_{product}$ and the topology induced by d as $\mathcal{T}_d$.

- $\mathcal{T}_{product} \subseteq \mathcal{T}_d$: For any $U \in \mathcal{T}_{product}$, any $(x_1, x_2) \in U$, there exist $\epsilon_1, \epsilon_2 > 0$, s.t.:

$$(x_1, x_2) \in B_{\epsilon_1}^{d_1}(x_1) \times B_{\epsilon_2}^{d_2}(x_2) \subseteq U$$

Take $\epsilon := \min\{\epsilon_1, \epsilon_2\}$. Note that

$$(x_1, x_2) \in B_{\epsilon}^d(x_1, x_2) \subseteq B_{\epsilon_1}^{d_1}(x_1) \times B_{\epsilon_2}^{d_2}(x_2) \subseteq U$$

Hence, $U \in \mathcal{T}_d$. Therefore, $\mathcal{T}_{product} \subseteq \mathcal{T}_d$.

- Similarly for $\mathcal{T}_d \subseteq \mathcal{T}_{product}$.

Therefore, $\mathcal{T}_d = \mathcal{T}_{product}$.

2. By 1, it suffices to show that the product metric on $\mathbb{R}^{n_1} \times \cdots \times \mathbb{R}^{n_m}$ and the Euclidean metric on $\mathbb{R}^{n_1 + \cdots + n_m}$ define the same topological.

□

Proposition 5.2. *Product Topology-Properties Proposition*

Let $X_1, \dots, X_k$ be non-empty topological spaces.

1. $X_1 \times \cdots \times X_k$ is compact $\Leftrightarrow$ each X_i is compact.
2. $X_1 \times \cdots \times X_k$ is (path-)connected $\Leftrightarrow$ each X_i is (path-)connected.
3. $X_1 \times \cdots \times X_k$ is Hausdorff $\Leftrightarrow$ each X_i is Hausdorff.

Proof. 1. • $(\Leftarrow)$: Directly by Tychonff Theorem.

- $(\Rightarrow)$: Consider the natural projections and then we're done.

2. • $(\Rightarrow)$: Also consider the natural projections and then we're done.

- $(\Leftarrow)$: It suffices to show that the case when $k = 2$ and then do induction on k .

– *Path-connected-ness*: Let (x, y) and (x', y') be two points in $X \times Y$. Since X and Y are path-connected, then there exist paths $p : I \rightarrow X$ from x to x' and $q : I \rightarrow Y$ from y to y' . Then

$$\begin{aligned} I &\rightarrow X \times Y \\ t &\mapsto (p(t), q(t)) \end{aligned}$$

is a path from (x, y) to (x', y') .

– *Connected-ness*: Pick $a \in X$ and $b \in Y$. Note that for any $x \in X$, $\{x\} \times Y$ and $X \times \{b\}$ are connected. Since $(x, b) \in \{x\} \times Y$ and $\in X \times \{b\}$, then

$$(\{x\} \times Y) \cup (X \times \{b\})$$

is connected. Since for any $x \in X$, $(a, b) \in (\{x\} \times Y) \cup (X \times \{b\})$, then

$$X \times Y = \bigcup_{x \in X} (\{x\} \times Y) \cup (X \times \{b\})$$

is connected.

3. • ($\Leftarrow$): $\forall (x, y) \neq (x', y') \in X \times Y$, WLOG, let $x \neq x'$. Then by the Hausdorff-ness of X , there exist U, U' open in X , s.t.:

$$x \in U, x' \in U', U \cap U' = \emptyset$$

Note that

$$(U \times Y) \cap (U' \times Y) = (U \cap U') \times Y = \emptyset \times Y = \emptyset$$

Then the disjoint open neighborhood of (x, y) and (x', y') can be $U \times Y$ and $U' \times Y$.

- ($\Rightarrow$): It suffices to show that X is Hausdorff. $\forall x \neq x' \in X$, let $y_0 \in Y$ and then there exist W, W' open in $X \times Y$, s.t.:

$$(x, y_0) \in W, (x', y_0) \in W', W \cap W' = \emptyset$$

By the def of the product topology, there exist open $U \ni x, U' \ni x', V \ni y_0, V' \ni y_0$, s.t.: $(U \times V) \cap (U' \times V') = \emptyset$. Hence,

$$(U \cap U') \times \{y_0\} = (U \cap \{y_0\}) \times (U' \cap \{y_0\}) \subseteq (U \times V) \cap (U' \times V') = \emptyset$$

Therefore, $U \cap U' = \emptyset$.

□

Proposition 5.3. *General-Product Topology-Properties Proposition*

Let $\{X_i\}_{i \in I}$ be a family of non-empty topological spaces.

1. $\prod_{i \in I} X_i$ is compact $\Leftrightarrow$ each X_i is compact.
2. $\prod_{i \in I} X_i$ is (path-)connected $\Leftrightarrow$ each X_i is (path-)connected.
3. $\prod_{i \in I} X_i$ is Hausdorff $\Leftrightarrow$ each X_i is Hausdorff.

Proof. The proof of all statements is same as the proof of the corresponding statements of the above proposition, except, for the " $\Leftarrow$ "-direction of 2 in the connected case.

Lemma 5.4. *Closure-Connected Lemma*

Let X be a topological space and let $A \subseteq X$ be a subset of X . If A is connected, then so is its closure $\bar{A}$.

Proof. Let $\bar{A} = C \cup D$ be a decomposition of $\bar{A}$ into two disjoint open subsets. Then $A = (A \cap C) \cup (A \cap D)$. Since A is connected, then WLOG $A = A \cap C$. Hence, $A \subseteq C$. Since $C = \bar{A} \setminus D$ where $\bar{A}$ is closed and D is open, then C is closed in $\bar{A}$. Hence, $\bar{A} \subseteq C$. Therefore, $\bar{A} \subseteq C \subseteq \bar{A} \Leftrightarrow \bar{A} = C$. □

$\forall i \in I$, we pick a point $P_i \in X_i$. Furthermore, for each finite subset $F \subseteq I$, we set

$$Y_F := \{(x_i)_{i \in I} \in \prod_{i \in I} X_i \mid x_i = P_i \forall i \notin F\}$$

- **Claim 1:** $Y := \bigcup_{F \subseteq I \text{ finite}} Y_F$ is connected.

Proof. Let $F \subseteq I$ be a finite subset. Note that there is an bijection between Y_F and $\prod_{i \in F} X_i$. It's easy to verify that the bijections are continuous and hence they are homeomorphisms. Then by Product Topology-Properties Proposition 2, $\prod_{i \in F} X_i$ is connected $\Rightarrow Y_F$ is connected. Since $(P_i)_{i \in I} \in Y_F$ for any F , then we're done. $\square$

- **Claim 2:** $Y := \bigcup_{F \subseteq I \text{ finite}} Y_F$ is dense in $\prod_{i \in I} X_i$.

Proof. For any open non-empty U in $\prod_{i \in I} X_i$, there exist open non-empty V_i in X_i , s.t.: $\prod_{i \in I} V_i \subseteq U$ and s.t.: for any $i \in F_0$, $V_i = X_i$ for some finite $F_0 \subseteq I$. Hence,

$$Y_{F_0} \cap \prod_{i \in I} V_i \neq \emptyset \Rightarrow Y \cap U \neq \emptyset$$

$\square$

$\square$

Definition 5.5. Isolated Points

Let X be a topological space. A point $x_0 \in X$ is called *isolated* if $\{x_0\}$ itself is an open subset of X .

Definition 5.6. Total Disconnected-ness

A topological space X is called *totally disconnected* if the only connected subsets are the empty set and the subsets consisting of a single point.

Proposition 5.7. Cantor Set Characterization Proposition

Let X be a topological space that is compact and totally disconnected. If X contains no isolated point, then X is homeomorphic to the Cantor set.

5.1.2 Application of the Product Topology

Proposition 5.8. Hausdorff-Diagonal Proposition

X is Hausdorff $\Leftrightarrow$ the diagonal $\Delta := \{(x, x) \mid x \in X\}$ is closed in $X \times X$.

Proof. • $(\Rightarrow)$: For any point in $X \times X \setminus \Delta$, it is (x, y) for some distinct $x, y \in X$. By the Hausdorff-ness of X , there exist open $U \ni x, V \ni y$, s.t.: $U \cap V = \emptyset$. Hence,

$$(U \times U) \cap (U \times V) = U \times (U \cap V) = \emptyset \Rightarrow (x, y) \in U \times V \subseteq X \times X \setminus U \times U$$

Therefore, $X \times X \setminus \Delta$ is open $\Rightarrow \Delta$ is closed.

- $(\Leftarrow)$: $\forall x \neq y \in X, (x, y) \in X \times X \setminus \Delta$. Since Δ is closed, then there exists open W in $X \times X$, s.t.:

$$(x, y) \in W, W \cap \Delta = \emptyset$$

Hence, by the def of the product topology, there exist open U, V , s.t.:

$$(x, y)] \in U \times V \subseteq W$$

If $U \cap V \neq \emptyset$, then there is a point z in $U \cap V \Rightarrow \emptyset \neq (z, z) \in W \cap \Delta$, contradicting the empty-ness of $W \cap \Delta$. Therefore, $U \cap V = \emptyset$. □

Corollary 5.9. *Maps-agree-one-Closure Proposition*

Let $f, g : X \rightarrow Y$ be two continuous maps between topological spaces. Suppose that Y is Hausdorff.

1. The set $\{x \in X \mid f(x) = g(x)\}$ is a closed subset of X .
2. If f and g agree on a subset A , then f and g also agree on $\overline{A}$.

Proof. 1. Consider the diagonal $\Delta = \{(y, y) \in Y \times Y \mid y \in Y\}$ and the map $(f, g) : X \rightarrow Y \times Y$ given by $x \mapsto (f(x), g(x))$. Note that by the Hausdorff-Diagonal Proposition, Δ is closed and that (f, g) is continuous. Hence,

$$\{x \in X \mid f(x) = g(x)\} = (f, g)^{-1}(\Delta)$$

is closed. □

5.1.3 Pullback of Topological Spaces

Proposition 5.10. *Topological-Pullback Proposition*

Let $p : X \rightarrow B$ be a continuous map between two topological spaces, let C be a topological space and let $f : C \rightarrow B$ be a continuous map. Consider the topological space

$$f^*X := \{(c, x) \in C \times X \mid f(c) = p(x)\}$$

and the maps

$$\varphi : f^*X \rightarrow C \quad \text{and} \quad \psi : f^*X \rightarrow X$$

$$(c, x) \mapsto c \quad \text{and} \quad (c, x) \mapsto x$$

The following statements holds:

1. (a) φ and ψ are continuous.
- (b) Given a topological space W and two continuous maps $\alpha : W \rightarrow C$ and $\beta : W \rightarrow X$, s.t.: $f \circ \alpha = p \circ \beta : W \rightarrow B$, the map $\Phi : W \rightarrow f^*X$ given by

$$\Phi(w) := (\alpha(w), \beta(w))$$

is continuous. In particular, the following diagram commutes:

$$\begin{array}{ccccc}
 W & & & & \\
 \searrow \alpha & & \xrightarrow{\beta} & & X \\
 & \searrow \exists! \Phi & & \nearrow \psi & \\
 & f^*X & \xrightarrow{\psi} & X & \\
 & \downarrow \varphi & \lrcorner & \downarrow p & \\
 & C & \xrightarrow{f} & B &
 \end{array}$$

- (c) $\Phi : W \rightarrow f^*X$ is continuous $\Leftrightarrow \varphi \circ \Phi : W \rightarrow C$ and $\psi \circ \Phi : W \rightarrow X$ are continuous.
2. f^*X with the natural maps $\varphi : f^*X \rightarrow C$ and $\psi : f^*X \rightarrow X$ is the pullback, in the category $\mathcal{Top}$ of topological spaces.

Example: Let $p : X \rightarrow B$ be a continuous map between two topological spaces, let $C \subseteq B$ be a subspace of B and let $\iota : C \hookrightarrow B$ be the inclusion map. One can verify that the maps

$$p^{-1}(C) \xrightarrow{\iota^*} \iota^*X \quad \text{and} \quad \iota^*X \xrightarrow{p^{-1}} p^{-1}(C)$$

$x \mapsto (p(x), x) \qquad (c, x) \mapsto x$

are continuous and inverses of one another. Hence, we see that ι^*X is naturally homeomorphic to $p^{-1}(C)$.

Proposition 5.11. Pullback-Properties Proposition

Let $p : X \rightarrow B$ be a continuous map between two topological spaces, let C be a topological space and let $f : C \rightarrow B$ be a continuous map.

1. If X and C are compact and if X, B, C are Hausdorff, then the pullback f^*X is compact.
2. If X and C are Hausdorff, then the pullback f^*X is also Hausdorff.

Proof. 1. Consider the map $\mu : C \times X \rightarrow B \times B$ given by

$$\mu(c, x) := (f(c), p(x))$$

By the topological-product prop.1, μ is continuous. Consider the diagram $\Delta := \{(b, b) \mid b \in B\} \subseteq B \times B$. Since B is Hausdorff, then by the Hausdorff-diagonal prop, Δ is closed in $B \times B$. By the def of the pullback, one has

$$f^*X = \mu^{-1}(\Delta) \subseteq C \times X$$

Since X and C are compact and Hausdorff, then by the product topology-properties prop.1,3, $C \times X$ is compact and Hausdorff. Therefore, f^*X is compact.

□

5.2 Disjoint Union of Topological Spaces

Definition 5.12. Disjoint Union

Let $\{A_i\}_{i \in I}$ be a family of sets.

- We define the *disjoint union* as the set

$$\bigsqcup_{i \in I} A_i := \bigcup_{i \in I} (A_i \times \{i\})$$

- Given $j \in I$, we refer to the map

$$\begin{aligned} A_j &\rightarrow \bigsqcup_{i \in I} A_i \\ a &\mapsto (a, j) \end{aligned}$$

as the *natural injection*.

Definition 5.13. Disjoint Union Topology

Let $\{X_i\}_{i \in I}$ be a family of topological spaces. Then we define the topology $\mathcal{T}$ on the disjoint union $\bigsqcup_{i \in I} X_i$ by

$$\mathcal{T} := \{U \subseteq \bigsqcup_{i \in I} X_i \mid U \cap X_i \in \mathcal{T}_{X_i}, \forall i \in I\}$$

Proposition 5.14. Topological-Coproduct Proposition

Let $\{X_i\}_{i \in I}$ be a family of topological spaces.

- (a) Each natural injection $\iota_j : X_j \rightarrow \bigsqcup_{i \in I} X_i$ is a continuous open closed embedding.
- (b) If Z is a topological space and if $\{f_i : X_i \rightarrow Z\}_{i \in I}$ is a family of continuous maps, then the map

$$F : \bigsqcup_{i \in I} X_i \rightarrow Z \\ (x, i) \mapsto f_i(x)$$

is continuous.

- (c) $f : \bigsqcup_{i \in I} X_i \rightarrow Z$ is continuous $\Leftrightarrow \forall i \in I, f \circ \iota_i : X_i \rightarrow Z$ is continuous.

- (d) $(\bigsqcup_{i \in I} X_i, \{\iota_i\}_{i \in I})$ is a coproduct of $\{X_i\}_{i \in I}$ in the category $\mathcal{Top}$ of topological spaces.

Proof. 2. Let Z be a topological space and let $\{f_i : X_i \rightarrow Z\}_{i \in I}$ be a family of continuous maps. By 1.(b), we've shown that there exists a continuous map F , s.t.: $f_i = F \circ \iota_i \forall i \in I$. It remains to show that such map is *unique*. $\square$

Proposition 5.15. Disjoint Union Topology-Properties Proposition

Let $\{X_i\}_{i \in I}$ be a family of topological spaces.

- $\bigsqcup_{i \in I} X_i$ compact $\Leftrightarrow$ each X_i is compact and there are only finitely many i 's such that $X_i \neq \emptyset$.
- (a) Each X_i is open and closed in $\bigsqcup_{i \in I} X_i$.
- (b) The (path-)components of $\bigsqcup_{i \in I} X_i$ are precisely the (path-)components of X_i .
- $\bigsqcup_{i \in I} X_i$ Hausdorff $\Leftrightarrow$ each X_i Hausdorff.

Proof. 1. • ($\Leftarrow$): Let $\{U_j\}_{j \in J}$ be an open cover of $\bigsqcup_{i \in I} X_i$ (in particular, it is also an open cover of X_i for any i). Let $i_1, \dots, i_k$ be such that $X_{i_l} \neq \emptyset$ and that $X_i = \emptyset \forall i \in I \setminus \{i_1, \dots, i_k\}$. Then $\bigsqcup_{i \in I} X_i = X_{i_1} \sqcup \dots \sqcup X_{i_k}$. By the compactness of $X_i, \forall i = 1, \dots, k$, $\exists$ finite $J_l \subseteq J$, s.t.:

$$\bigcup_{j \in J_l} (U_j \cap X_{i_l}) = X_{i_l}$$

Hence, $\{U_j \mid j \in J_l\}_{l=1}^k$ is a finite subcover, which gives the compactness of $\bigsqcup_{i \in I} X_i$.

• ($\Rightarrow$): Note that $\{X_i\}_{i \in I}$ is an open cover of $\bigsqcup_{i \in I} X_i$. By the compactness of $\bigsqcup_{i \in I} X_i$, $\exists i_1, \dots, i_k \in I$, s.t.:

$$\bigcup_{l=1}^k X_{i_l} = \bigsqcup_{i \in I} X_i$$

Then it's easy to show that X_{i_l} 's are compact and $X_i = \emptyset \forall i \neq i_1, \dots, i_k$. □

Proposition 5.16. *Disjoint Union Embedding Proposition*

Let X be a topological space and let $\{A_i\}_{i \in I}$ be a family of disjoint subsets of X . The followings are equivalent:

1. $\forall i \in I, \exists$ an open neighborhood U_i of A_i , s.t.: for any $j \neq i$

$$U_i \cap A_j = \emptyset$$

2. The map $\Phi : \bigsqcup_{i \in I} A_i \rightarrow \bigcup_{i \in I} A_i$ given by

$$\Phi(a, i) := a$$

is a homeomorphism.

Proof. First, note that Φ is a bijection since A_i 's are disjoint, and that Φ is continuous, since the inclusion maps $\iota_i : A_i \hookrightarrow \bigcup_{i \in I} A_i$ are continuous.

• ($1 \Rightarrow 2$): It suffices to show that Φ is open. For any open U in $\bigsqcup_{i \in I} A_i$, $U \cap A_i$ is open in A_i for any $i \in I$. Hence, for any $i \in I$, there exists W_i open in X , s.t.: $A_i \cap U = A_i \cap W_i$. Therefore,

$$\Phi(U) = \Phi\left(\bigsqcup_{i \in I} A_i \cap U\right) := \bigcup_{i \in I} (A_i \cap U) = \bigcup_{i \in I} ((A_i \cap U) \cap U_i) = \bigcup_{i \in I} (A_i \cap (W_i \cap U_i))$$

Since $A_i \subseteq W_i \cap U_i$ and $A_i \cap (U_j \cap W_j) = \emptyset \forall j \neq i$, then

$$\Phi(U) = \left(\bigcup_{i \in I} A_i\right) \cap \left(\bigcup_{j \in I} (U_j \cap W_j)\right)$$

Since $U_j \cap W_j$ is open in X , then so is $\bigcup_{j \in I} (U_j \cap W_j)$ and hence $\Phi(U)$ is open in $\bigsqcup_{i \in I} A_i$.

- (2 $\Rightarrow$ 1): Since Φ is homeomorphism, then for any $j \in I$, $\Phi(A_j) := A_j$ is open in $\bigcup_{i \in I} A_i$. Hence, $\exists$ open V_j in X , s.t.:

$$\bigcup_{i \in I} (A_i \cap V_j) = V_j \cap \left(\bigcup_{i \in I} A_i \right) = A_j$$

Note that $\forall k \neq j$, $A_k \cap V_j \subseteq A_k \cap A_j = \emptyset$. Hence, $A_j \cap V_j = A_j$. Therefore, we get $V_i = U_i$ $\forall i \in I$ is the desired open neighborhoods.

□

5.3 Topological Groups

Definition 5.17. Topological Groups

A *topological group* is a topological space with a group structure, s.t.: the maps

$$\begin{array}{c} X \times X \rightarrow X \\ (x, y) \mapsto xy \end{array} \quad \text{and} \quad \begin{array}{c} X \rightarrow X \\ x \mapsto x^{-1} \end{array}$$

are continuous.

Proposition 5.18. $\forall n \in \mathbb{N}_+$, the matrix groups $GL(n, \mathbb{R})$ and $GL(n, \mathbb{C})$ are topological groups.

Definition 5.19. The category $\mathcal{TopGr}$ of topological groups consists of

- $Ob(\mathcal{TopGr}) :=$ all topological groups
- $Mor_{\mathcal{TopGr}}(X, Y) :=$ all continuous homomorphisms from X to Y

with the usual composition of maps.

Proposition 5.20. *Topological Subgroup Proposition*

1. Let X be a topological group and let Y be a subgroup of X . If we equip Y with the subspace topology, then Y is also a topological group.
2. Let $\{X_i\}_{i \in I}$ be a family of topological groups. If we equip $\prod_{i \in I} X_i$ with the product topology and the obvious group structure, then $\prod_{i \in I} X_i$ is again a topological group.

Proposition 5.21. *Components-of-Topological Group Proposition*

Let G be a topological group.

1. Given $g \in G$, the maps

$$l_g : G \rightarrow G \quad \text{and} \quad r_g : G \rightarrow G$$

$$h \mapsto gh \quad \text{and} \quad h \mapsto hg$$

are continuous.

2. If G is a topological group, then any two path-components of G are homeomorphic.

Proof. 1. Note that $l_g = f \circ k$ where

$$f : G \rightarrow G \times G \quad \text{and} \quad k : G \times G \rightarrow G$$

$$h \mapsto (g, h) \quad \text{and} \quad (x, y) \mapsto xy$$

By the def of topological group, k is continuous. f is obviously continuous. Hence, l_g is continuous. Similarly for r_g .

2. Let X, Y be two path-components of G . Fix $x \in X$ and $y \in Y$. Consider the map $\varphi : G \rightarrow G$ given by

$$\varphi(g) := yx^{-1}g$$

By 1, φ is continuous. Similarly, the map $\psi : G \rightarrow G$ with

$$\psi(g) := xy^{-1}g$$

is continuous. Note that

$$\varphi \circ \psi = \psi \circ \varphi = id_G$$

Since $\varphi(X) \subseteq Y$ and $\psi(Y) \subseteq X$, then $X \cong Y$ are homeomorphic.

□

Chapter 6

Quotient Space and Simplicial Complex

6.1 Equivalence Relations and Equivalence Classes

Definition 6.1. • **(Equivalence Relation):** The equivalence relation on a set X is a relation such that:

- (Reflexive): $x \sim x$ for all $x \in X$;
 - (Symmetric): $x \sim y$ implies $y \sim x$;
 - (Transitive): $x \sim y, y \sim z$ implies $x \sim z$.
- **(Partition):** Let X be a non-empty set. A partition $\mathcal{P} := \{P_i \mid i \in I\}$ of X is a collection of subsets such that:
- $P_i \subseteq X$ is non-empty;
 - $P_i \cap P_j = \emptyset$ if $i \neq j$;
 - $\bigcup_{i \in I} P_i = X$.
- **(Equivalence Class):** Let X be a set with equivalence relation $\sim$. The equivalence class of $x \in X$ is defined by:

$$[x] := \{y \in X \mid x \sim y\}.$$

- **(Quotient Space):** The collection of all equivalence classes is called the equivalence space:

$$X/\sim := \{[x] : x \in X\}.$$

Remark: If $\{p_j \mid j \in J\}$ is a partition of X , we can define an equivalence relation by:

$$x \sim y \Leftrightarrow x, y \in p_j \text{ for some } j \in J.$$

Example (Möbius Strip):

Let $X = [0, 1] \times [0, 1]$. We define $(x_1, y_1) \sim (x_2, y_2)$ if

- $x_1 = x_2, y_1 = y_2$ with $x_i \in (0, 1)$;
- $x_1 = 0, x_2 = 1$ and $y_1 = 1 - y_2$;
- $x_1 = 1, x_2 = 0$ and $y_1 = 1 - y_2$.

Then the partition $X = \{(x, y)\}_{x \in (0, 1), y \in [0, 1]} \cup \{(1, y), (0, 1 - y)\}_{y \in [0, 1]}$.

6.2 Quotient Topology

Definition 6.2. Quotient Topology

Suppose $(X, \mathcal{T})$ is a topological space, and that $\sim$ is an equivalence relation on X . Define the *canonical projection map* by

$$p : X \rightarrow X/\sim \text{ with } x \mapsto [x].$$

Define a family of subsets $\tilde{\mathcal{T}}$ on $X/\sim$ by:

$$\tilde{U} \subseteq X/\sim \text{ is in } \tilde{\mathcal{T}} \text{ if } p^{-1}(\tilde{U}) \in \mathcal{T}.$$

Then $\tilde{\mathcal{T}}$ is a topology for $X/\sim$ and we say $(X/\sim, \tilde{\mathcal{T}})$ the quotient topology of the quotient space.

Verification:

- Since $p^{-1}(X/\sim) = X \in \mathcal{T}$ and $p^{-1}(\emptyset) = \emptyset \in \mathcal{T}$, then $X/\sim, \emptyset \in \tilde{\mathcal{T}}$.
- Suppose that $\tilde{U}, \tilde{V} \in \tilde{\mathcal{T}}$, i.e.: $p^{-1}(\tilde{U}), p^{-1}(\tilde{V}) \in \mathcal{T}$. Then $p^{-1}(\tilde{U} \cap \tilde{V}) = p^{-1}(\tilde{U}) \cap p^{-1}(\tilde{V}) \in \mathcal{T}$. Hence, $\tilde{U} \cap \tilde{V} \in \tilde{\mathcal{T}}$.
- Suppose that $\tilde{U}_i \in \tilde{\mathcal{T}} \forall i \in I$, i.e.: $p^{-1}(\tilde{U}_i) \in \mathcal{T}$ for all $i \in I$. Hence,

$$p^{-1}\left(\bigcup_{i \in I} \tilde{U}_i\right) = \bigcup_{i \in I} p^{-1}(\tilde{U}_i) \in \mathcal{T} \Rightarrow \bigcup_{i \in I} \tilde{U}_i \in \tilde{\mathcal{T}}.$$

Definition 6.3. Induced Map

Let $f : X \rightarrow Y$ be a map such that $f(x) = f(y)$ whenever $x \sim y$. Then there exists a well-defined unique map $g : X/\sim \rightarrow Y$, s.t.:

$$g([x]) := f(x)$$

i.e.: $f = g \circ p$ where p denotes the canonical projection map. We refer to this map g as the induced map, often denoted by $g = \bar{f}$.

Definition 6.4. Open Map

A function $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ is an open map if for each open $U \in \mathcal{T}_X$, $f(U) \in \mathcal{T}_Y$.

Proposition 6.5. $\tilde{V} \subseteq X/\sim$ is closed $\Leftrightarrow p^{-1}(\tilde{V})$ is closed in X .

Proof.

$$p^{-1}((X/\sim) - \tilde{V}) = X - p^{-1}(\tilde{V})$$

□

Proposition 6.6. *Let X, Z be two topological spaces, and let $\sim$ be an equivalence relation on X . Let $g : X/\sim \rightarrow Z$ be a function and let $p : X \rightarrow X/\sim$ be a projection map. Then:*

$$g \text{ is cont} \Leftrightarrow g \circ p : X \rightarrow Z \text{ is cont.}$$

Proof. • $(\Rightarrow)$: $g, p \text{ cont} \Rightarrow g \circ p \text{ cont}$.

- $(\Leftarrow)$: For any open $U \subseteq Z$, $(g \circ p)^{-1}(U) = p^{-1}g^{-1}(U)$ is open in X . Then by the def of quotient topology, $g^{-1}(U)$ is open in $X/\sim$. Hence, g is cont.

□

Definition 6.7. Quotient Map

A map $q : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ is a quotient map if

- q is surjective;
- q is cont;
- for any $U \subseteq Y$ satisfying that $q^{-1}(U)$ is open in X , U is open in Y .

Remark: It's easy to show that the canonical map is a quotient map.

Proposition 6.8. *Let $q : X \rightarrow Y$ be a quotient map, and let $g : Y \rightarrow Z$ be any map. Then g is cont $\Leftrightarrow g \circ q$ is cont.*

Proof. Similar for proposition 4.5.

□

Example: Consider $X = [0, 1]$ and define $x_1 \sim x_2$ if

$$(x_1, x_2) = (0, 1) \text{ or } (1, 0).$$

Then $X = \{0, 1\} \cup (\bigcup_{x \in (0, 1)} \{x\})$. Then intuitively, $X \cong S^1$ are homeomorphic.

Proof. Define $f : [0, 1] \rightarrow S^1$ by $f(t) := (\cos 2\pi t, \sin 2\pi t)$. Then $f(0) = f(1) = (1, 0)$. Hence, f can induce a well-defined fct $g : [0, 1]/\sim \rightarrow S^1$ with $g([t]) := f(t)$, i.e.: $f = g \circ p$. By the continuity of f and proposition 4.7, we have g is cont.

Therefore, now we have: g is cont and bijective, S^1 is Hausdorff, and $[0, 1]$ is cpt. Then by theorem 3.30, $[0, 1]/\sim \cong S^1$.

□

Proposition 6.9. *Let $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ be a surjective cont map, and let $\sim$ be an equivalence relation on X defined by the partition $\{f^{-1}(y) \mid y \in Y\}$ (i.e.: $x \sim x'$ iff $f(x) = f(x')$). If X is compact and Y is Hausdorff, then $X/\sim \cong Y$ are homeomorphic.*

Example:

- Consider $X = [0, 1] \times [0, 1]$ and define $(s_1, t_1) \sim (s_2, t_2)$ if
 - $(s_1, t_1) = (s_2, t_2)$;
 - $\{s_1, s_2\} = \{0, 1\}$, $t_1 = t_2$;
 - $\{t_1, t_2\} = \{0, 1\}$, $s_1 = s_2$;
 - $\{s_1, s_2\} = \{t_1, t_2\} = \{0, 1\}$.

Define $f : X \rightarrow \mathbb{T}^2$ by $f(t_1, t_2) := (e^{2\pi i t_1}, e^{2\pi i t_2})$. Then we have: f is surjective and cont, $\mathbb{T}^2$ is Hausdorff, X is cpt, and

$$\begin{cases} (s_1, t_1) \sim (s_2, t_2) \Rightarrow f(s_1, t_1) = f(s_2, t_2) \\ f(s_1, t_1) = f(s_2, t_2) \Rightarrow (s_1, t_1) \sim (s_2, t_2). \end{cases}$$

Then by proposition 4.8, $X/\sim = [0, 1] \times [0, 1]/\sim \cong \mathbb{T}^2$.

- Consider $\mathbb{D}^2 := \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$, and define $(x_1, y_1) \sim (x_2, y_2)$ by:
 - $(x_1, y_1) = (x_2, y_2)$;
 - $(x_1, y_1), (x_2, y_2) \in \partial\mathbb{D}^2$.

Define $f : \mathbb{D}^2 \rightarrow \mathbb{S}^2$, where $\mathbb{S}^2 := \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$, by:

$$\begin{aligned} f(0, 0) &:= (0, 0, 1) \\ f(x, y) &:= \left(\frac{x}{\sqrt{x^2 + y^2}} \sin(\pi\sqrt{x^2 + y^2}), \frac{y}{\sqrt{x^2 + y^2}} \sin(\pi\sqrt{x^2 + y^2}), \cos(\pi\sqrt{x^2 + y^2}) \right). \end{aligned}$$

Then we have: f is surjective and cont, $\mathbb{S}^2$ is Hausdorff, $\mathbb{D}^2$ is cpt, and

$$\begin{cases} (x_1, y_1) \sim (x_2, y_2) \Rightarrow f(x_1, y_1) = f(x_2, y_2) \\ f(x_1, y_1) = f(x_2, y_2) \Rightarrow (x_1, y_1) \sim (x_2, y_2). \end{cases}$$

Then by proposition 4.8, $\mathbb{D}^2 \cong \mathbb{S}^2$ are homeomorphic.

Proposition 6.10. *Suppose $q : X \rightarrow Y$ is a quotient map, and that $\sim$ is an equivalence relation on X , given by the partition $\{q^{-1}(y) \mid y \in Y\}$. Then $X/\sim \cong Y$ are homeomorphic.*

Proof. Construct $h : X/\sim \rightarrow Y$ by $h([x]) := q(x)$. Then h is bijective and well-defined. By proposition 4.7, since $q = h \circ p$ is cont, then h is cont.

Now we need to show that h^{-1} is cont.

For any open $\tilde{U} \subseteq X/\sim$, $q^{-1}(h(\tilde{U})) = p^{-1}h^{-1}(h(\tilde{U})) = p^{-1}(\tilde{U})$ is open, by the def of quotient topology. Then by the def of quotient map, $h(\tilde{U}) = (h^{-1})^{-1}(\tilde{U})$ is open, i.e.: h^{-1} is cont. $\square$

Example: Consider $X = \mathbb{R}$ and define $\sim$ by $x \sim y \Leftrightarrow x - y \in \mathbb{Z}$. Then consider $q : \mathbb{R} \rightarrow \mathbb{S}^1$ with

$q(x) := (\cos 2\pi x, \sin 2\pi x)$. Then q is a cont surjective open map. Therefore, $\mathbb{R}/\mathbb{Z} := \mathbb{R}/\sim \cong \mathbb{S}^1$.

Definition 6.11. Closed Equivalence Relation

Let X be a topological space and let $\sim$ be an equivalence relation on X . The relation is called *closed* if

$$\sim = \{(x, y) \in X \times X \mid x \sim y\}$$

is closed in $X \times X$.

Proposition 6.12. *Quotient Topology-Properties Proposition*

Let $\sim$ be an equivalence relation on a topological space X .

1. If X is compact, then so is $X/\sim$.
2. If X is (path-)connected, then so is $X/\sim$.
3. If X is Hausdorff and compact and if $\sim$ is closed, then $X/\sim$ is Hausdorff.

Proof. 1,2 immediately holds by the continuity of p . 3.

- **Claim 1:** Given any closed subset C of X , $p(C)$ is closed in $X/\sim$.
- *Proof.* By the def of quotient topology, it suffices to show that $p^{-1}(p(C))$ is closed in X . Let $p_1, p_2 : X \times X \rightarrow X$ be the projection maps onto the first respectively the second factor. Note that

$$p^{-1}(p(C)) = \{y \in X \mid y \sim x \text{ for some } x \in C\} = p_2(p_1^{-1}(C) \cap \sim)$$

Since $p_1^{-1}(C) = C \times X$ is closed in $X \times X$ (C and X are closed in X), then $p_1^{-1}(C) \cap \sim$ is closed in $X \times X$. Since X is compact, then so is $X \times X$ and hence so is $p_1^{-1}(C) \cap \sim$. Therefore, $p^{-1}(p(C)) = p_2(p_1^{-1}(C) \cap \sim)$ is compact (by the continuity of p_2) and hence it is closed in X . $\square$

Now let $[x], [y] \in X/\sim$ be two disjoint points in $X/\sim$. By the Hausdorff-ness of X , $\{x\}$ and $\{y\}$ are closed in X . Hence, by claim 1,

$$\{[x]\} = p(\{x\}) \text{ and } \{[y]\} = p(\{y\})$$

are closed in $X/\sim$. By the continuity of p ,

$$p^{-1}([x]) = p^{-1}(\{[x]\}) \text{ and } p^{-1}([y]) = p^{-1}(\{[y]\})$$

are two disjoint closed subsets of X . Since X is Hausdorff and compact, then X is normal and hence $\exists$ open neighborhoods U, V of $p^{-1}([x]), p^{-1}([y])$ respectively, s.t.: $U \cap V = \emptyset$.

- **Claim 2:** $(X/\sim) \setminus p(V^C)$ and $(X/\sim) \setminus p(U^C)$ are disjoint open neighborhoods of $[x]$ and $[y]$ in $X/\sim$.
- *Proof.* By claim 1, U^C, V^C closed can imply that $p(V^C), p(U^C)$ are closed. As for the "disjoint-

ness",

$$\begin{aligned} ((X/\sim) \setminus p(U^C)) \cap ((X/\sim) \setminus p(V^C)) &= (X/\sim) \setminus (p(U^C) \cup p(V^C)) = (X/\sim) \setminus p(U^C \cap V^C) \\ &= (X/\sim) \setminus p(X \setminus (U \cap V)) = (X/\sim) \setminus p(X) \stackrel{p \text{ surj}}{=} \emptyset \end{aligned}$$

Since $p^{-1}([x]) \cap V^C = \emptyset$, then $[x] \in (X/\sim) \setminus p(V^C)$. Similarly, $[y] \in (X/\sim) \setminus p(U^C)$. Therefore, by claim 2, we're done. $\square$

$\square$

Proposition 6.13. *Twice Quotient Proposition*

1. Let X be a topological space, let $\sim_1$ be an equivalence relation on X and let $\sim_2$ be an equivalence relation on $X/\sim_1$. Given $x, y \in X$, we define

$$x \sim y :\Leftrightarrow [x]_{\sim_1} \sim_2 [y]_{\sim_1}$$

(a) $\sim$ gives an equivalence relation.

(b) The map $X/\sim \rightarrow (X/\sim_1)/\sim_2$ given by $[x] \mapsto [[x]_{\sim_1}]_{\sim_2}$ is a homeomorphism.

2. Let $f : X \rightarrow Y$ be a continuous map between two topological spaces and let $\sim_X$ and $\sim_Y$ be equivalence relations on X and Y respectively. If $x_1 \sim_X x_2$ implies $f(x_1) \sim_Y f(x_2)$, then the map

$$\begin{array}{c} X/\sim_X \rightarrow Y/\sim_Y \\ [x] \mapsto [f(x)] \end{array}$$

is well-defined and continuous.

Definition 6.14. Quotients by subspace

Let X be a topological space and let $A \subseteq X$ be a subset. If $A = \emptyset$, then we define

$$X/\emptyset := X \sqcup \{*\}$$

Otherwise, we define $\sim_A$ by

$$x \sim_A y :\Leftrightarrow x = y \text{ or } x, y \in A$$

which can easily be verified an equivalence relation on X . Then we define

$$X/A := X/\sim_A$$

equipped with the quotient topology.

Remark:

- $X/A \cong (X \setminus A) \cup \{[A]\}$ are homeomorphic.

- Given any subset $Z \subseteq X/A = (X \setminus A) \cup \{[A]\}$, we have

$$p^{-1}(Z) = \begin{cases} Z & , \text{if } [A] \notin Z \\ Z \cup A & , \text{if } [A] \in Z \end{cases}$$

Proposition 6.15. *Cylinder Quotient-Ball Proposition*

- The map

$$([0,1] \times \mathbb{S}^n) / (\{0\} \times \mathbb{S}^n) \rightarrow \overline{\mathbb{B}}^{n+1} \\ [(r,v)] \mapsto rv$$

is a homeomorphism.

- Let X be a topological space and let $f : [0,1] \times \mathbb{S}^n \rightarrow X$ be a continuous map. If $f(0,v) = f(0,w)$ for any $v, w \in \mathbb{S}^n$, then the map

$$\overline{\mathbb{B}}^{n+1} \rightarrow X \\ r \cdot v \mapsto f(r,v)$$

is well-defined and continuous.

6.3 Group Actions on Topological Spaces

Definition 6.16. Let X be a set and let G be a group.

- An (*group*) *action* of G on X is a map

$$G \times X \rightarrow X \\ (g,x) \mapsto g \cdot x$$

with the following axioms: for any $x \in X$, any $g, h \in G$,

$$e \cdot x = x, (g \cdot h) \cdot x = g \cdot (h \cdot x)$$

- The action is called *free* if $g \cdot x = x$ for some $x \in X$ implies that $g = e$.
- If X is a topological space, then we say that the action is *continuous* if for any $g \in G$, the map

$$X \rightarrow X \\ x \mapsto g \cdot x$$

is continuous.

- Let X be a topological space and let G be a group that acts on X . We define an equivalence relation on X by

$$x \sim_G y :\Leftrightarrow g \cdot x = y \text{ for some } g \in G$$

Then we define

$$X/G := X / \sim_G$$

equipped with the quotient topology.

Proposition 6.17. *Torus-as-Quotient Proposition*

1. The map

$$\begin{aligned} \mathbb{Z}^n \times \mathbb{R}^n &\rightarrow \mathbb{R}^n \\ (z, v) &\mapsto z + v \end{aligned}$$

defines a free and continuous action of the group $(\mathbb{Z}^n, +)$ on the topological space $X = \mathbb{R}^n$.

2. The map

$$\varphi : \begin{aligned} \mathbb{R}^n / \mathbb{Z}^n &\rightarrow (\mathbb{S}^1)^n \\ [(t_1, \dots, t_n)] &\mapsto (\exp(2\pi i t_1), \dots, \exp(2\pi i t_n)) \end{aligned}$$

is a homeomorphism.

Definition 6.18. Proper Action

Let X be a topological space and let G be a group that acts on X . We say that G acts *properly* if $\forall$ two points $x, y \in X$, $\exists$ open $U \ni x, V \ni y$, s.t.:

$$|\{g \in G \mid gU \cap V \neq \emptyset\}| < \infty$$

Example:

- The action of any finite group is proper.
- Consider the action of $G = \mathbb{Z}$ on $X = \mathbb{R} \times [0, 1]$ given by

$$\begin{aligned} \mathbb{Z} \times (\mathbb{R} \times [0, 1]) &\rightarrow \mathbb{R} \times [0, 1] \\ (n, (x, y)) &\mapsto (x + n, y) \end{aligned}$$

which is proper. For example, given $(x, s), (y, t) \in \mathbb{R} \times [0, 1]$,

$$U := \{(a, b) \in \mathbb{R} \times [0, 1] : |a - x| < 1\} \text{ and } V := \{(a, b) \in \mathbb{R} \times [0, 1] : |b - y| < 1\}$$

have the desired properties in the def of proper action.

Proposition 6.19. *Group Action-Hausdorff Quotient Proposition*

Let G be a group that acts on a Hausdorff topological space X . If the action is proper and continuous, then X/G is Hausdorff.

Proof.

Lemma 6.20. *Group Action-Canonical Projection Map Lemma*

Let X be a topological space and let G be a group that acts continuously on X .

The canonical projection map p is open.

- If $A \subseteq X$ is a closed subset with $GA = A$, then $p(A)$ is closed in X/G .

Proof. 1. For any open U in X , it suffices to show that $p^{-1}(p(U))$ is open in X . Note that

$$p^{-1}(p(U)) = \bigcup_{g \in G} gU$$

Since G acts contly, then the map $x \mapsto gx$ is a homeomorphism. Hence, gU is open for any $g \in G$. Therefore, $p^{-1}(p(U))$ is open in X .

2. Note that

$$(X/G) \setminus p(A) = p(X \setminus A)$$

Since $X \setminus A$ is open, then by 1, so is $p(X \setminus A)$.

□

Lemma 6.21. *Disjoint Translated Neighborhoods Lemma*

Let G be a group that acts continuously and properly on a topological space. If X is Hausdorff, then $\forall$ two points $a, b \in X$, $\exists$ open $A \ni a, B \ni b$, s.t.: for any $g \in G$ with $ga = b$, we have

$$gA \cap B = \emptyset$$

Proof. Since the action is proper, then there exists open $U \ni a, V \ni b$, s.t.:

$$|\{g \in G \mid gU \cap V \neq \emptyset\}| < \infty$$

Denote the above set as $\{g_1, \dots, g_n\}$ and then set $I := \{i \in \{1, \dots, n\} \mid g_i a = b\}$. By the Hausdorff-ness of X , for any $i \in I$, there exists open $U_i \ni g_i a, V_i \ni b$, s.t.: $U_i \cap V_i = \emptyset$. Now take

$$A := U \cap \left(\bigcap_{i \in I} g_i^{-1} U_i \right) \text{ and } B := V \cap \left(\bigcap_{i \in I} V_i \right)$$

which are the intersection of finitely many open subsets and hence which are open. Therefore, A is an open neighborhood of a and B is an open neighborhood of b .

Let $g \in G$ with $ga \neq b$. If $g \notin \{g_1, \dots, g_n\}$, then $gU \cap V = \emptyset$ and hence $gA \cap B = \emptyset$. Otherwise, $g = g_i$ for some $i \in I$. Then

$$g_i(g_i^{-1}U) \cap V_i = U_i \cap V_i = \emptyset \Rightarrow gA \cap B = g_i A \cap B = \emptyset$$

□

Lemma 6.22. *Disjoint Quotient Lemma*

Let G be a group acts on a set X . Let A and B be subsets of X . Then

$$p(A) \cap p(B) = \emptyset \Leftrightarrow \forall g \in G, \text{ we have } gA \cap B = \emptyset$$

or equivalently

$$p(A) \cap p(B) \neq \emptyset \Leftrightarrow \exists g \in G, \text{ s.t. : } gA \cap B \neq \emptyset$$

Proof. We prove the second statement:

$$\begin{aligned} p(A) \cap p(B) \neq \emptyset &\Leftrightarrow p(a) = p(b) \text{ for some } a \in A, b \in B \\ &\Leftrightarrow ga = b \text{ for some } a \in A, b \in B, g \in G \\ &\Leftrightarrow gA \cap B \neq \emptyset \text{ for some } g \in G \end{aligned}$$

□

Now we back to the proof of the Group Action-Hausdorff Quotient Proposition. For any $x \neq y \in X/G$, there exist $a, b \in X$, s.t.:

$$p(a) = x \neq p(b) = y$$

Hence, by the def of $\sim_G$, for any $g \in G$, $ga \neq b$. By the Disjoint Translated Neighborhoods Lemma, there exist open $A \ni a, B \ni b$, s.t.: $\forall g \in G$

$$gA \cap B = \emptyset$$

Then by the Disjoint Quotient Lemma,

$$p(A) \cap p(B) = \emptyset \quad (*)$$

By the open-ness of p (Group Action-Canonical Projection Map Lemma.1) and A, B , then $p(A)$ and $p(B)$ are open. Therefore, $p(A)$ and $p(B)$ are respectively the disjoint open neighborhoods of $x = p(a)$ and $y = p(b)$. □

Remark for the Group Action-Canonical Projection Map Lemma: Even if G acts contly on X , it is NOT true in general that the projection $p : X \rightarrow X/G$ is closed. For example, let $X = \mathbb{R}$ and $G = \mathbb{Z}$ with the additive action on $\mathbb{R}$. The set $A := \{n + \frac{1}{n} \mid n \in \mathbb{Z}\}$ is closed in $X = \mathbb{R}$ but $p(A)$ is **NOT** closed in $X/G = \mathbb{R}/\mathbb{Z}$.

Proposition 6.23. *Mobius Band-as-Quotient Proposition*

1. The map

$$\begin{aligned} \mathbb{Z} \times (\mathbb{R} \times [-1, 1]) &\rightarrow \mathbb{R} \times [-1, 1] \\ (n, (x, y)) &\mapsto (x+n, (-1)^n y) \end{aligned}$$

defines an action of the group $(\mathbb{Z}, +)$ on the topological space $X = \mathbb{R} \times [-1, 1]$ that is free, continuous and proper.

2. Consider $([0, 1] \times [0, 1]) / \sim$ with $(0, y) \sim (1, 1 - y)$, giving a mobius band. The map

$$\begin{aligned} ([0, 1] \times [0, 1]) / \sim &\rightarrow (\mathbb{R} \times [-1, 1]) / \mathbb{Z} \\ [(x, y)] &\mapsto [(x, 2y - 1)] \end{aligned}$$

is a homeomorphism.

3. The mobius band $M = ([0, 1] \times [0, 1]) / \sim$ is compact and Hausdorff.

6.4 Projective Spaces

Definition 6.24. Projective Spaces

- The map

$$\begin{aligned} (\mathbb{R} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\}) &\rightarrow \mathbb{R}^{n+1} \setminus \{0\} \\ (z, P) &\mapsto z \cdot P \end{aligned}$$

defines a continuous action of $(\mathbb{R} \setminus \{0\}, \cdot)$ on $\mathbb{R}^{n+1} \setminus \{0\}$. The quotient space $\mathbb{R}^{n+1} \setminus \{0\} / \mathbb{R} \setminus \{0\}$ is called the *n-dimensional real projective space* $\mathbb{R}P^n$. Particularly, we often call the *real projective plane* the 2-dimensional real projective space $\mathbb{R}P^2$.

- We define the *n-dimensional complex projective space* $\mathbb{C}P^n := (\mathbb{C}^{n+1} \setminus \{0\}) / (\mathbb{C} \setminus \{0\})$ by replacing " $\mathbb{R}$ " by " $\mathbb{C}$ ".

Remark (Notation): For $(x_0, \dots, x_n) \in \mathbb{R} \setminus \{0\}$, we denote by $[x_0 : \dots : x_n]$ the corresponding point in $\mathbb{R}P^n$. Similarly for points in $\mathbb{C}P^n$.

Proposition 6.25. Projective Space-Sphere Quotient Proposition

Let $n \in \mathbb{N} \cup \{0\}$.

1. The map

$$\begin{aligned} \{\pm 1\} \times \mathbb{S}^n &\rightarrow \mathbb{S}^n \\ (w, (x_1, \dots, x_{n+1})) &\mapsto (w \cdot x_1, \dots, w \cdot x_{n+1}) \end{aligned}$$

defines a continuous action of $\{\pm 1\}$ on $\mathbb{S}^n$. The maps

$$\begin{aligned} \mathbb{S}^n / \{\pm 1\} &\rightarrow \mathbb{R}P^n & \text{and} & & \mathbb{R}P^n &\rightarrow \mathbb{S}^n / \{\pm 1\} \\ [x] &\mapsto [x] & & & [x] &\mapsto [\frac{x}{\|x\|}] \end{aligned}$$

are inverses of one another and they are homeomorphisms.

2. The map

$$\begin{aligned} \mathbb{S}^1 \times \mathbb{S}^{2n+1} &\rightarrow \mathbb{S}^{2n+1} \\ (w, (z_1 \in \mathbb{C}, \dots, z_{n+1} \in \mathbb{C})) &\mapsto (w \cdot z_1, \dots, w \cdot z_{n+1}) \end{aligned}$$

defines a continuous action of $(\mathbb{S}^1, \cdot)$ on $\mathbb{S}^{2n+1}$. The maps

$$\begin{aligned} \mathbb{S}^{2n+1} / \mathbb{S}^1 &\rightarrow \mathbb{C}P^n & \text{and} & & \mathbb{C}P^n &\rightarrow \mathbb{S}^{2n+1} / \mathbb{S}^1 \\ [z] &\mapsto [z] & & & [z_1 : \dots : z_{n+1}] &\mapsto [\frac{(z_1, \dots, z_{n+1})}{\sqrt{|z_1|^2 + \dots + |z_{n+1}|^2}}] \end{aligned}$$

are inverses of one another and they are homeomorphisms.

Proposition 6.26. *Projective Spaces Properties Proposition*

$\mathbb{R}P^n$ and $\mathbb{C}P^n$ are compact, path-connected and Hausdorff.

Proof. By the previous proposition, it suffices to show that $\mathbb{S}^n/\{\pm 1\}$ and $\mathbb{S}^{2n+1}/\mathbb{S}^1$ are compact, path-connected and Hausdorff. It follows immediately from the fact that $\mathbb{S}^n$ and $\mathbb{S}^{2n+1}$ are compact and path-connected together with the Quotient Topology-Properties 1,2 that both quotients are compact and path-connected.

As for the Hausdorff-ness, note that the action of $\{\pm 1\}$ on $\mathbb{S}^n$ is proper and then by the Group Action-Hausdorff Quotient Prop, $\mathbb{S}^n/\{\pm 1\}$ is Hausdorff.

Now it remains to show that $\mathbb{S}^{2n+1}/\mathbb{S}^1$ is Hausdorff. It suffices to show that $\sim_{\mathbb{S}^1}$ is closed by the Quotient Topology-Properties Prop.3, i.e.:

$$G := \{(v_1, v_2) \in \mathbb{S}^{2n+1} \times \mathbb{S}^{2n+1} \mid w \cdot v_1 = v_2 \text{ for some } w \in \mathbb{S}^1\}$$

is closed in $\mathbb{S}^{2n+1} \times \mathbb{S}^{2n+1}$. Consider the map $\Phi : \mathbb{S}^1 \times \mathbb{S}^{2n+1} \rightarrow \mathbb{S}^{2n+1} \times \mathbb{S}^{2n+1}$ given by

$$\Phi(z, v) := (v, z \cdot v)$$

Note that $G = \Phi(\mathbb{S}^1 \times \mathbb{S}^{2n+1})$ and that Φ is continuous and $\mathbb{S}^1 \times \mathbb{S}^{2n+1}$ is compact. Therefore, G is compact and hence it is closed. $\square$

Proposition 6.27. *Projective Space-Avatar Proposition*

1. The maps

$$z \mapsto \begin{cases} [z : 1] & , \text{if } z \in \mathbb{C} \\ [1 : 0] & , \text{if } z = \infty \end{cases} \quad \text{and} \quad [z_0 : z_1] \mapsto \begin{cases} z_0 \cdot z_1^{-1} & , \text{if } z_1 \neq 0 \\ 0 & , \text{otherwise} \end{cases}$$

are inverses of one another and both are homeomorphisms. Similarly we can see that

$$\mathbb{R}P^1 \cong \mathbb{R} \cup \{\infty\} \cong \mathbb{S}^1$$

are homeomorphic.

2. Let $n \in \mathbb{N} \cup \{0\}$. If we denote by $\sim$ the equivalence relation on $\overline{B}^n$ that is generated by $P \sim -P$ for $P \in \mathbb{S}^{n-1} = \partial \overline{B}^n$, then the maps

$$\begin{aligned} \overline{B}^n / \sim &\rightarrow \mathbb{R}P^n & \text{and} & & \mathbb{R}P^n &\rightarrow \overline{B}^n / \sim \\ [x] &\mapsto [(x, \sqrt{1-||x||^2})] & & & [(v, x_{n+1})] &\mapsto \begin{cases} [v] & , \text{if } x_{n+1} \geq 0 \\ [-v] & , \text{if } x_{n+1} < 0 \end{cases} \end{aligned}$$

are homeomorphisms that are inverses of one another.

3. As in 2, let $\sim$ be the equivalence relation on $\overline{B}^3$ that is generated by $x \sim -x$ for $x \in \mathbb{S}^2$. The

map

$$F := \begin{array}{c} \bar{B}^3 / \sim \rightarrow SO(3) \\ [r \cdot v] \mapsto A_{r,v} \end{array}$$

where $A_{r,v}$ denotes the matrix in $SO(3)$ which represents rotation by the angle $\pi \cdot r$ around the axis $\mathbb{R}_{\geq 0} \cdot v$ where the rotation direction is given by the right-hand rule, is a homeomorphism. In particular, $SO(3) \cong \mathbb{RP}^3$ are homeomorphic.

Remark: Now we get the identifications:

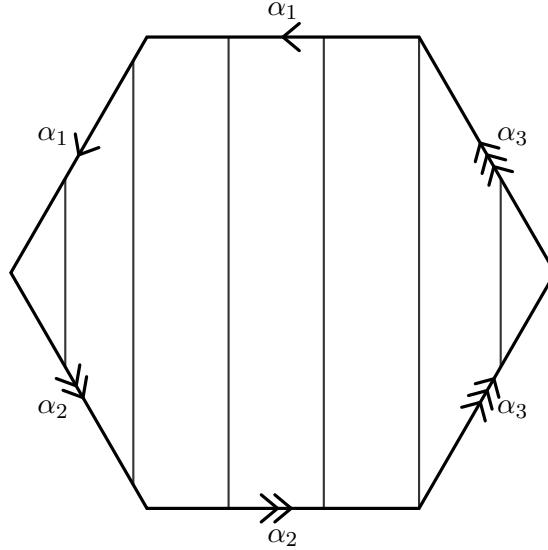
$$\mathbb{S}^1 =_i \mathbb{RP}^1, \mathbb{S}^2 =_i \mathbb{CP}^1, \mathbb{RP}^n =_i \bar{B}^n / \sim$$

with $\sim$ defined in the second statement of the above proposition (Projective Space-Avatar Proposition).

6.5 Surfaces

Definition 6.28. Given $g \in \mathbb{N}_+$, take a regular $4g$ -gon E_{4g} in $\mathbb{R}^2 =_i \mathbb{C}$ and for $j = 0, \dots, g-1$, we use reflections to identify the $(4j+1)$ -st edge with the $(4j+3)$ -rd edge and the $(4j+2)$ -nd edge with the $(4j+4)$ -th edge. We refer to the quotient space $E_{4g} / \sim$ as the *surface Σ_g of genus g* . Furthermore, we refer to $\mathbb{S}^2$ as the *surface of genus 0*.

Definition 6.29. Given $g \in \mathbb{N}_{\geq 2}$, consider the regular $2g$ -gon E_{2g} with the identification of the boundary segments shown for example in the figure below. We refer to this topological space as the *non-orientable surface N_g of genus g* . Furthermore, we refer to $\mathbb{RP}^2$ as the *non-orientable surface of genus 1*.



6.6 Simplicial Complex

Definition 6.30. n-simplex

$$\Delta^n := \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid x_i \geq 0, \sum_{i=1}^{n+1} x_i = 1\}$$

- Vertices of $\Delta^n := \{(x_1, \dots, x_{n+1}) \in \Delta^n \mid x_i = 1 \text{ for some } i\}$.
- For any given $\mathcal{A} \subseteq \{1, \dots, n+1\}$, the face of Δ^n is defined by $\{(x_1, \dots, x_{n+1}) \in \Delta^n \mid x_i = 0, \forall i \in \mathcal{A}\}$.
- Interior of $\Delta^n := \{(x_1, \dots, x_{n+1}) \in \Delta^n \mid x_i \neq 0, \forall i\}$.

Definition 6.31. Face Inclusion

Let $m < n$. A face inclusion of Δ^m into Δ^n is an injective map $f : \Delta^m \rightarrow \Delta^n$ which maps vertices to vertices and is the restriction of an injective linear map $T : \mathbb{R}^{m+1} \rightarrow \mathbb{R}^{n+1}$.

Remark:

{all face inclusion} $\xleftrightarrow{1:1}$ {injective map $g : \{1, \dots, m+1\} \rightarrow \{1, \dots, n+1\}$ }.

Example: Linear map $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ with $T(1, 0) = (0, 1, 0)$ and $T(0, 1) = (0, 0, 1)$ is a face inclusion.

Definition 6.32. Simplicial Complex

A simplicial complex is a pair $K = (V, \Sigma)$, where

- V is a (finite most of times) set of vertices;
- Σ is a collection of non-empty finite subsets of V , called simplices, satisfying:
 - For any $v \in V$, $\{v\} \in \Sigma$;
 - If $\sigma \in \Sigma$, then $(P(\sigma) - \{\emptyset\}) \subseteq \Sigma$.

Example: If $V = \{1, 2, 3, 4\}$, then one can take:

$$\Sigma = \{\{1\}, \dots, \{4\}, \{1, 3, 4\}, \{1, 3\}, \{3, 4\}, \{1, 4\}\}.$$

Definition 6.33. Topological Realization

Let $K = (V, \Sigma)$ be a simplicial complex. The topological realization of K is a topological space, denoted by $|K|$ or $|(V, \Sigma)|$, given by identifying Δ^n simplices, i.e.:

- For any $\sigma \in \Sigma$ with $|\sigma| = n+1$, take a copy of n-simplex and denote it as Δ_σ ;
- Whenever $\sigma \subseteq \tau \in \Sigma$, identify Δ_σ with a face of Δ_τ through face inclusion.

Or equivalently, $|K|$ is a quotient space of the disjoint union $\bigsqcup_{\sigma \in \Sigma} \Delta_\sigma$, by the equivalence relation which identifies a point $y \in \sigma$ with its image under the face inclusion $\sigma \rightarrow \tau$, for any $\sigma \subseteq \tau$.

Definition 6.34. Triangulation

A triangulation of a topological space X is a simplicial complex $K = (V, \Sigma)$ together with a choice homeomorphism $f : |K| \xrightarrow{\cong} X$.

Definition 6.35. Euler Characteristic

For any simplicial complex (V, Σ) , the Euler characteristic is defined by

$$\chi(V, \Sigma) := \sum_{i=1}^{\infty} (-1)^i \cdot \#\{\text{subsets in } \Sigma \text{ with } (i+1) - \text{element}\}.$$

Proposition 6.36.

$$\chi(V_1, \Sigma_1) = \chi(V_2, \Sigma_2) \Leftrightarrow |(V_1, \Sigma_1)| = |(V_2, \Sigma_2)|$$

Definition 6.37. For any topological space X with $X \cong |(V, \Sigma)|$, by proposition 4.16, we can define

$$\chi(X) := \chi(V, \Sigma).$$

6.7 Properties of Simplicial Complex

Definition 6.38. Simplicial Subcomplex

A subcomplex of a simplicial complex $K = (V, \Sigma)$ is a simplicial complex $K' = (V', \Sigma')$ such that

$$V' \subseteq V, \Sigma' \subseteq \Sigma.$$

Proposition 6.39. Let $K' = (V', \Sigma')$ be a subcomplex of $K = (V, \Sigma)$ and let $i : (X' :=) \bigsqcup_{\sigma' \in \Sigma'} \sigma' \rightarrow (X :=) \bigsqcup_{\sigma \in \Sigma} \sigma$ be an inclusion map. Then we can get a map

$$\tilde{i} := (p')^{-1} \circ i \circ p \text{ with } \tilde{i}([x']) = [i(x')].$$

Then the map $\tilde{i}$ is injective, well-defined, and continuous.

Moreover, $\tilde{i}(|K'|)$ is closed in $|K|$.

Proof. Note that for $x', y' \in X'$,

$$x' \sim y' \Leftrightarrow i(x') \sim i(y').$$

Hence, $\tilde{i}$ is injective and well-defined. Also, since $p \circ i : X' \rightarrow X/\sim$ is cont, then $\tilde{i}$ is cont.

To see that $\tilde{i}(|K'|)$ is closed in $|K|$, by the def of $\tilde{i}$, we only need to show that $p \circ i(X')$ is closed in $|K|$:

For any $y \in p^{-1}((p \circ i)(X'))$, $y \in \Delta_\tau$ for some $\tau \in \Sigma$. Then $y \in \Delta_\tau$ must lie on a face of Δ_τ which is "glued" to $\Delta_{\tau \cap \sigma'}$ by face inclusion for some $\sigma' \in \Sigma'$.

Let $\Delta_{\tau \cap \Sigma'}$ be the union of faces of Δ_τ given by face inclusion $\Delta_{\tau \cap \sigma'} \xrightarrow{i} \Delta_\tau$ for some $\sigma' \in \Sigma'$. Then $\Delta_{\tau \cap \Sigma'}$ is closed in Δ_τ (a face on Δ_τ is the zero set of some linear equations $f_1, \dots, f_n$, so it is equal to $\bigcap_{i=1}^n f_i^{-1}(0)$,) and $p^{-1}(p \circ i(X')) = \bigsqcup_{\tau \in \Sigma} \Delta_{\tau \cap \Sigma'}$ is closed. Then by the def of p , canonical projection map, we're done $\square$

Definition 6.40. Let $K = (V, \Sigma)$ be a simplicial complex and let $V' \subseteq V$. Then the subcomplex spanned by V' is (V', Σ') such that

- V' denotes the vertex set;
- the simplices Σ' is given by

$$\{\sigma \in \Sigma \mid \sigma \subseteq V'\}.$$

Definition 6.41. Let $K = (V, \Sigma)$ be a simplicial complex.

- **(Link):** The link of $v \in V$, denoted as $lk(v)$, is the subcomplex with vertex set

$$\{w \in V - \{v\} \mid \{v, w\} \in \Sigma\}$$

and simplices

$$\{\sigma \in \Sigma \mid v \notin \sigma, \sigma \cup \{v\} \in \Sigma\}.$$

- **(Star):** The star of v , denoted as $st(v)$, is

$$\bigcup_{\sigma \in \Sigma, v \in \sigma} \text{inside}(\sigma),$$

where $\text{inside}(\Delta^n) := \{(x_1, \dots, x_{n+1}) \in \Delta^n \mid x_i > 0, \forall i\}$ for n -simplex Δ^n .

Proposition 6.42. $st(v)$ is open and $v \in st(v)$.

Proposition 6.43. Let $K = (V, \Sigma)$ with $|V| < \infty$. Then K is compact.

Proof. Let $p : D \rightarrow K$ be a canonical map, which is cont. Since D , the finite disjoint union of Δ_σ 's, is compact, then $p(D) = |K|$ is compact. $\square$

Proposition 6.44. For any $K = (V, \Sigma)$ with $|V| < \infty$, there is a continuous injection: for some n ,

$$f : |K| \rightarrow \mathbb{R}^n.$$

Proof. Let $K^\Pi := (V, \Sigma^\Pi)$ where $\Sigma^\Pi = P(V)$. Then

$$|K^\Pi| = \Delta^{|V|-1} \subseteq \mathbb{R}^{|V|}.$$

Consider $D := \bigsqcup_{\sigma \in \Sigma} \Delta_\sigma$ and $D' := \bigsqcup_{\Delta_{\Sigma^\Pi} \in \Sigma^\Pi} \Delta_{\Sigma^\Pi}$ in (V, Σ) and (V, Σ^Π) . Then construct the map $\tilde{i} : D \xrightarrow{i} D' \xrightarrow{p'} |K|$, which can descend to an inclusion $i : D/\sim \rightarrow |K^\Pi|$. Therefore, $|K| \xrightarrow{i} |K^\Pi|$. $\square$

Corollary 6.45. If $K = (V, \Sigma)$ with $|V| < \infty$, then $|K|$ is Hausdorff.

Proof. Let $g : |K| \rightarrow \mathbb{R}^n$. Consider the bijective $g : |K| \rightarrow g(|K|)$, which is cont. Since $|K|$ is compact, and $g(|K|)$ is Hausdorff, then $|K| \cong g(|K|)$. Therefore, $|K|$ is Hausdorff. $\square$

Definition 6.46. Edge Path

An edge path of $K = (V, \Sigma)$ is a sequence of vertices $(v_1, \dots, v_n)$, $v_i \in V$, s.t.: $\{v_i, v_{i+1}\} \in \Sigma$ for all i .

Proposition 6.47. *Let $K = (V, \Sigma)$ be a simplicial complex. The followings are equivalent:*

1. $|K|$ is connected;
2. $|K|$ is path-connected;
3. Any two vertices V can be joined by an edge path, i.e.: $\forall u, v \in V, \exists v_1, \dots, v_k \in V$, s.t.: $(u, v_1, \dots, v_k, v)$ is an edge path.

Proof. • $(2 \Rightarrow 1)$: Clearly.

- $(3 \Rightarrow 2)$: For any $y, x \in |K| = |(V, \Sigma)|$, there exist $\sigma, \tau \in \Sigma$, s.t.:

$$x \in \Delta_\sigma, y \in \Delta_\tau.$$

Since $\Delta_\sigma, \Delta_\tau$ path-connected can join x to a vertex v_x in Δ_σ , and y to a vertex v_y in Δ_τ . By 3, there exists a edge path joining v_x to v_y . Hence, there exists a path joining x to y .

- $(1 \Rightarrow 3)$: Suppose on the contrary that fix $v \in V$, and there exists $w \in V$, s.t.: there are NO edge path joining v, w . Define:

$V' := \{v' \in V \mid v, v' \text{ can be joined by an edge path} \}$ and

$V'' := \{v'' \in V \mid v, v'' \text{ can NOT be joined by any edge path} \}$.

Then we have $V', V'' \neq \emptyset$ (by assumption) is disjoint (by the def of V', V'') and $V = V' \cup V''$.

Consider $K' :=$ subcomplex of K spanned by V' and

$K'' :=$ subcomplex of K spanned by V'' .

Then it's clear that $|K'|$ and $|K''|$ are disjoint in $|K|$.

– **Claim:** $|K| = |K'| \cup |K''|$.

Proof of claim: Suppose on the contrary that there exists $x \in |K| - (|K'| \cup |K''|)$. Then there exists $\pi \in \Sigma$, s.t.: $x \in \Delta_\pi$. Then π must contain vertices in both V' and V'' , otherwise Δ_π is in $|K'|$ or $|K''|$. Hence, there exist $w' \in V', w'' \in V''$, s.t.:

$$w', w'' \in \Delta_\sigma.$$

But then $\{w', w''\} \in \Sigma$, which means that there exists an edge path joining $w' \in V'$ and $w'' \in V''$, contradicting to the def of V' and V'' . $\square$

Since $|K'|$ and $|K''|$ are closed in $|K|$, then by claim, $|K''| = |K| - |K'|$ and $|K'| = |K| - |K''|$ are open in $|K|$. Therefore, $|K|$ is disconnected, contradicting 1.

$\square$

Chapter 7

Second-Countability and the Urysohn Metrization Theorem

7.1 Second-Countable Topological Space

Lemma 7.1. *Second Countability Lemma*

1. $\mathbb{R}^n$ is second-countable.
2. If X is a topological space that is second-countable, then so is every subspace of X .
3. (a) If $X_1, \dots, X_k$ are topological spaces that are second-countable, then $X_1 \times \dots \times X_k$ is second-countable.
(b) Let $\{X_i\}_{i \in I}$ be a countable family of topological spaces. If each X_i is second-countable, then so is $\prod_{i \in I} X_i$.
4. Let X be a topological space. If there exists a countable family of open subsets $\{U_i\}_{i \in I}$ such that $\bigcup_{i \in I} U_i = X$ and such that each U_i is second-countable, then X itself is second-countable.
5. Let X be a topological space and let $\sim$ be an equivalence relation on X . If X is second-countable and if the canonical projection map $p : X \rightarrow X/\sim$ is an open map, then $X/\sim$ is also second-countable.

Proof.

Lemma 7.2. *Quotient Topology-Basis Lemma*

Let X be a topological space and let $\sim$ be an equivalence relation on X . Furthermore, let $\mathcal{B}$ be a basis for the topology of X . If the canonical projection map $p : X \rightarrow X/\sim$ is an open map, then

$$p(\mathcal{B}) = \{p(B) \mid B \in \mathcal{B}\}$$

is a basis for the topology of $X/\sim$.

Proof. of the Quotient Topology-Basis Lemma:

- Since p is open, then every sets in $p(\mathcal{B})$ is open in $X/\sim$.
- Let $U \subseteq X/\sim$ be an open subset and let $y \in U$. Since p is surjective, then $\exists x \in X$, s.t.: $p(x) = y$. Since $\mathcal{B}$ is a basis of X , then $\exists B \in \mathcal{B}$, s.t.:

$$x \in B \subseteq p^{-1}(U)$$

Therefore, $y = p(x) \in p(B) \subseteq U$.

□

Now we return to the *proof of the Second Countability Lemma*:

1. Omitted.
2. Omitted.
3. (a) Omitted.
- (b) For each $i \in I$, let $\mathcal{B}_i$ be a countable basis of X_i . Then

$$\mathcal{C} := \bigcup_{i_1, \dots, i_r \in I} \{\Pi_{i \in I} U_i \mid \forall i \notin \{i_1, \dots, i_r\}, U_i = X_i; \forall j = 1, \dots, r, U_{i_j} \in \mathcal{B}_{i_j}\}$$

is a basis for $\Pi_{i \in I} X_i$. Since I is countable, then there are countably many subsets of I . Note that given any $\{i_1, \dots, i_r\}$, the $\Pi_{i \in I} U_i$'s with the properties in $\mathcal{C}$ are countably many. Therefore, $\mathcal{C}$ is countable.

4. For each $i \in I$, let $\mathcal{B}_i$ be a countable basis of U_i . Then $\bigcup_{i \in I} \mathcal{B}_i$ is a basis of X and hence we're done.
5. holds immediately the Quotient Topology-Basis Lemma.

□

Proposition 7.3. *Countable Covering Proposition*

Let X be a topological space and let $\{U_i\}_{i \in I}$ be an open cover of X . If X is second-countable, then there exists a countable subset $J \subseteq I$, s.t.:

$$X = \bigcup_{j \in J} U_j$$

Proof. Let $\mathcal{B}$ be a countable basis of X . Define

$$\mathcal{B}' := \{V \in \mathcal{B} \mid V \subseteq U_i, \text{ for some } i \in I\}$$

and then define J by

$$J := \{j \in I \mid V \subseteq U_j, \text{ for some } V \in \mathcal{B}'\}$$

Since $\mathcal{B}'$ is countable, then so is J . For any $x \in X$, by the hypothesis, there is an $i \in I$, s.t.: $x \in U_i$. Since $\mathcal{B}$ is a basis and since U_i is open, then there exists $V \in \mathcal{B}$, s.t.:

$$x \in V \subseteq U_i$$

Hence, $V \in \mathcal{B}'$. Then by the def of J , $i \in J$. □

Definition 7.4. Exhaustion

Let X be a topological space. We say that a sequence $\{P_i\}_{i \in \mathbb{N}_+}$ of subsets of X is an *exhaustion* if the followings are satisfied:

- $X = \bigcup_{i \in \mathbb{N}_+} P_i$
- for any $i \in \mathbb{N}_+$, we have $\overline{P_i} \subseteq P_{i+1}^\circ$.

Furthermore, we say that the exhaustion is *compact* if each P_i is compact.

Proposition 7.5. *Compact Exhaustion Proposition*

Let X be a topological space which is second-countable, Hausdorff and regionally compact.

1. There is an exhaustion of X by compact subsets.
2. Given any compact subset $K \subseteq X$, there exists an open neighborhood P of K , s.t.: $\overline{P}$ is compact.

Proof. 1. Recall that we say $A \subseteq X$ is *precompact* in X if $\overline{A}$ is compact in X . Since X is Hausdorff and regionally compact, then by the previous proposition, X is locally precompact, which implies that X admits a basis of precompact open subsets. Since X is second-countable, then by the Countable Covering Proposition, X is covered by a countable family $\{U_i\}_{i \in \mathbb{N}_+}$ of precompact open subsets. It suffices to prove the following claim:

- **Claim:** There exists a sequence $\{K_i\}_{i \in \mathbb{N}_+}$ of compact subsets of X , s.t.: for any $i \in \mathbb{N}_+$, $U_i \subseteq K_i$ and s.t.: for any $i \geq 2$, $K_{i-1} = \overline{K_{i-1}} \subseteq K_i^\circ$.
- *Proof. of claim:* First set $K_1 := \overline{U_1}$. Now suppose that we've already constructed $K_1, \dots, K_n$ with the properties stated in the claim. Since K_n is compact, then there exists $r \in \mathbb{N}_+$, s.t.:

$$K_n \subseteq U_1 \cup \dots \cup U_r$$

Set $K_{n+1} := \overline{U_1} \cup \dots \cup \overline{U_r} \cup \overline{U_{r+1}}$. Since U_i 's are precompact, then $\overline{U_i}$'s are compact. Hence, K_{n+1} is also compact. Now we check whether K_{n+1} satisfies the properties:

- It's clear that $U_{n+1} \subseteq \overline{U_{n+1}} \subseteq K_{n+1}$.
- $\overline{K_n} = K_n \subseteq U_1 \cup \dots \cup U_r \subseteq K_{n+1}^\circ \subseteq K_{n+1}$.

□

Let K be a compact subset of X and continue with the notations from 1. The sets K_n° form an open cover of K . By the compact-ness of K and the property that $K_n^\circ \subseteq K_{n+1}^\circ$, $K \subseteq K_n^\circ$

for some $n \in \mathbb{N}_+$. Then set $P := K_n^\circ$ and then we're done. □

7.2 Urysohn Metrization Theorem

Theorem 7.6. *Urysohn Metrization Theorem*

Let X be a topological space. If X is second-countable, then the followings are equivalent:

1. X is normal and every one-point subset is closed.
2. X is metrizable.

Proof.

Definition 7.7.

$$\mathbb{R}^{\mathbb{N}} := \{\text{all maps } \mathbb{N}_+ \rightarrow \mathbb{R}\} \cong \{(x_1, x_2, \dots) \mid x_i \in \mathbb{R}\}$$

We equipped $\mathbb{R}^{\mathbb{N}}$ with the product topology, i.e.: the topology induced by the basis given by the sets

$$U_1 \times \dots \times U_k \times \mathbb{R} \times \mathbb{R} \times \dots \subseteq \mathbb{R}^{\mathbb{N}}$$

where $U_i \subseteq \mathbb{R}$ is open in $\mathbb{R}$.

Lemma 7.8. *The topological space $\mathbb{R}^{\mathbb{N}}$ is metrizable.*

Proof. of lemma: Throughout the proof we need adopt the following notations:

- Given $a, b \in \mathbb{R}$, we write $d(a, b) := \min\{|b - a|, 1\}$.
- Given $x \in \mathbb{R}^{\mathbb{N}}$, we denote by $x_1, x_2, \dots$ the "coordinates" of x .
- Given $x, y \in \mathbb{R}^{\mathbb{N}}$, we set

$$D(x, y) := \sup_{k \in \mathbb{N}_+} \left\{ \frac{1}{k} \cdot d(x_k, y_k) \right\} \in \mathbb{R}$$

- **Claim 1:** The map D defines a metric on $\mathbb{R}^{\mathbb{N}}$.
- *Proof. of claim 1:* It's clear that D is symmetric and reflective. As for the triangle inequality, let $x, y, z \in \mathbb{R}^{\mathbb{N}}$.

$$D(x, z) := \sup_{k \in \mathbb{N}_+} \left\{ \frac{1}{k} \cdot d(x_k, z_k) \right\} \stackrel{d \text{ metric}}{\leq} \sup_{k \in \mathbb{N}_+} \left\{ \frac{1}{k} \cdot d(x_k, y_k) + \frac{1}{k} \cdot d(y_k, z_k) \right\} \leq D(x, y) + D(y, z)$$

□

It remains to prove the following claim:

- **Claim 2:** $\mathcal{T}_D = \mathcal{T}_{\text{product}}$ on $\mathbb{R}^{\mathbb{N}}$.
- *Proof. of claim 2:*

- $\mathcal{T}_D \subseteq \mathcal{T}_{product}$: For any $U \in \mathcal{T}_D$, any $x \in U$, there exists $\epsilon > 0$, s.t.: $x \in B_\epsilon^D(x) \subseteq U$. Pick $n \in \mathbb{N}_+$ with $\frac{1}{n} < \epsilon$ and then set

$$V := (x_1 - \epsilon, x_1 + \epsilon) \times \cdots \times (x_n - \epsilon, x_n + \epsilon) \times \mathbb{R} \times \mathbb{R} \times \cdots \in \mathcal{T}_{product}$$

It's clear that $V \subseteq B_\epsilon^D(x)$. Hence, $x \in V \subseteq U$. Therefore, $U \in \mathcal{T}_{product}$.

- $\mathcal{T}_{product} \subseteq \mathcal{T}_D$: It suffices to show that $\mathcal{B} \subseteq \mathcal{T}_D$. For any

$$U := U_1 \times \cdots \times U_k \times \mathbb{R} \times \mathbb{R} \times \cdots \in \mathcal{B}$$

with $U_1, \dots, U_k$ open in $\mathbb{R}$, any $x \in U$, any $i \in \{1, \dots, k\}$, there exists $\mu_i \in (0, 1)$, s.t.: $(x_i - \mu_i, x_i + \mu_i) \subseteq U_i$. Set

$$\epsilon := \min\left\{\frac{\mu_1}{1}, \frac{\mu_2}{2}, \dots, \frac{\mu_k}{k}\right\}$$

Then $x \in B_\epsilon^D(x) \subseteq U \Rightarrow U \in \mathcal{T}_D$.

□

□

Now we return to the *proof of the Urysohn Metrization Theorem*:

- (2 $\Rightarrow$ 1): Since X is metrizable, then it is normal and Hausdorff where the Hausdorff-ness implies that every one-point subset is closed.
- (1 $\Rightarrow$ 2):
 - **Claim 3:** There exists a family $\{\varphi_n : X \rightarrow [0, 1]\}_{n \in \mathbb{N}_+}$ of continuous maps with the property that for any $x \in X$, any open $U \ni x$ in X , there exists $n \in \mathbb{N}_+$, s.t.:

$$\varphi_n(x) = 1, \varphi_n|_{X \setminus U} \equiv 0$$

- *Proof. of claim 3:* Let $\{B_j\}_{j \in J}$ be a countable basis of X . For any $i, j \in J$ with $\overline{B_i} \subseteq B_j$, applying Urysohn's Lemma to the disjoint closed subsets $\overline{B_i}$ and $X \setminus B_j$, we obtain a map $g_{i,j} : X \rightarrow [0, 1]$ with

$$g_{i,j}|_{\overline{B_i}} \equiv 1, g_{i,j}|_{X \setminus B_j} \equiv 0$$

Since J is countable, then so is the set

$$K := \{(i, j) \in J \times J \mid \overline{B_i} \subseteq B_j\} \subseteq J \times J$$

and hence in bijection with a subset of $\mathbb{N}_+$. Then it suffices to show that $\{g_{i,j}\}_{(i,j) \in K}$ has the desired property. For any $x \in X$, any open $U \ni x$ open in X , there exists $j \in J$, s.t.: $x \in B_j \subseteq U$. By the hypothesis and the Normal-Open-Closed-Neighborhood prop,

there exists an open V , s.t.:

$$x \in \{x\} \subseteq V \subseteq \overline{V} \subseteq B_j$$

Also, since V is open, then there exists $i \in J$, s.t.: $x \in B_i \subseteq V$. Hence,

$$\overline{B_i} \subseteq \overline{V} \subseteq B_j$$

Therefore, by the def of $g_{i,j}$, we're done. $\square$

– **Claim 4:** The map $\Phi : X \rightarrow \mathbb{R}^{\mathbb{N}}$ given by $x \mapsto (\varphi_1(x), \varphi_2(x), \dots)$ is an embedding.

– *Proof.* * Since φ_i 's are continuous, then so is Φ .

* Φ is injective: For any $x \neq y \in X$, since X is normal and since $\{x\}$ and $\{y\}$ are closed in X by hypothesis, then there is an open U , s.t.: $x \in U$ and $y \notin U$. By the property in claim 3, there is an $n \in \mathbb{N}_+$, s.t.:

$$\varphi_n(x) = 1 \neq 0 = \varphi_n(y)$$

Hence, $\Phi(x) \neq \Phi(y)$.

* Φ is open: For any open U in X , any $z_0 \in \Phi(U)$, there is a $x_0 \in U$, s.t.: $\Phi(x_0) = z_0$. By the property in claim 3, $\varphi_n(x_0) = 1, \varphi_n|_{X \setminus U} \equiv 0$. Denote by $\pi_n : \mathbb{R}^{\mathbb{N}} \rightarrow \mathbb{R}$ the projection onto the n -coordinate, which is continuous. Set

$$V := \pi_n^{-1}((0, \infty)), W := V \cap \Phi(X)$$

By the continuity of π_n , V is open in $\mathbb{R}^{\mathbb{N}}$ and hence W is open in $\Phi(X)$. It remains to show that $z_0 \in W$ and that $W \subseteq \Phi(U)$.

$$\cdot \pi_n(z_0) = (\pi_n \circ \Phi)(x_0) = \varphi_n(x_0) = 1 > 0 \Rightarrow z_0 \in W.$$

$$\cdot \text{ For any } z \in W, \text{ there exists } x \in X, \text{ s.t.: } \Phi(x) = z. \text{ Note that } \pi_n(z) = (\pi_n \circ \Phi)(x) = \varphi_n(x) = 1 > 0. \text{ Since } \varphi_n|_{X \setminus U} \equiv 0, \text{ then } x \in U \text{ and hence } z = \Phi(x) \subseteq \Phi(U).$$

$\square$

$\square$

7.3 Regular Topological Spaces

Definition 7.9. Regular

A topological space X is called *regular* if given any $x \in X$ and any closed subset $A \subseteq X$ with $x \notin A$, there are two disjoint open subsets U and V , s.t.: $x \in U$ and $A \subseteq V$.

Proposition 7.10. *Regular-Open-Closed-Neighborhood Proposition*

Let X be a topological space, let $x \in X$ and let $U \subseteq X$ be an open neighborhood of x . If X is regular, then there exists open $V \subseteq X$ in X , s.t.:

$$x \in V \subseteq \bar{V} \subseteq U$$

Proof. Similarly as the proof of the Normal-Open-Closed-Neighborhood Proposition. $\square$

Proposition 7.11. *Regular + Second Countable Proposition*

Let X be a topological space. If X is a regular and if X is second-countable, then X is normal.

Proof. Let $\{B_j\}_{j \in J}$ be a countable basis of X . Pick two disjoint closed subsets of X .

- **Claim 1:** There are open subsets $\{U_m\}_{m \in \mathbb{N}_+}$, s.t.:

$$C \subseteq \bigcup_{m \in \mathbb{N}_+} U_m \subseteq \bigcup_{m \in \mathbb{N}_+} \bar{U}_m \subseteq X \setminus D$$

- *Proof. of claim 1:* Since J is countable, then so is $J' = \{j \in J \mid \bar{B}_j \subseteq X \setminus D\} \subseteq J$. It suffices to show that $C \subseteq \bigcup_{j \in J'} B_j$. For any $c \in C$, since X is regular and D is closed, then by the Regular-Open-Closed-Neighborhood Proposition, there exists an open $U \ni c$, s.t.: $\bar{U} \subseteq X \setminus D$. By the def of basis, there exists $j \in J$, s.t.:

$$c \in B_j \subseteq U \Rightarrow \bar{B}_j \subseteq \bar{U} \subseteq X \setminus D$$

and then we're done. $\square$

Similarly, there exist open subsets $\{V_n\}_{n \in \mathbb{N}_+}$, s.t.:

$$D \subseteq \bigcup_{n \in \mathbb{N}_+} V_n \subseteq \bigcup_{n \in \mathbb{N}_+} \bar{V}_n \subseteq X \setminus C$$

Set $\tilde{U}_m := U_m \setminus \bigcup_{i=1}^m \bar{V}_i$ and $\tilde{V}_n := V_n \setminus \bigcup_{i=1}^n \bar{U}_i$, which are open. Since $\bar{V}_i \cap C = \emptyset$, then $\{\tilde{U}_m\}_{m \in \mathbb{N}_+}$ covers C and similarly $\{\tilde{V}_n\}_{n \in \mathbb{N}_+}$ covers D . Hence it suffices to prove the following claim:

- **Claim 2:** $(\bigcup_{m \in \mathbb{N}_+} \tilde{U}_m) \cap (\bigcup_{n \in \mathbb{N}_+} \tilde{V}_n) = \emptyset$.
- *Proof. of claim 2:* It suffices to show that for any m, n , $\tilde{U}_m \cap \tilde{V}_n = \emptyset$. If $m \leq n$, then by the construction, we have

$$\tilde{V}_n \subseteq V_n \subseteq \bar{V}_n \subseteq X \setminus \tilde{U}_m$$

Similarly if $n \leq m$, then $\tilde{U}_m \subseteq X \setminus \tilde{V}_n$. $\square$

$\square$

Theorem 7.12. *The standard formulation of the Urysohn Metrization Theorem*

Let X be a topological space. If X is second-countable, then the followings are equivalent:

1. X is regular and every one-point subset is closed.
2. X is metrizable.

Theorem 7.13. *Nagata-Smirnov Metrization Theorem*

Let X be a topological space. The followings are equivalent:

1. X is metrizable.
2. *The following three conditions are satisfied:*
 - (a) X is regular.
 - (b) X is Hausdorff.
 - (c) *There exists a countable set I , s.t.: for any $i \in I$, there is a locally finite family $\mathcal{B}_i$ of subsets of X , s.t.: $\bigcup_{i \in I} \mathcal{B}_i$ is a basis of X .*

Chapter 8

Homotopy

8.1 Basic Concepts for Homotopy

General Idea: To understand a space X , we can study continuous maps $f : A \rightarrow X$ or $g : X \rightarrow B$. For example, $A = [0, 1] \leftrightarrow$ path-connected; $A = \mathbb{S}^1 \leftrightarrow$ loops.

Definition 8.1. Homotopy

Let X, Y be two topological spaces.

- A *homotopy* between two continuous maps $f, g : X \rightarrow Y$ is a continuous map $H : X \times [0, 1] \rightarrow Y$, s.t.:

$$H(x, 0) = f(x) \quad \text{and} \quad H(x, 1) = g(x).$$

If such H exists, we say f and g are *homotopic*, denoted as

$$f \simeq g.$$

- A continuous map $f : X \rightarrow Y$ is called *null-homotopic* if it is homotopic to a constant map.

Proposition 8.2. *Convex Homotopy Proposition*

Let X be a topological space. Given any two continuous maps $f, g : X \rightarrow Y (\subseteq \mathbb{R}^n)$ to a convex subspace of $\mathbb{R}^n$, the map $H : X \times [0, 1] \rightarrow Y$ given by

$$H(x, t) := f(x)(1 - t) + g(x)t$$

is a homotopy between f and g .

Example: $Y = \mathbb{R}^2$. Consider two arbitrary cont maps $f, g : X \rightarrow Y$. Then define:

$$H(x, t) := (1 - t)f(x) + tg(x),$$

an interpolate. The well-defined-ness of H is by the convexity of $\mathbb{R}^2$, i.e.: for all x, t , $H(x, t) \in Y$.

Proposition 8.3. *Homotopy is an equivalence relation.*

Proof. • Take $H(x, t) = f(x)$ for all t . Then $f \simeq f$.

- If $f \simeq g$, then there exists H in the definition of homotopy. Then define $H' : X \times [0, 1] \rightarrow Y$ by $H'(x, t) := H(x, 1 - t)$. Hence, $g \simeq f$.
- It $f \simeq g$ and $g \simeq f$, then there are two continuous maps $H : X \times [0, 1] \rightarrow Y$ and $K : X \times [0, 1] \rightarrow Y$ with $H(x, 0) = f(x)$, $H(x, 1) = K(x, 0) = g(x)$, and $K(x, 1) = h(x)$. Then define

$$J(x, t) := \begin{cases} H(x, 2t), & 0 \leq t \leq \frac{1}{2} \\ K(x, 2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

It's clear that J is cont. Then $f \simeq h$.

□

Definition 8.4. Homotopy Class

Let $f : X \rightarrow Y$ be a continuous map. A homotopy class $[f]$ is the collection of continuous maps homotopic to f :

$$[f] := \{g : X \rightarrow Y \mid g \simeq f\}.$$

Example: If $Y = \mathbb{R}^2$, then $[f] = \mathcal{C}(X \rightarrow \mathbb{R}^2)$.

Proposition 8.5. Consider four continuous maps

$$W \xrightarrow{f} X, X \xrightarrow{g} Y, X \xrightarrow{h} Y, Y \xrightarrow{k} Z.$$

If $g \simeq h$, then

$$g \circ f \simeq h \circ f \quad \& \quad k \circ g \simeq k \circ h.$$

Proof. Let $H : X \times [0, 1] \rightarrow Y$ be a homotopy between g and h . Then

- $(H \circ id_{[0,1]}) : W \times [0, 1] \rightarrow Y$ gives a homotopy between $g \circ f$ and $h \circ f$;
- $k \circ H : X \times [0, 1] \rightarrow Z$ gives a homotopy between $k \circ g$ and $k \circ h$.

□

Definition 8.6. Homotopy Equivalence

Two topological spaces X and Y are homotopy equivalent if there exist two continuous maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$, s.t.:

$$\begin{aligned} g \circ f &\simeq id_X \\ f \circ g &\simeq id_Y \end{aligned}$$

denoted by $X \simeq Y$.

Proposition 8.7. Let X and Y be topological spaces. Then

1. if $X \cong Y$, then $X \simeq Y$;

2. the homotopy equivalence $X \simeq Y$ gives a bijection between $\{h : W \rightarrow X \mid h \text{ cont}\}$ and $\{k : W \rightarrow Y \mid k \text{ cont}\}$, for any topological space W ;
3. the homotopy equivalence $X \simeq Y$ forms an equivalence relation between topological spaces.

Proof. 1. If $X \cong Y$, then $g \circ f = id_X \simeq id_X$ and $f \circ g = id_Y \simeq id_Y$. Hence, $X \simeq Y$.

2. Construct $\alpha : \{h : W \rightarrow X \mid h \text{ cont}\} / \sim \rightarrow \{k : W \rightarrow Y \mid k \text{ cont}\} / \sim$ by $\alpha([h]) := [f \circ h]$ and $\beta : \{k : W \rightarrow Y \mid k \text{ cont}\} / \sim \rightarrow \{h : W \rightarrow X \mid h \text{ cont}\} / \sim$ by $\beta([k]) := [g \circ k]$.

Check: α and β are well-defined.

- If $h_1 \sim h_2$, then there exists a homotopy $H : W \times [0, 1] \rightarrow X$. Then by proposition 6.4, $f \circ H$ gives a homotopy between $f \circ h_1$ and $f \circ h_2$. Hence,

$$f \circ h_1 \simeq f \circ h_2 \Rightarrow \alpha([h_1]) = \alpha([h_2]).$$

Therefore, α is well-defined.

- Similarly for the well-defined-ness of β .

Then:

- for any $k \in \{k : W \rightarrow Y \mid k \text{ cont}\}$, we have

$$(\alpha \circ \beta)([k]) := [f \circ g \circ k] = [id_Y \circ k] = [k]$$

i.e.: $\alpha \circ \beta = id$;

- similarly for $\beta \circ \alpha = id$.

Hence, $\{h : W \rightarrow X \mid h \text{ cont}\}$ and $\{k : W \rightarrow Y \mid k \text{ cont}\}$ have one-to-one correspondence.

3. • It's easy to check reflexivity and symmetry.
- If $X \simeq Y, Y \simeq Z$, then there exist cont maps

$$\begin{array}{ccc} X & \xrightarrow{f} & Y, Y \xrightarrow{g} X \\ Y & \xrightarrow{h} & Z, Z \xrightarrow{k} Y \end{array}$$

s.t.:

$$\begin{array}{l} f \circ g \simeq id_Y, g \circ f \simeq id_X \\ h \circ k \simeq id_Z, k \circ h \simeq id_Y. \end{array}$$

Then $(h \circ f) : X \rightarrow Z$ and $(g \circ k) : Z \rightarrow X$ satisfy that

—

$$\begin{aligned}
(h \circ f) \circ (g \circ k) &= h \circ (f \circ g) \circ k \\
&\simeq h \circ id_Y \circ k \\
&= h \circ k \\
&\simeq id_Z
\end{aligned}$$

—

$$\begin{aligned}
(g \circ k) \circ (h \circ f) &= g \circ (k \circ h) \circ f \\
&\simeq g \circ id_Y \circ f \\
&= g \circ f \\
&\simeq id_X.
\end{aligned}$$

Therefore, $X \simeq Z$.

□

Definition 8.8. Contractible

A topological space X is called contractible if for any single point $*$, $X \simeq \{*\}$.

In other words, there exist continuous maps f, g , s.t.:

$$\begin{aligned}
\{*\} &\xrightarrow{f} X \xrightarrow{g} \{*\}, g \circ f \simeq id_{\{*\}} \\
X &\xrightarrow{g} \{*\} \xrightarrow{f} X, f \circ g \simeq id_X.
\end{aligned}$$

Proposition 8.9. *The def of X being contractible can be simplified by:*

1. X is contractible if there exists $\{*\}$, s.t.: $X \simeq \{*\}$; or,
2. X is contractible if $id_X \simeq$ some constant map.

Proof. 1. By proposition 6.6.3, it suffices to show that $\{c_1\} \simeq \{c_2\}$ for all single points c_1, c_2 , which is easy to show.

2.
 - $g \circ f : \{*\} \rightarrow \{*\}$ must be a identity map.
 - For any $x \in X$,

$$x \xrightarrow{g} * \xrightarrow{f} y$$

for the same $y \in X$. Hence, if we define $c_y : X \rightarrow Y$ by $c_y(x) := y$ for any $x \in X$, then

$$c_y = f \circ g \simeq id_X.$$

□

Proposition 8.10. $X = \mathbb{R}^2$ is contractible.

Proof. By proposition 6.8.2, it suffices to show that $id_X \simeq c_{(0,0)}$. Consider $H : \mathbb{R}^2 \times [0, 1] \rightarrow \mathbb{R}^2$ be given by

$$H(\mathbf{x}, t) := t \cdot id_X = t \cdot \mathbf{x}.$$

Then

$$H(\mathbf{x}, 0) := (0, 0) = c_{(0,0)}(\mathbf{x})$$

$$H(\mathbf{x}, 1) := id_X(\mathbf{x}).$$

Therefore, $c_{(0,0)} \simeq id_X$. □

Corollary 8.11. Any convex subset of $\mathbb{R}^2$ is contractible.

Example: $\mathbb{S}^1$ is NOT contractible.

Definition 8.12. Homotopy Retract

Let X be a topological space and $A \subseteq X$. Let $i : A \rightarrow X$ be an inclusion. We say A is a homotopy retract of X if there exists a continuous map $r : X \rightarrow A$, s.t.:

$$\begin{aligned} r \circ i : A &\xrightarrow{i} X \xrightarrow{r} A = id_A \\ i \circ r : X &\xrightarrow{r} A \xrightarrow{i} X \simeq id_X. \end{aligned}$$

Moreover, we call r here a retraction.

Example:

- The 1-sphere $\mathbb{S}^1 = [0, 1] / \sim$ is a homotopy retract of Mobius band $M = [0, 1]^2 / \sim$. Consider the inclusion $i : A \rightarrow M$ and the retraction $r : X \rightarrow A$ ve given by

$$\begin{aligned} i([x]) &:= [(x, \frac{1}{2})] \\ r([x, y]) &:= [x]. \end{aligned}$$

Then we have $r \circ i = id_{\mathbb{S}^1}$ and $(i \circ r)([(x, y)]) = [(x, \frac{1}{2})]$. Construct a cont map $H : M \times [0, 1] \rightarrow M$ by

$$H([(x, y)], t) := [(x, (1-t)y + \frac{t}{2})].$$

Check: H is well-defined.

- The equivalence relation on M is defined by

$$(x, y) \sim (x', y') \Leftrightarrow \begin{cases} x = x', y = y' & \text{if } x, x' \in (0, 1) \\ \{x, x'\} = \{0, 1\}, y' = 1 - y & \text{if } x, x' \in \{0, 1\} \end{cases}.$$

Hence, it suffices to show that

$$H([(0, y)], t) = H([(1, 1 - y)], t) \quad \forall y \in [0, 1],$$

which holds immediately by a easy computation:

$$(1 - t)y + \frac{t}{2} + (1 - t)(1 - y) + \frac{t}{2} = y - ty + \frac{t}{2} + 1 - y - t + ty + \frac{t}{2} = 1.$$

Then we have

$$\begin{aligned} H([(x, y)], 0) &:= [(x, y)] = id_M([x, y]) \\ H([(x, y)], 1) &:= [(x, \frac{1}{2})] = (i \circ r)([(x, y)]). \end{aligned}$$

Hence, $id_M \simeq i \circ r$.

- The $(n - 1)$ -sphere $\mathbb{S}^{n-1} = \{\mathbf{x} \in \mathbb{R}^n \mid \sum_i x_i^2 = 1\}$ is a homotopy retract of $\mathbb{R}^n - \{0\}$. Consider the usual inclusion $i : \mathbb{S}^{n-1} \rightarrow \mathbb{R}^n - \{0\}$ and $r : \mathbb{R}^n - \{0\} \rightarrow \mathbb{S}^{n-1}$ be given by

$$r(\mathbf{x}) := \frac{\mathbf{x}}{\|\mathbf{x}\|}.$$

Then $r \circ i = id_{\mathbb{S}^{n-1}}$ and $(i \circ r)(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|}$. Consider $H : (\mathbb{R}^n - \{0\}) \times [0, 1] \rightarrow \mathbb{R}^n - \{0\}$ be given by

$$H(\mathbf{x}, t) := (1 - t)\mathbf{x} + t \frac{\mathbf{x}}{\|\mathbf{x}\|}.$$

It's necessary but easy to check that H is well-defined, i.e.: for all $t \in [0, 1]$, $H(\mathbf{x}, t) \neq 0$. Then we have

$$\begin{aligned} H(\mathbf{x}, 0) &= \mathbf{x} = id_{\mathbb{R}^n - \{0\}} \\ H(\mathbf{x}, 1) &= \frac{\mathbf{x}}{\|\mathbf{x}\|} = (i \circ r)(\mathbf{x}) \end{aligned}$$

which gives $i \circ r \simeq id_{\mathbb{R}^n - \{0\}}$.

8.2 Simplicial Approximation

Goal/Question: For any $f : X \rightarrow Y$ continuous, if K_1, K_2 are triangulations of X, Y , respectively, i.e.: $X \cong |K_1|$ and $Y \cong |K_2|$, then the question is: can we understand f by just looking at K_1, K_2 ?

Definition 8.13. Simplicial map

Let $K_1 = (V_1, \Sigma_1)$ and $K_2 = (V_2, \Sigma_2)$ be simplicial complices. A simplicial map between K_1 and K_2 is a map $f : K_1 \rightarrow K_2$ satisfying that

- f maps vertices to vertices, i.e.: $f(v) \in V_2$ for any $v \in V_1$;

- f maps simplices to simplices, i.e.: $f(\sigma) \in \Sigma_2$ for any $\sigma \in \Sigma_1$.

Example: Let $K_1 = (V_1, \Sigma_1)$ where $V_1 = \{1, 2, 3, 4\}$ and $\Sigma_1 = \{\{1\}, \dots, \{4\}, \{1, 2\}, \dots, \{3, 4\}, \{1, 2, 3\}, \dots, \{2, 3, 4\}\}$ and let $K_2 = (V_2, \Sigma_2)$ where $V_2 = \{1, 2, 3\}$ and $\Sigma_2 = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$. In particular, $\{1, 2, 3, 4\} \notin \Sigma_1$ and $\{1, 2, 3\} \in \Sigma_2$. Then define $f : K_1 \rightarrow K_2$ by

$$f(1) := 1, f(2) := 2, f(3) = f(4) := 3.$$

Then f is a simplicial map. In particular, $f(\{1, 2, 4\}) = \{1, 2, 3\}$.

Now we want to define the simplicial map between the topological realization. There are several observations:

- $|K| \subseteq \mathbb{R}^{|V|-1}$;
- for any $x \in |K|$, there exists a unique $\sigma \in \Sigma$, s.t.: $x \in \Delta_\sigma$;
- let $V_1 = \{v_1, \dots, v_n\} \subseteq \mathbb{R}^m$ be the vertices of K_1 . Then for any $x \in K_1$, x can be uniquely written as

$$x = \sum_{i=1}^k \alpha_i v_{\sigma_i}$$

where $\alpha_i \geq 0$, $\sum_i \alpha_i = 1$ and $\sigma = \{v_{\sigma_1}, \dots, v_{\sigma_k}\} \in \Sigma_1$ is the unique simplex where $x \in \text{interior of } \Delta_\sigma$, e.g.: the center of an equilateral triangle can be uniquely written as $\frac{1}{3}(\mathbf{1} + \mathbf{2} + \mathbf{3}) \in \text{interior of } \Delta_{\{\mathbf{1}+\mathbf{2}+\mathbf{3}\}}$;

- let $f : K_1 \rightarrow K_2$ be a simplicial map, where $V_2 = \{u_1, \dots, u_l\} \subseteq \mathbb{R}^m$. Then for any i , we have

$$f(v_i) = u_j$$

for some $j \in \{1, \dots, l\}$.

Definition 8.14. Map induced from Simplicial Map

The simplicial map $f : K_1 \rightarrow K_2$ induces a map $|f| : |K_1| \rightarrow |K_2|$ between the topological realizations satisfying:

- $|f|$ maps vertices to vertices, i.e.: $|f|(v_1) := f(v_1)$ for any $v_1 \in V_1$;
- $|f|(\sum_{i=1}^k \alpha_i v_i) = \sum_{i=1}^k \alpha_i |f|(v_i)$.

Remark: $|f|$ is continuous.

Definition 8.15. Subdivision

Let K be a simplicial complex. A simplicial complex K' is called a subdivision of K if

- each simplex of K' is contained in K ;
- each simplex of K equals the union of *finitely* many simplices of K' .

As a result,

- there exists a homeomorphism $h : |K'| \xrightarrow{\cong} |K|$, s.t.: for any $\sigma' \in \Sigma'$,

$$h([\Delta_{\sigma'}]) \subseteq [\Delta_{\sigma}]$$

for some $\sigma \in \Sigma$;

- for any $\sigma \in \Sigma$,

$$\Delta_{\sigma} = \bigcup_{\sigma'' \in \Sigma''} [\Delta_{\sigma''}]$$

for a *finite* subset $\Sigma'' \subseteq \Sigma'$.

Example (*Barycentric Subdivision*):

Let $K = (V, \Sigma)$. Define $V' := \Sigma = \{\sigma_1, \dots, \sigma_n\}$ and define Σ' by

$$\{\sigma_{i_1}, \dots, \sigma_{i_k} \mid i_j \in \{1, \dots, n\}\} \in \Sigma' \Leftrightarrow \sigma_{i'_1} \subseteq \dots \subseteq \sigma_{i'_k} \text{ by reordering } \{i_1, \dots, i_k\}.$$

Definition 8.16. Star

The star of v , denoted as $st_K(v)$, is

$$\bigcup_{\sigma \in \Sigma, v \in \sigma} \Delta_{\sigma}^{\circ},$$

where $\Delta_{\sigma}^{\circ} := \text{inside}(\Delta_{\sigma}) := \{(x_1, \dots, x_{n+1}) \in \Delta_{\sigma} \mid x_i > 0, \forall i\}$ for simplex Δ_{σ} .

Remark (obvious propositions):

- $st_K(v)$ is open in $|K|$, since its complement is the simplicial subcomplex of K , spanned by $V \setminus \{v\}$.
- Every $w \in st_K(v)$ is path-connected to v . In particular, $st_K(v)$ is path-connected.

Proposition 8.17. Suppose $f : |K| \rightarrow |L|$ is continuous. If there exists a map $g : K \rightarrow L$, s.t.: for any $v \in V_K$,

$$f(st_K(v)) \subseteq st_L(g(v))$$

then g is simplicial and

$$|g| \simeq f.$$

Proof. • **Claim:** Suppose that $\sigma = \{v_0, \dots, v_n\} \in \Sigma_K$ and that $x \in \sigma^{\circ} \subseteq |K|$. If $f(x) \in |L|$ lies in the inside of the (unique) simplex $\tau \in \Sigma_L$ (i.e.: $f(x)$ can be uniquely expressed as

$$\sum_{u_i \in \tau} \beta_i u_i$$

where $\beta_i > 0$ and $\sum_i \beta_i = 1$), then

$$g(v_0), \dots, g(v_n)$$

are vertices of τ .

Proof of claim: By the def of σ° , $x = \sum_{i=1}^n \alpha_i v_i$ where $\alpha_i > 0$ and $\sum_i \alpha_i = 1$. Hence, $x \in st_K(v_i)$ for $i = 1, \dots, n$, where

$$st_K(v_i) = \{av_i + \sum_{j=1}^m b_j w_j \mid a, b_j > 0, a + \sum_j b_j = 1, \{v_i, w_j, \dots, w_m\} \in \Sigma_K\}.$$

Therefore, $f(x) \in f((st_K(v_i))^\circ) \subseteq st_L(g(v))$. Then we get

$$f(x) = ag(v_i) + \sum_{j=1}^m b_j u_j$$

where $a, b_j > 0$, $a + \sum_j b_j = 1$ and $\{g(v_i), u_1, \dots, u_m\} \in \Sigma_L$. Then by the hypothesis on $f(x)$, $g(v_i)$ is a vertex of simplex τ , for all i .

Therefore, the map g maps simplices to simplices which is a simplicial map. Then we can construct a homotopy between f and $|g|$:

- (*Construction of homotopy between f and $|g|$*): Consider $x \in |K|$ and let $\tau \in \Sigma_L$ be such that $f(x) \in \tau^\circ$. Let $\sigma = \{v_0, \dots, v_n\} \in \Sigma_K$ be such that $x \in \sigma^\circ$. Then

$$|g|(x) := g(x) = \sum_{i=1}^n \lambda_i g(v_i)$$

where by claim $g(v_0), \dots, g(v_n)$ are vertices of τ . Since $f(x) = \sum_i \mu_i \tau_i$, then we can rewrite $|g|(x) = \sum_i \lambda'_i \tau_i$ (by adding some $\lambda_i := 0$ if necessary). Then we can define the map $H : |K| \times I \rightarrow |L|$ by

$$H(x, t) := \sum (t\lambda'_i + (1-t)\mu_i) \tau_i$$

which follows that H is a homotopy between $|g| \simeq f$.

□

Theorem 8.18. *Simplicial Approximation Theorem*

Let K, L be two simplicial complexes with V_K finite and let $f : |K| \rightarrow |L|$ be continuous. Then there exists a subdivision K' of K and a simplicial map $g : K' \rightarrow L$, s.t.:

$$|g| : |K'| \rightarrow |L| \simeq f.$$

Proof. Since $\{st_L(w) \mid w \in L\}$ is an open cover of L , then

$$|K| \subseteq \bigcup_{i=1}^n f^{-1}(st_L(w_i)).$$

Then there exists a small $\delta > 0$, s.t.: for any $x, y \in |K|$ with $d(x, y) < \delta$ and $x, y \in f^{-1}(st_L(w_i))$ for some i . Then we can construct a simplicial subdivision K' of K with the coarseness less than

δ , i.e.:

$$x, y \in st_{K'}(v), d(x, y) < \delta.$$

Therefore, for any $v \in V_K$, $st_{K'}(v) \subseteq f^{-1}(st_L(w_i))$ for some i , i.e.:

$$f(st_{K'}(v)) \subseteq st_L(w_i).$$

Then set $g(v) := w_i$ and then apply proposition 6.18.

□

Chapter 9

Graph

9.1 (Abstract) Graphs

Definition 9.1. (Abstract) Graph

A *graph* $\Gamma = (V, E, l, \tau)$ is defined by

- vertex set V is a finite (or countably infinite) set;
- edge set E is a finite (or countably infinite) set;
- a function $\delta : E \rightarrow V \times V$ is such that

$$\delta(e) := (l(e), \tau(e))$$

where l denotes the leading part of e and τ denotes the terminal part.

Definition 9.2. Category of (Abstract) Graphs

- Let $\Gamma = (V, E, l, \tau)$ and $\Gamma' = (V', E', l', \tau')$ be two (abstract) graphs. We define a *morphism* from Γ to Γ' by a pair $(\alpha : V \rightarrow V', \beta : E \rightarrow E')$ of maps such that for each $e \in E$, we have

$$l'(\beta(e)) = \alpha(l(e)), \tau'(\beta(e)) = \alpha(\tau(e))$$

- We refer to the category $\mathcal{AbsGraph}$ with
 - $Ob(\mathcal{AbsGraph}) :=$ all (abstract) graphs
 - $Mor_{\mathcal{AbsGraph}}(\Gamma, \Gamma') :=$ all maps from Γ to Γ' in the above sense

with the usual composition of maps as the *category of (abstract) graph*.

Definition 9.3. Realization of graph

The *realization* of a graph $\Gamma = (V, E, l, \tau)$ is given by

$$|\Gamma| := \frac{(\bigsqcup_{|V|} 0 - \text{simplicies}) \sqcup (\bigsqcup_{|E|} 1 - \text{simplicies})}{\sim_\delta} \cong (V \sqcup \bigsqcup_{e \in E} (\{e\} \times [0, 1])) / \sim_{(e,0) \sim l(e), (e,1) \sim \tau(e)}$$

Remark:

- We later will use the second quotient space to construct maps but we can better understand the realization of graph the first quotient space.
- Even though Γ may NOT come from a simplicial complex, we can do "subdivision" Γ' of Γ so that

$$|\Gamma'| \cong |\Gamma|$$

and Γ' comes from a simplicial complex.

Definition 9.4. Let $\Gamma = (V, E, l, \tau)$ be a graph.

- (*Inverse of edges*):

$$E^{-1} := \{e^{-1} \mid e \in E\}$$

and a map $\delta^{-1} : E^{-1} \rightarrow V \times V$ with $\delta^{-1}(e^{-1}) := (\tau(e), l(e))$.

- (*Edge Path on Γ*): a sequence of elements $p = (e_1, e_2, \dots, e_n)$ where $e_i \in E \cup E^{-1}$, s.t.:

$$\tau(e_i) = l(e_{i+1})$$

Furhermore, we say p is an *edge path of length n* .

- (*Parametrization of edge path*): for each edge path $p = (e_1, e_2, \dots, e_n)$, we can define a path $|p| : [0, 1] \rightarrow \Gamma$, s.t.: for any $i = 1, \dots, n$

$$|p|[\frac{i-1}{n}, \frac{i}{n}] := e_i$$

called the *parametrization* of p .

- (*Edge Loop*): an edge path $p = (e_1, e_2, \dots, e_n)$ satisfying

$$l(e_1) = \tau(e_n)$$

- (*Embedded edge loop*): an edge loop α is *embedded* if $|\alpha|(0, 1)$ is injective.

Definition 9.5. Let $\Gamma = (V, E, l, \tau)$ be a graph and let $e \in E$.

- We refer to the map

$$\Phi_e : [0, 1] \xrightarrow{t \mapsto (e, t)} \{e\} \times [0, 1] \xrightarrow{p} |\Gamma|$$

where p denotes the canonical projection map, as the *characteristic map* of the edge e .

- We write $|e| := \Phi_e([0, 1])$ and $\langle e \rangle := \Phi_e((0, 1))$.

Proposition 9.6. *Graph-Topology Proposition*

Let $\Gamma = (V, E, l, \tau)$ be a graph.

1. For each $e \in E$, the characteristic map $\Phi_e : [0, 1] \rightarrow |\Gamma|$ is continuous.
2. Let $U \subseteq |\Gamma|$ be a subset.
 - (a) $U \subseteq |\Gamma|$ is open in $|\Gamma| \Leftrightarrow \forall e \in E, \Phi_e^{-1}(U)$ is open in $[0, 1]$.
 - (b) $U \subseteq |\Gamma|$ is closed in $|\Gamma| \Leftrightarrow \forall e \in E, \Phi_e^{-1}(U)$ is closed in $[0, 1]$.
3. Let $f : |\Gamma| \rightarrow X$ be a map to a topological space X .

$$f \text{ is continuous} \Leftrightarrow \forall e \in E, f \circ \Phi_e : [0, 1] \rightarrow X \text{ is continuous.}$$

Proposition 9.7. *Realization-Functor Proposition*

The followings are equivalent:

1. Let $\Gamma = (V, E, l, \tau)$ and $\Gamma' = (V', E', l', \tau')$ be two graphs, let $f := (\alpha : V \rightarrow V', \beta : E \rightarrow E')$ be a map between Γ and Γ' . Consider $|f| : |\Gamma| \rightarrow |\Gamma'|$ given by

$$|f|([x]) := \begin{cases} [\alpha(x)] & , \text{if } x \in V \\ [\beta(x)] & , \text{if } x = (e, t) \text{ for some } e \in E \end{cases}$$

This map is well-defined and continuous.

2. The map $F : \text{AbsGraph} \rightarrow \text{Top}$ given by the map

$$\begin{array}{ccc} \text{Ob}(\text{AbsGraph}) & \rightarrow & \text{Ob}(\text{Top}) \\ \Gamma & \mapsto & |\Gamma| \end{array}$$

together with the maps

$$\begin{array}{ccc} \text{Mor}_{\text{AbsGraph}}(\Gamma, \Gamma') & \rightarrow & \text{Mor}_{\text{Top}}(|\Gamma|, |\Gamma'|) \\ f & \mapsto & |f| \end{array}$$

forms a covariant functor from AbsGraph to Top .

Proposition 9.8. *Realization-Properties Proposition*

Let $\Gamma = (V, E, l, \tau)$ be a graph.

1. (a) Every compact subspace K of $|\Gamma|$ is contained in the realization of a finite subgraph.
 (b) $|\Gamma|$ is compact $\Leftrightarrow \Gamma$ is finite.
2. The followings are equivalent:
 - (a) Γ is connected.
 - (b) $\forall v \neq w \in V, \exists$ an embedding $\gamma : [0, 1] \rightarrow |\Gamma|$ with $\gamma(0) = v, \gamma(1) = w$, s.t.: $\gamma([0, 1]) \subseteq |\Gamma|$ is the union of edges.
 - (c) $|\Gamma|$ is path-connected.
 - (d) $|\Gamma|$ is connected.

3. $|\Gamma|$ is Hausdorff.

Proof. 1. (a) Set

$$\tilde{E} := \{e \in E \mid K \cap \langle e \rangle \neq \emptyset\}, \tilde{V} := \{v \in V_K \mid v \notin |e|, \forall e \in E\}$$

Furthermore,

- for each $e \in \tilde{E}$, we pick $P_e \in K \cap \langle e \rangle$ and set $A_e := \langle e \rangle$
- for each $v \in \tilde{V}$, we set $B_v := \{v\} \cup (\bigcup_{e \in E, l(e)=v} \bigcup_{e \in E, \tau(e)=v} \langle e \rangle)$
- we set $C := (|\Gamma| \setminus (\bigcup_{e \in \tilde{E}} \{P_e\})) \setminus \tilde{V}$.

By the Realization-Properties Proposition, A_e, B_v and C are open in $|\Gamma|$. Note that $\{A_e\}_{e \in \tilde{E}}, \{B_v\}_{v \in \tilde{V}}$ and C form an open cover of K . Since K is compact, then we can cover K by finitely many of these open subsets.

But each A_e contains a point, namely P_e , of K , that does NOT lie in any of B_v and C . Similarly, B_v contains v where $v \notin A_e \cap C$. These show that $\tilde{E}$ and $\tilde{V}$ are finite. Therefore, K must be contained in the realization of a finite subgraph.

(b) If $|\Gamma|$ is compact, then by 1.(a) Γ is finite. If Γ is finite, by the compact-ness of each $|e|$, we're done.

2. Note that every graph is a simplicial complex. Hence, $(a) \Leftrightarrow (c) \Leftrightarrow (d)$. So we just need to prove $(a) \Rightarrow (b)$ and $(b) \Rightarrow (c)$.

- $((a) \Rightarrow (b))$: $\forall v \neq w \in V$, it's clear that there is an injective path $e_1, \dots, e_k$ from v to w , i.e.: the vertices on this path are distinct. Consider the map $\gamma : [0, 1] \rightarrow |\Gamma|$ given by $\gamma(0) = v$ such that on each $[\frac{i-1}{n}, \frac{i}{n}]$, γ is just a reparametrization of Φ_{e_i} , which goes from $l(e_i)$ to $\tau(e_i)$. By the Graph-Topology Proposition and the Pasting Proposition.2, γ is continuous. By the choice of the injective path, γ is an embedding.
- $((b) \Rightarrow (c))$: $\forall P, Q \in |\Gamma|, \exists e, f \in E$, s.t.:

$$P \in |e|, Q \in |f|$$

Pick one of the vertices e, f , say v of e , w of f . By the continuity of Φ_e and Φ_f , there exist a path from P to v and a path w to Q . Furthermore, by 2.(b), there is a path from v to w and hence we're done.

□

9.2 Topological Graphs

Definition 9.9. Topological Graphs

A *topological graph* is a triple $(X, \Gamma = (V, E, l, \tau), \Theta : |\Gamma| \rightarrow X)$ where

- X is a topological space
- $\Gamma = (V, E, l, \tau)$ is a graph
- $\Theta : |\Gamma| \rightarrow X$ is a homeomorphism between the realization $|\Gamma|$ of Γ and X .

Furthermore, given a topological graph $(X, \Gamma = (V, E, l, \tau), \Theta : |\Gamma| \rightarrow X)$, we introduce the following definitions:

- *Vertex:* Given $v \in V$, we refer to $\Theta(v)$ as a *vertex of the topological graph*.
- *Edge:* Given $e \in E$, we refer to $\Theta(e)$ as a *edge of the topological graph*.
- *Loop:* Given a loop α , we refer to $\Theta(|\alpha|)$ as a *loop of the topological graph*.
- We call Γ the *underlying graph*.
- Sometimes we refer, somewhat incorrectly, to X as a *realization of Γ* .

Proposition 9.10. *Realization-Embedding Proposition*

Let $\Gamma = (V, E, l, \tau)$ be a graph and let $n \in \mathbb{N} \cup \{0\}$. Suppose we're given the followings:

1. An injective map $\alpha : V \rightarrow \mathbb{R}^n$.
2. $\forall e \in E$, we're given a continuous map $\beta_e : [0, 1] \rightarrow \mathbb{R}^n$ which is injective on $(0, 1)$, which satisfies $\beta_e(0, 1) \subseteq \mathbb{R}^n \setminus \alpha(V)$ and which satisfies $\beta_e(0) = \alpha(l(e)), \beta_e(1) = \alpha(\tau(e))$.
3. $\forall e \neq f \in E$, $\beta_e((0, 1)) \cap \beta_f((0, 1)) = \emptyset$.

If Γ is infinite, then we also need that

4. $\forall r \in \mathbb{R}_{\geq 0}$, there are finitely many $v \in V$ with $\alpha(v) \cap B_r^n(0) \neq \emptyset$ and there are finitely many $e \in E$ with $\beta_e([0, 1]) \cap B_r^n(0) \neq \emptyset$.

Then the map $\varphi : |\Gamma| \rightarrow \mathbb{R}^n$ given by

$$\varphi([x]) := \begin{cases} \alpha(x) & , \text{if } x \in V \\ \beta_e(t) & , \text{if } x = (e, t) \text{ for some } e \in E \end{cases}$$

is a closed embedding.

Remark:

- This proposition gives us a way to construct topological graphs with a given underlying graph.
- Given any finite graph, there exists an embedding of $|\Gamma|$ into $\mathbb{R}^3$.

9.3 Undirected Graphs

Definition 9.11. Undirected (Abstract) Graphs

An *undirected (abstract) graph* Γ is a triple (V, E, φ) where V is a set, E is a set and φ is a map from E to $\mathcal{P}(V) := \{\text{subsets of } V\}$, s.t.: $\varphi(e)$ consists of one or two elements in V . Furthermore, we introduce the following definitions:

- The elements of V are called the *vertices* of Γ and the elements of E are called the *edges* of Γ .
- We say that an edge $e \in E$ a *loop* if $\#\varphi(e) = 1$.
- Given $e \in E$, the points in $\varphi(e)$ are called the *vertices of e* .
- We say that Γ has *multi-edges* if $\exists e \neq e' \in E$, s.t.:

$$\varphi(e) = \varphi(e')$$

- We say that Γ is *finite (countable)* if both V and E are finite (countable).

Definition 9.12. Category of Undirected (Abstract) Graphs

- Let $\Gamma = (V, E, \varphi)$ and $\Gamma' = (V', E', \varphi')$ be two undirected (abstract) graphs. We define a *morphism* from Γ to Γ' by a pair $(\alpha : V \rightarrow V', \beta : E \rightarrow E')$ of maps such that for any $e \in E$,

$$\varphi'(\beta(e)) = \alpha(\varphi(e))$$

- We refer to the category $UndirAbsGraph$ with
 - $Ob(UndirAbsGraph) :=$ all undirected (abstract) graph
 - $Mor_{UndirAbsGraph}(\Gamma, \Gamma') :=$ all maps from Γ to Γ' in the above sense

with the usual composition of maps as the *category of undirected (abstract) graphs*.

Definition 9.13. • Let $n \in \mathbb{N} \cup \{0\}$. We denote by I_n the graph that is given by $V = \{0, \dots, n\}$, $E = \{0, \dots, n-1\}$ and $\varphi(i) := \{i, i+1\}$ for $i \in E$.

- We say that an undirected graph $\Gamma = (V, E, \varphi)$ is *connected* if given any vertices $v, w \in V$, there exists a map $p : I_n \rightarrow \Gamma$, s.t.: $p(0) = v, p(n) = w$.

Proposition 9.14. *Combinatorial-Connected-to-Discrete Proposition*

Let $\Gamma = (V, E, l, \tau)$ be a graph or let $\Gamma = (V, E, \varphi)$ be an undirected graph. Let $f : V \rightarrow X$ be a map to a set X , s.t.: for any $e \in E$, the values of f on the vertices of e agree. If Γ is connected, then f is constant.

Definition 9.15. Realization of Undirected Graphs

Let $\Gamma = (V, E, \varphi)$ be an undirected graph. We define the *realization* of Γ as the topological space

$$|\Gamma| := (V \sqcup (\bigsqcup_{e \in E, \#\varphi(e)=1} [0, 1]) \sqcup (\bigsqcup_{e \in E, \#\varphi(e)=2} [0, 1] \times Bij(\{0, 1\}, \varphi(e)))) / \sim$$

where $\sim$ is generated by the followings:

- $\forall e \in E$ with $\#\varphi(e) = 1$, we identify 0 and 1 in the corresponding $[0, 1]$ with the unique point $\varphi(e)$ in V .

- $\forall e \in E$ with $\#\varphi(e) = 2$, we make the following identifications:
 - $\forall \gamma \in \text{Bij}(\{0, 1\}, \varphi(e))$, set $(0, \gamma) \sim \gamma(0)$ and $(1, \gamma) \sim \gamma(1)$
 - $\forall \alpha \neq \beta \in \text{Bij}(\{0, 1\}, \varphi(e))$, $\forall t \in [0, 1]$, set $(t, \alpha) \sim (1 - t, \beta)$.

Definition 9.16. Undirected Topological Graphs

An *undirected topological graph* is a triple $(X, \Gamma = (V, E, \varphi), \Theta : |\Gamma| \rightarrow X)$ with the followings:

- X is a topological space
- Γ is a undirected graph
- $\Theta : |\Gamma| \rightarrow X$ is a homeomorphism between the realization $|\Gamma|$ of Γ and X .

Proposition 9.17. *Realization-Homeomorphism Proposition*

Suppose we are given a graph $\Gamma = (V, E, l, \tau)$. As above, we consider the undirected graph $\Gamma' = (V, E, \varphi)$ where $\varphi : E \rightarrow \mathcal{P}(V)$ is the map given by

$$\varphi(e) := \{l(e), \tau(e)\}$$

The map $\Theta_\Gamma : |\Gamma| \rightarrow |\Gamma'|$ given by

$$\Theta_\Gamma([x]) := \begin{cases} [x] & , \text{if } x \in V \\ [t] & , \text{if } x = (e, t) \text{ for some } e \in E \text{ with } l(e) = \tau(e) \\ [(t, \gamma)] & , \text{if } x = (e, t) \text{ for some } e \in E \text{ with } l(e) \neq \tau(e) \end{cases}$$

where $\gamma \in \text{Bij}(\{0, 1\}, \varphi(e))$ is given by $\gamma(0) := l(e), \gamma(1) := \tau(e)$, is a homeomorphism.

9.4 Tree

Definition 9.18. Let $\Gamma = (V, E, l, \tau)$ be a graph.

(*Tree*): a connected graph $\Gamma = (V, E, l, \tau)$ is a *tree* if Γ has NO embedded edge loop.

(*Subgraph*): a subgraph $\Gamma' = (V', E', l, \tau)$ of Γ is given by

- $V' \subseteq V$
- $E' \subseteq E$
- $\delta' : E' \rightarrow V' \times V' := \delta|_{E'}$.

(*Subtree*): a subgraph that is a tree is called a *subtree*.

(*Maximal Tree*): a *maximal tree* $T = (V_T, E_T, l, \tau)$ of Γ is defined by

- T is a subgraph of Γ
- T is a tree
- if we add an edge $e \in E \setminus E_T$ to T , then the new graph $T \cup \{e\}$ is NOT a tree.

Proposition 9.19. *Tree-Basics Proposition*

1. Any connected non-empty subgraph of a tree is again a tree.
2. Let Γ be a graph. Let $T = (V, E)$ be a subtree of Γ and let $\{e_i\}_{i \in I}$ be a family of edges of Γ with the following properties:

- (a) one of $l(e_i)$ and $\tau(e_i)$, say t_i , lies in T and the other one, say v_i , does NOT lie in T
- (b) the vertices v_i are distinct.

Then $(V \cup \bigcup_{i \in I} \{v_i\}, E \cup \bigcup_{i \in I} \{e_i\})$ is also a subtree of Γ .

3. Let Γ be a graph. If $T_1 \subseteq T_2 \subseteq \dots$ is a sequence of subtrees of Γ , then $\bigcup_{i \in \mathbb{N}_+} T_i$ is also a subtree of Γ .

Proof. 1. is obvious.

2. Set $W := V \cup \bigcup_{i \in I} \{v_i\}$ and $F := E \cup \bigcup_{i \in I} \{e_i\}$. By the property (a), (W, F) has no loops. Now, for any $w, w' \in W$, we want to show that there is a unique injective path from w to w' . We only treat the case when $w = v_i, w' = v_j$ for some $i, j \in I$ and the other cases are treated in the similar way.

- *Existence:* Since (V, E) is a subtree, then $\exists!$ an injective path α in (V, E) from t_i to t_j . Then we find an injective path in (W, F) from $w = v_i$ going through α to $w' = v_j$
- *Uniqueness:* Let $f_1, \dots, f_k$ be an injective path in (W, F) from w to w' . By the properties (a),(b), $f_1 = e_i$ and $f_k = e_j$ and then $f_2, \dots, f_{k-1}$ forms an injective path in (V, E) from t_i to t_j . Hence, by the uniqueness of α , we're done.

3. Note that $\bigcup_{i \in \mathbb{N}_+} T_i = (\bigcup_{i \in \mathbb{N}_+} V_{T_i}, \bigcup_{i \in \mathbb{N}_+} E_{T_i})$. For any $v_1, v_2 \in \bigcup_{i \in \mathbb{N}_+} V_{T_i}$, there exist $i_1, i_2 \in I$, s.t.: $v_1 \in V_{T_{i_1}}, v_2 \in V_{T_{i_2}}$. WLOG, let $i_1 \leq i_2$. Then by the hypothesis, $T_{i_1} \subseteq T_{i_2}$ and hence $v_1 \in V_{T_{i_1}} \subseteq V_{T_{i_2}}$. Since T_{i_2} is a subtree, then $\exists!$ an injective path β in T_{i_2} from v_1 to v_2 . Suppose on the contrary that there is a natural $j \geq i_2$, s.t.: there is an injective path $\gamma \neq \beta$ in T_j from v_1 to v_2 . Then $\gamma \circ \beta^{-1}$ gives a loop in T_j , contradicting that T_j is a tree.

□

Proposition 9.20. *If $\Gamma = (V, E, l, \tau)$ is a tree and $|V| < \infty$, then*

$$|E| = |V| - 1$$

Theorem 9.21. *Let $\Gamma = (V, E, l, \tau)$ be a connected graph with $|V| < \infty$. Then T is maximal tree of Γ if and only if*

$$|V_T| = |V|$$

Proof. • $(\Rightarrow)$: Suppose on the contrary that $|V_T| < |V|$. Take a vertex $v \in V \setminus V_T$. Since Γ is connected and $|V| < \infty$, then we can find a *shortest* path from $v' \in V_T \subseteq V$ to v . Then this

path begins in V_T and ends at $v \notin V_T$, say $p = (e_1, \dots, e_n)$. Since

$$(V_T \ni) v' = l(e_1) \neq \tau(e_n) = v (\in V \setminus V_T)$$

then p is NOT an edge loop. Hence, $e_n \notin E_T$. Meanwhile, since p is the shortest path between v' and v , then $T \cup \{e_n\}$ is a larger tree.

- ($\Leftarrow$): Since $|V_T| = |V|$, then $V_T = V$ and hence for any $v_1, v_2 \in V = V_T$, there exists an edge path $p = (e_1, \dots, e_n)$ with

$$l(e_1) = v_2 \text{ and } \tau(e_n) = v_1$$

Hence, if we add (v_1, v_2) in T , then $\alpha = (e_1, \dots, e_n, (v_1, v_2))$ is an edge loop in $T \cup (v_1, v_2)$ which is NOT embedded in $T \cup (v_1, v_2)$ and hence $T \cup (v_1, v_2)$ is NOT a tree. Therefore, T is a maximal tree. □

Corollary 9.22. *All graphs have maximal tree.*

Corollary 9.23. *For any maximal tree T of connected graph $\Gamma = (V, E, l, \tau)$, we always have: T has $|V|$ vertices and $|V| - 1$ edges.*

Definition 9.24. Spanning Tree

Let Γ be a(n undirected) graph. A *spanning tree* for Γ is a subtree of Γ that contains all vertices of Γ .

Definition 9.25. Let Γ be a(n undirected) graph and let Γ' be a subtree graph.

- We say that an *edge* e of Γ is *adjacent* to Γ' if e doesn't lie in Γ' and if at least one vertex of e lies in Γ' .
- We say that an *vertex* v of Γ is *adjacent* to Γ' if v doesn't lie in Γ' and if v is a vertex of an edge that is adjacent of Γ' .

Proposition 9.26. *Spanning Tree-Characterization Proposition*

Let Γ be a connected (undirected) graph and let $T \subseteq \Gamma$ be a tree. The followings are equivalent:

1. T is spanning tree.
2. T is maximal in the sense that if T is contained in a subtree T' of Γ , then $T = T'$.
3. T doesn't admit an adjacent vertex.

Proof. • ($1 \Rightarrow 2$): Let H be a subgraph of Γ with $T \subsetneq H$. Now we want to show that H is NOT a tree. Take an edge e lying in H but not in T . If e is a loop, then we're done immediately. Otherwise, let $v \neq w$ be the vertices of e . By the def of spanning tree, v, w lie in T . Since T is a tree, then there exists an injective path α in T from v to w . On the other hand, e is an injective path in H from w to v . Then α and e form a loop in H and hence H is NOT a tree

- (2 $\Rightarrow$ 3): It suffices to show that if $T = (W, F)$ is a subtree that admits an adjacent vertex x , then it is NOT maximal in the sense stated in 2. Let f be an edge such that one vertex of f is x and the other one lies in W . By the Tree-Basics Prop.2, $(W \cup \{x\}, F \cup \{f\})$ is a subtree of Γ that contains T .
- (3 $\Rightarrow$ 1): It suffices to show that if T is NOT a spanning tree, then it admits an adjacent vertex. Suppose that $\exists$ a vertex v , s.t.: v does NOT lie in T . Let w be a vertex in T . Since Γ is connected, then there is a path $e_1, \dots, e_n$ from w to v . Then one of vertices of e_i is adjacent to T for some $i \in \{1, \dots, n\}$.

□

Proposition 9.27. *Spanning Tree-Existence Proposition*

Every connected non-empty (undirected) graph admits a spanning tree.

9.5 Euler Characteristic

Definition 9.28. Euler Characteristic

Let $\Gamma = (V, E)$ be a finite (undirected) graph. The *Euler characteristic* of Γ is defined as

$$\chi(\Gamma) := \#V - \#E$$

Proposition 9.29. *Tree Characteristic Proposition*

Let $\Gamma = (V, E)$ be a finite (undirected) graph. The followings are equivalent:

1. Γ is a tree.
2. Γ is connected and $\chi(\Gamma) = 1$.

Proof.

Definition 9.30. Valence

If $\Gamma = (V, E, l, \tau)$ is a graph, then the *valence* of a vertex v is defined as

$$\text{valence}(v) := \#\{e \in E \mid l(e) = v\} + \#\{e \in E \mid \tau(e) = v\}$$

- If $\Gamma = (V, E, \varphi)$ is an undirected graph, then the *valence* of a vertex v is defined as

$$\text{valence}(v) := \#\{e \in E \mid v \in \varphi(e), \#\varphi(e) = 2\} + 2 \cdot \#\{e \in E \mid v \in \varphi(e), \#\varphi(e) = 1\}$$

In a slightly less formal but more convenient way, the *valence* of a vertex v is the number of edges at v , but hereby we have to count any loop at v twice.

Lemma 9.31. *Tree-Valence One Edge Lemma*

Let $\Gamma = (V, E)$ be a finite (undirected) graph. If $\chi(\Gamma) = 1$ and if Γ has at least one edge, then Γ

admits at least two vertices of valence one.

Proof. of lemma: We prove this lemma for graphs and the proof for undirected graphs is the same. Since $\chi(\Gamma) = 1$ and $\Gamma = (V, E, l, \tau)$ has at least one edge, then Γ must have NO vertices of valence 0, i.e.: $\text{valence}(v) \geq 1$ for any $v \in V$. Since

$$\sum_{v \in V} \text{valence}(v) := \sum_{v \in V} (\#\{e \in E \mid l(e) = v\} + \#\{e \in E \mid \tau(e) = v\}) = 2 \cdot \#E \stackrel{\chi(\Gamma)=1}{=} 2 \cdot \#V - 2$$

then $2 \cdot \#V - 2 \geq \#V \Leftrightarrow \#V \geq 2$. □

Now we return to the *proof of the Tree Characteristic Proposition*:

- (1 $\Rightarrow$ 2): We've proven it in the previous proposition.
- (2 $\Rightarrow$ 1): Do induction on $\#E$.
 - *Basic Case:* $\#E = 0$. Then by the connectivity of Γ (assumption in 2), $\#V = 1 \Rightarrow \Gamma$ is a trivial tree.
 - *Inductive Step:* Suppose that the statement holds for $\#E = n$. Let $\Gamma = (V, E)$ be a connected (undirected) graph with $(n+1)$ -edges and with $\chi(\Gamma) = 1$. By the Tree-Valence One Edge Lemma, Γ admits a vertex v of valence one. Let e be the corresponding edge. Note that $\Gamma' = (V \setminus \{v\}, E \setminus \{e\})$ is connected and that

$$\chi(\Gamma') = \#(V \setminus \{v\}) - \#(E \setminus \{e\}) = \#V - \#E = \chi(\Gamma) = 1$$

Therefore, by the inductive hypothesis, Γ' is a tree and hence by the Tree-Basic Proposition.2, Γ is tree. □

9.6 Quotients of Graphs

Definition 9.32. Let $G = (V, E, l, \tau)$ be a graph and let $H = (W, F)$ be a non-empty subgraph.

- We define $\sim$ by the equivalence relation generated by $v \sim v'$ if v and v' are both vertices in H , i.e.: $v \sim v' \Leftrightarrow v, v' \in W$. Furthermore, we denote by $*$ the equivalence class defined by the vertices in H .
- We denote by $\pi : V \rightarrow V/\sim$ the map that is given by $v \mapsto [v]$.
- Finally, we define the graph

$$G/H := (V/\sim, E \setminus F, \pi \circ l : E \setminus F \rightarrow V/\sim, \pi \circ \tau : E \setminus F \rightarrow V/\sim)$$

We refer to G/H as the *quotient of the graph G by H* .

Proposition 9.33. *Graph-Quotient Proposition*

Let $G = (V, E, l, \tau)$ be a graph and let $H = (W, F)$ be a non-empty subgraph.

1. If G is finite, then $\chi(G/H) = \chi(G) - \chi(H) + 1$.
2. Consider the map $p : |G| \rightarrow |G/H|$ given by

$$p([x]) := \begin{cases} * & , \text{if } x \in H \\ [\pi(x)] & , \text{if } x \in V \setminus W \\ [(e, t)] & , \text{if } x = (e, t) \text{ for some } e \in E \end{cases}$$

The induced map $\bar{p} : |G|/|H| \rightarrow |G/H|$ is well-defined and is a homeomorphism.

Proof. 1.

$$\chi(G/H) = \#(V/\sim) - \#(E \setminus F) = (\#V - \#W + 1) - (\#E - \#F) = \chi(G) - \chi(H) + 1$$

□

Chapter 10

Group Theory

Definition 10.1. Let $S := \{s_i \mid i \in I\}$ be a set of "alphabets".

- Define $S^{-1} := \{s^{-1} \mid s \in S\}$ such that
 - $S \cap S^{-1} = \emptyset$;
 - $(s^{-1})^{-1} \in (S^{-1})^{-1}$ satisfies $(s^{-1})^{-1} = s$. In particular,

$$(S^{-1})^{-1} = S.$$

- A **word** in S is a *finite* sequence

$$w = s_1 \cdots s_n$$

where $n \in \mathbb{N}_+$ and $s_i \in S \cup S^{-1}$.

In particular, if $n = 0$, then $w = \emptyset$.

- The *concatenation* of two words $x_1 \cdots x_n$ and $y_1 \cdots y_m$ is the word

$$x_1 \cdots x_n y_1 \cdots y_m.$$

- Two words w, w' are *equivalent*, denoted as $w \sim w'$, if there are words $w = w_1 \cdots w_n = w'$, s.t.: for any i ,

$$w_i = \cdots y_1 s s^{-1} y_2 \cdots, w_{i+1} = \cdots y_1 y_2 \cdots$$

OR

$$w_i \cdots y_1 y_2 \cdots, w_{i+1} = \cdots y_1 s s^{-1} y_2 \cdots$$

for some $s \in S \cup S^{-1}$.

Definition 10.2. Free Group

Let S be a set. A *free group with generator S* is given by

$$F(S) := \{[w] \mid w \text{ is a word in } S\}.$$

Check: $F(S)$ has group structure:

- $[w] * [u] = [wu]$: If $w_1 \sim w_2$ and $u_1 \sim u_2$, then check $w_1 u_1 \sim w_2 u_2$.
- $e_{F(S)} = [\emptyset]$.
- $[x_1 \cdots x_n]^{-1} = [x_1^{-1} \cdots x_n^{-1}]$ where $x_i \in S \cup S^{-1}$.

Example:

- $S = \{a\}$. All elements in $F(S)$ are of the form $[a]^i$, $i \in \mathbb{Z}$. Therefore, $F(S) \cong \mathbb{Z}$.

Proof. Define a map $\phi : F(S) \rightarrow \mathbb{Z}$ by

$$\phi([a]^i) := i.$$

Then check that ϕ is a group isomorphism. □

- $S = \{a, b\}$ and $S^{-1} = \{a^{-1}, b^{-1}\}$.
 - $F(\{a_1, \dots, a_n\}) \subseteq F(\{a, b\})$.
 - $[ab] \neq [ba]$ in general, and hence $F(S)$ is **NOT** abelian, which follows that

$$F(\{a, b\}) \not\cong \mathbb{Z} \times \mathbb{Z}.$$

– $[ab^{-1}a^2b^2a^{-2}b]$ can **NOT** be reduced further.

- Every subgroup of free group is a free group.

Definition 10.3. Group with Relations

Let S be a set and let $R(S)$ be a set of words in S . A *group with relations* $R(S)$ is defined by

$$G = \langle S \mid R(S) \rangle$$

also called the group presentation, where elements in G are equivalence class of words in S given by:

$$w \sim w' \Leftrightarrow$$

- w and w' differ by some xx^{-1} , $x \in S \cup S^{-1}$; **OR**
- w and w' differ by some element $z \in R(S)$, or its inverse.

Example:

- $F(\{a, b\}) = \langle a, b \mid \emptyset \rangle$
- $\langle a \mid a^5 \rangle \cong \mathbb{Z}_5$
- $\langle a, b \mid a^2, b^2, abab^{-1}a^{-1}b^{-1} \rangle \cong S_3$

Chapter 11

Fundamental Group

11.1 Definition of $\Pi_1(X, b)$

Definition 11.1. Homotopic Relative

Let $A \subseteq X$ be topological spaces. We say $f, g : X \rightarrow Y$ are homotopic relative to A if there exists $H : X \times [0, 1] \rightarrow Y$, s.t.:

$$\begin{cases} H(x, 0) = f(x) \\ H(x, 1) = g(x) \end{cases} \quad \text{and} \quad H(a, t) = f(a) = g(a) \quad \forall a \in A, t \in I.$$

Definition 11.2. Fundamental Group

The *fundamental group of topological space X at base point $b \in X$* is defined by

$$\Pi_1(X, b) := \frac{\{l : I \rightarrow X \text{ loop based at } b, \text{ i.e. : } f(0) = f(1) = b\}}{\sim \text{ homotopy relative to } \{0, 1\}}$$

which is also called the 1-st homotopy group.

Proposition 11.3. *The fundamental group $\Pi_1(X, b)$ has group structure.*

Proof. Denote $\{l : I \rightarrow X \text{ loop based at } b, \text{ i.e. : } f(0) = f(1) = b\}$ by L .

- We need to firstly define an operation on the fundamental group $\Pi_1(X, b)$:
 - For two loops $l, l' \in L$, define the binary operation $\cdot : L \times L \rightarrow L$ by

$$(l \cdot l')(s) := \begin{cases} l(2s) & 0 \leq s \leq 1/2 \\ l'(2s - 1) & 1/2 \leq s \leq 1 \end{cases}.$$

It remains to show that $l \cdot l' \in L$ by showing $(l \cdot l')(0) = l(0) = b$ and $(l \cdot l')(1) = l'(0) = b$.

- Define $*$: $\Pi_1(X, b) \times \Pi_1(X, b) \rightarrow \Pi_1(X, b)$ by

$$[l] * [l'] := [l \cdot l'].$$

It remains to check the well-defined-ness of $*$, i.e.: if $l_1 \simeq l_2$ and $l'_1 \simeq l'_2$ relative to $\{0, 1\}$, then $l_1 \cdot l_2 \simeq l'_1 \cdot l'_2$ relative to $\{0, 1\}$:

Take the homotopies H between l_1 and l_2 and K between l'_1 and l'_2 . Then define $G : I \times I \rightarrow X$ by

$$G(s, t) := \begin{cases} H(2s, t) & 0 \leq s \leq 1/2 \\ K(2s - 1, t) & 1/2 \leq s \leq 1 \end{cases}.$$

One can check that $G(s, 0) = (l_1 \cdot l_2)(s)$ and $G(s, 1) = (l'_1 \cdot l'_2)(s)$.

– It remains to show that $*$ is associative:

Take $[u], [v], [w] \in \Pi_1(X, b)$. Define $H' : I \times I \rightarrow X$ by

$$H'(s, t) := \begin{cases} u(\frac{4s}{2-t}) & 0 \leq s \leq \frac{1}{2} - \frac{1}{4}t \\ v(4s - 2 + t) & \frac{1}{2} - \frac{1}{4}t \leq s \leq \frac{3}{4} - \frac{1}{4}t \\ w(4s - 3 + \frac{t}{1+t}) & \frac{3}{4} - \frac{1}{4}t \leq s \leq 1 \end{cases}$$

which is easy to check that H' is the homotopy between $(u \cdot v) \cdot w \simeq u \cdot (v \cdot w)$. Since $u(0) = u(1) = v(0) = v(1) = w(0) = w(1) = b$, then we have

$$([u] * [v]) * [w] = [u] * ([v] * [w]).$$

•

$$e_{\Pi_1(X, b)} = [c_b]$$

where $c_b(s) = b$ for all s .

• For each loop $l \in L$, define:

$$l^{-1}(s) := l(1 - s)$$

for any $s \in I$. Then it remains to show that $l \cdot l^{-1} \simeq c_b$ by constructing $H'' : I \times I \rightarrow X$ as

$$H''(s, t) := \begin{cases} l(2s(1 - t)) & 0 \leq s \leq 1/2 \\ l((2 - 2s)(1 - t)) & 1/2 \leq s \leq 1 \end{cases}$$

which is the homotopy between $l \cdot l^{-1} \simeq c_b$ and that similarly for $l^{-1} \cdot l \simeq c_b$.

□

Definition 11.4. Simple connectivity

X is simply connected if for any $b \in X$, s.t.:

$$\Pi_1(X, b) \cong \{e\}.$$

Example:

- $\mathbb{R}^2$ is simply connected:

Since $\mathbb{R}^2$ is convex, then we can use the linear homotopy between any loop and some point.

- $\mathbb{R}^2 \setminus \{0\}$ is **NOT** simply connected, since the loops who encircle 0 can not be homotopic to any c_b .

Proposition 11.5. *If $b, b' \in X$ are connected by a path $\omega : [0, 1] \rightarrow X$, then one has*

$$\Pi_1(X, b) \cong \Pi_1(X, b')$$

Proof. Construct a map $\omega_* : \Pi_1(X, b) \rightarrow \Pi_1(X, b')$ by

$$\omega_*[l] := [\omega^{-1} \cdot l \cdot \omega]$$

- ω_* is a group homomorphism:

$$\begin{aligned} \omega_*([l] \cdot [l']) &= \omega_*[l \cdot l'] := [\omega^{-1} \cdot l \cdot l' \cdot \omega] = [\omega^{-1} \cdot l \cdot \omega \cdot \omega^{-1} \cdot l' \cdot \omega] \\ &= [\omega^{-1} \cdot l \cdot \omega \cdot \omega^{-1}] \cdot [l' \cdot \omega] := \omega_*[l] \cdot \omega_*[l'] \end{aligned}$$

- ω_* is invertible: Take $\alpha : \Pi_1(X, b') \rightarrow \Pi_1(X, b)$ given by

$$\alpha[l'] := [\omega \cdot l' \cdot \omega^{-1}]$$

Then check that $\alpha \cdot \omega_* = id$ and $\omega_* \cdot \alpha = id$, i.e.: $\alpha = \omega_*^{-1}$.

Therefore, ω_* is a group isomorphism. □

Remark: If X is *path-connected*, then for any $x_1, x_2 \in X$, one has

$$\Pi_1(X, x_1) \cong \Pi_1(X, x_2)$$

Moreover, in such case, we'll write $\Pi_1(X)$ instead of $\Pi_1(X, b)$. But when computing, we need to pick a based point.

Definition 11.6. Induced Homomorphism

Let (X, x_0) and (Y, y_0) be two topological spaces with basepoint x_0 and y_0 , respectively, and let $f : X \rightarrow Y$ be a continuous map with $f(x_0) = y_0$. Then every loop $l : I \rightarrow X$ based at x_0 gives a loop $f \circ l : I \rightarrow Y$, i.e.: f induces a group homomorphism $f_* : \Pi_1(X, x_0) \rightarrow \Pi_1(Y, y_0)$ satisfying

$$f_*[l] := [f \circ l]$$

Check: f_* is well-defined: If $l \simeq l'$ relative to $\{0, 1\}$ based at x_0 , then there exists a homotopy

$H : I \times I \rightarrow X$, s.t.: for any $t \in I$,

$$\begin{aligned} H(t, 0) &= l(t), H(t, 1) = l'(t) \\ H(a, t) &= l(a) = l'(a) = x_0, \forall a \in \{0, 1\} \end{aligned}$$

Then consider $K : I \times I \xrightarrow{H} X \xrightarrow{f} Y$, i.e.:

$$K = f \circ H$$

Then one has

$$\begin{aligned} K(t, 0) &:= f(H(t, 0)) = f(l(t)) = (f \circ l)(t) \\ K(t, 1) &:= f(H(t, 1)) = f(l'(t)) = (f \circ l')(t) \\ K(a, t) &:= f(H(a, t)) = (f \circ l)(a) = (f \circ l')(a), \forall a \in \{0, 1\} \end{aligned}$$

Hence, $f \circ l \simeq f \circ l'$ relative to $\{0, 1\}$ based at x_0 .

Proposition 11.7. 1. $(id_X)_* = id_{\Pi_1(X, x_0)}$

2. $(g \circ f)_* = g_* \circ f_*$

3. If $f \simeq f'$ relative to $x_0 \in X$, then $f_* = (f')_*$

Corollary 11.8. Let X, Y be path-connected and let $X \simeq Y$. Then one has

$$\Pi_1(X) \cong \Pi_1(Y)$$

Proof. Since $X \simeq Y$, then by def, there exist continuous functions f, g , s.t.:

$$\begin{aligned} X &\xrightarrow{f} Y \xrightarrow{g} X \simeq id_X \\ Y &\xrightarrow{g} X \xrightarrow{f} Y \simeq id_Y \end{aligned}$$

WARNING: We can NOT use $(g \circ f)_* = (id_X)_*$, since $g \circ f \simeq id_X$ is not relative to x_0 .

Denote $(g \circ f)(x_0)$ by x_1 . By the hypothesis of corollary, there exists a path $\omega : I \rightarrow X$ from x_0 to x_1 . Then consider $H : X \times I \rightarrow X$ to be the homotopy between $g \circ f \simeq id_X$ given by

$$H(x_0, t) := \omega(t)$$

and the induced group homomorphism is

$$\Pi_1(X, x_0) \xrightarrow{f_*} \Pi_1(Y, y_0) \xrightarrow{g_*} \Pi_1(X, x_1)$$

For any loop l based at x_0 , consider $K : I \times I \xrightarrow{l \times id} X \times I \xrightarrow{H} X$, i.e.:

$$K(t, s) := H(l(t), s)$$

Then K gives the homotopy between $g \circ f \circ l$ and l . Note that $[\omega^{-1}(g \circ f \circ l)\omega] = [l]$. Then by the def of ω_* in proposition 8.5, one has

$$[l] = \omega_*[g \circ f \circ l] = \omega_* \circ g_* \circ f_*[l]$$

Hence $\omega_* \circ g_* \circ f_* = id_{\Pi_1(X, x_0)}$. Since ω_* is an group isomorphism which is proved in proposition 8.5, then we have: g_* is surjective and f_* is injective. Similarly, g_* is injective and f_* is surjective. Then we're done. $\square$

11.2 Computation of $\Pi_1(X, b)$

Definition 11.9. Let $K = (X, \Sigma)$ be a simplicial complex and let $\alpha := (v_0, v_1, \dots, v_n)$ and $\beta := (w_0, w_1, \dots, w_m)$ be two edge paths.

- **Edge Loop:** α is an edge loop based at v_0 if $v_n = v_0$
- **Concatenation:** If $v_n = w_0 = w_m = v_0$, then we define the *concatenation of loops based at v_0* by

$$\alpha \cdot \beta = (v_0, \dots, v_n = v_0) \cdot (w_0 = v_0, \dots, w_m = v_0) := (v_0, \dots, v_n, w_1, \dots, w_m)$$

Definition 11.10. Elementary Contraction/Expansion

Let α, β be edge paths.

- An *elementary contraction* of α is given by one of the 3 operations below:
 1. replacing $\dots v_{i-1}v_i \dots$ by $\dots v_{i-1} \dots$ if $v_{i-1} = v_i$;
 2. replacing $\dots v_{i-1}v_i v_{i+1} \dots$ by $\dots v_{i-1} \dots$ if $v_{i-1} = v_{i+1}$;
 3. replacing $\dots v_{i-1}v_i v_{i+1} \dots$ by $\dots v_{i-1}v_{i+1} \dots$ if $\{v_{i-1}, v_i, v_{i+1}\} \in \Sigma$.
- An *elementary expansion* is the reverse of an elementary contraction.
- α and β are *equivalent*, denoted as $\alpha \sim \beta$, if they differ by a finite number of elementary contractions or expansions.

Example: Let $K = (V, \Sigma)$ where $V = \{1, 2, 3\}$ and $\Sigma = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}\}$.

$$11123213211231 \sim 12321321231 \sim 1323231 \sim 13231 \sim 131 \sim 1$$

Remark: If $\alpha \sim \beta$, then the topological realization of α and β are homotopic.

Definition 11.11. Edge Loop Group

Let $K = (V, \Sigma)$ and $b \in V$. The *edge loop group based at b* is defined by

$$\Pi_1(K, b) := \frac{\{\text{edge loops based at } b\}}{\sim \text{ elementary contractions or expansions}}$$

Proposition 11.12. $(\Pi_1(K, b), *)$ has a group structure where

$$[\alpha] * [\beta] := [\alpha \cdot \beta]$$

Proof. • *Well-defined-ness of $*$:* If $\alpha \sim \alpha'$ and $\beta \sim \beta'$, i.e.: $[\alpha] = [\alpha']$ and $[\beta] = [\beta']$, then

$$[\alpha] * [\beta] = [\alpha'] * [\beta'].$$

- *Associativity of $*$:* Clearly.
- *Identity:* $e = [b]$, by operation 1.
- *Inverse:* $[bv_1 \cdots v_{n-1}b]^{-1} = [bv_{n-1} \cdots v_1b]$, by operation 2.

□

Proposition 11.13. Let $K = (V, \Sigma)$ where $V = \{1, 2, 3\}$ and $\Sigma = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}\}$.

Then

$$\Pi_1(K) \cong (\mathbb{Z}, +)$$

Proof. Intuition: Since $\{1, 2, 3\} \notin \Sigma$, then we can NOT apply operation 3, i.e.: $[\cdots (123)^{m_1} \cdots] \neq [\cdots (123)^{m_2} \cdots]$ if $m_1 \neq m_2$.

Define the *winding number of α* , $\Phi : \{\text{edge loops based at } 1\} \rightarrow \mathbb{Z}$, by

$$\Phi(\alpha) := \#\{23 \text{ appearing in } \alpha\} - \#\{32 \text{ appearing in } \alpha\}$$

and then define a map $\phi : \Pi_1(K, 1) \rightarrow \mathbb{Z}$ by

$$\phi[\alpha] := \Phi(\alpha)$$

- ϕ is a group homomorphism:

$$\begin{aligned} \phi([\alpha] * [\beta]) &= \phi[\alpha \cdot \beta] = \Phi(\alpha \cdot \beta) \\ &= \#\{23 \text{ appearing in } \alpha \cdot \beta\} - \#\{32 \text{ appearing in } \alpha \cdot \beta\} \\ &= \#\{23 \text{ appearing in } \alpha\} + \#\{23 \text{ appearing in } \beta\} - \#\{32 \text{ appearing in } \alpha\} - \#\{32 \text{ appearing in } \beta\} \\ &= \#\{23 \text{ appearing in } \alpha\} - \#\{32 \text{ appearing in } \alpha\} + \#\{23 \text{ appearing in } \beta\} - \#\{32 \text{ appearing in } \beta\} \\ &= \Phi(\alpha) + \Phi(\beta) = \phi[\alpha] + \phi[\beta] \end{aligned}$$

- ϕ is surjective: For any $n \in \mathbb{Z}$,

$$\begin{cases} n > 0 \Rightarrow \phi((123)^n 1) = n \\ n = 0 \Rightarrow \phi(1) = n \\ n < 0 \Rightarrow \phi((132)^n 1) = n \end{cases}$$

- ϕ is injective:

$$\begin{aligned} \ker \phi &:= \{0\} = \{[\beta] \mid \#\{23 \text{ appearing in } \alpha\} = \#\{32 \text{ appearing in } \alpha\}\} \\ &= \{[1]\} \end{aligned}$$

Therefore, ϕ is a group isomorphism, i.e.:

$$\Pi_1(K) = \Pi_1(K, 1) \cong \mathbb{Z}$$

□

Remark: $\Pi_1(K)$ is infinite cyclic.

Theorem 11.14.

$$\Pi_1(K, b) \cong \Pi_1(|K|, b)$$

where $\Pi_1(K, b)$ denotes the edge loop group and $\Pi_1(|K|, b)$ denotes the fundamental group.

Recall: Simplicial Approximation Theorem: for any $f : I \rightarrow |K|$, there is a simplicial complex (subdivision) of I

$$I_n := \{[0, \frac{1}{n}], \dots, [\frac{n-1}{n}, 1]\}$$

so that $V_{I_n} = \{0, \frac{1}{n}, \dots, \frac{n-1}{n}, 1\}$, $\Sigma_{I_n} = \{\{0, \frac{1}{n}\}, \dots, \{\frac{n-1}{n}, 1\}\}$ and $|I_n| \cong I$ are homeomorphic. Then take a simplicial map $g : I_n \rightarrow K$, s.t.:

$$|g| \simeq f$$

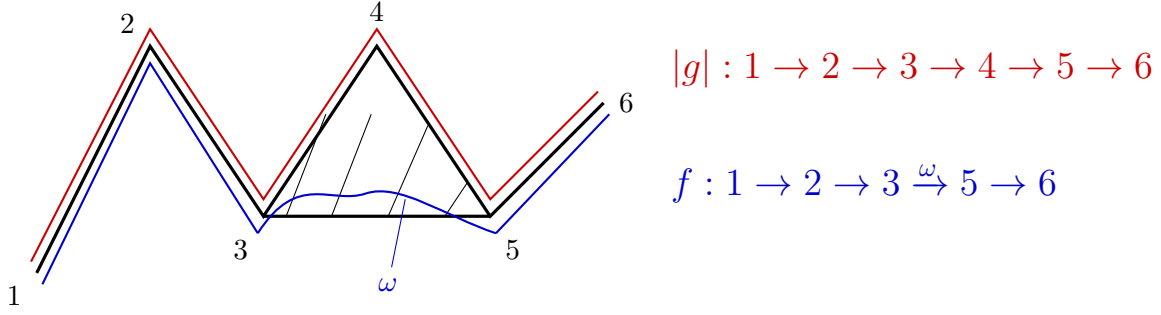
relative to "overlapping part", e.g.: where the "overlapping part" of $|g|$ and f are $1 \rightarrow 2 \rightarrow 3$ and $5 \rightarrow 6$.

Proof. (of Theorem 8.14) For any edge loop α , define the realization of α , $g_\alpha : I \rightarrow \Pi_1(|K|, b)$. Then define $\Theta : \Pi_1(K, b) \rightarrow \Pi_1(|K|, b)$ by

$$\Theta[\alpha] := [g_\alpha]$$

The well-defined-ness of Θ is easy to check, since if $\alpha \sim \beta$, then $g_\alpha \sim g_\beta$. Then for two edge loops α_1 and α_2 ,

$$\Theta[\alpha_1] \cdot \Theta[\alpha_2] = [g_{\alpha_1}] \cdot [g_{\alpha_2}] = [g_{\alpha_1} \cdot g_{\alpha_2}] \stackrel{(*)}{=} [g_{\alpha_1 \alpha_2}] = \Theta([\alpha] \cdot [\beta])$$

Figure 11.1: Paths of mappings $|g|$ and f

where $(*)$ holds since $[g_{\alpha_1} \cdot g_{\alpha_2}]$ and $[g_{\alpha_1 \alpha_2}]$ have the same path with the different speeds, which follows that Θ is a group homomorphism.

Now it remains to show that Θ is an group isomorphism:

- Θ is *surjective*: Take a loop $f : I \rightarrow |K|$ based at b , by Simplicial Approximation Theorem, take I_n to be in **Recall**, i.e.: $I_n \cong I$. Then there exists a simplicial map $g : I_n \rightarrow K$, s.t.:

$$|g| \simeq f$$

Now take $\alpha := (b = g(0), g(1/n), \dots, g((n-1)/n), g(1) = b)$, which is an edge loop in K based at b , since for any $i \in \{0, \dots, n\}$, $g(i/n)$ is either a vertex in K or a point on one of the edges in K , with $|g| = g_\alpha : I \rightarrow |K|$. Hence, $[g_\alpha] = [f] \in \Pi_1(|K|, b)$.

- Θ is *injective*: For any edge loop α based at b such that $[\alpha] \in \ker \Theta$, i.e.: $\Theta[\alpha] := [g_\alpha] = e_{\Pi_1(|K|, b)}$, i.e.:

$$g_\alpha \simeq c_b$$

Hence, there exists a homotopy $H : I \times I \rightarrow |K|$ between g_α and c_b . Then apply Simplicial Approximation Theorem on H . Namely, there exists a sufficiently large positive integer r , s.t.: $(I \times I)_r$ is a subdivision of $I \times I$ with $r \times r$ squares (every square contains two 2-simplices), i.e.:

$$|(I \times I)_r| \cong I \times I$$

and a simplicial map $G : (I \times I)_r \rightarrow K$, s.t.: $|G| \simeq H$, as figure 8.2 shown.

where $\alpha = (b = v_0, v_1, \dots, v_{n-1}, v_n = b)$. Hence, one has

$$\alpha = bv_1 \cdots v_{n-1}b \sim \cdots \sim bb \cdots bb \sim b$$

which is equivalent to $[\alpha] = [b]$, which is the trivial edge loop class in $\Pi_1(|K|, b)$, i.e.:

$$\ker \Theta = \{[b]\} = \{e_{\Pi_1(|K|, b)}\}$$

which implies the injectivity of Θ .

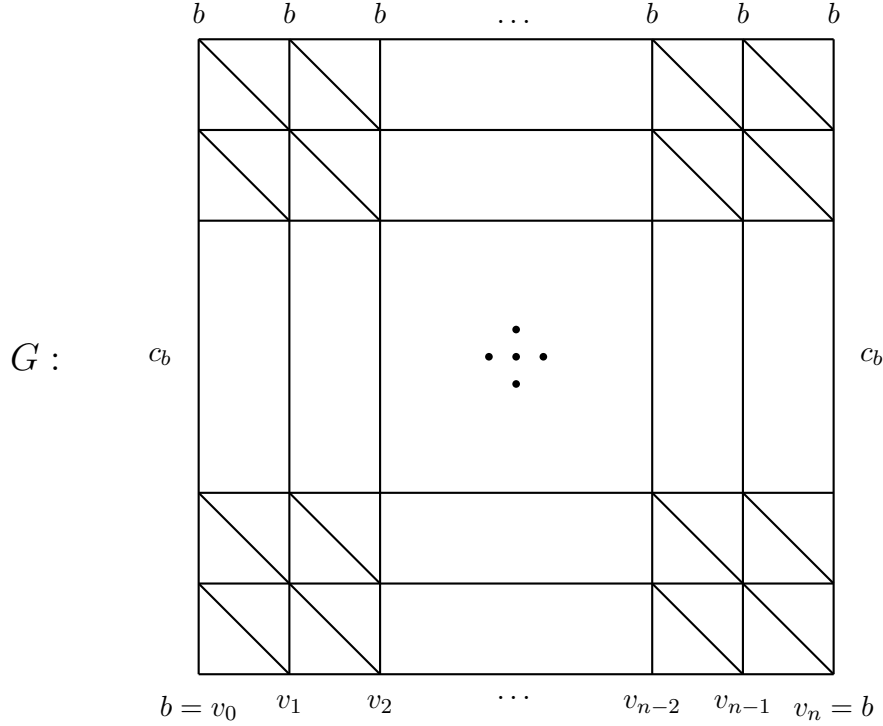


Figure 11.2:

□

Corollary 11.15.

$$\Pi_1(|K|) \cong (\mathbb{Z}, +)$$

where K is the simplicial complex defined in proposition 8.13.

Definition 11.16. Skeleton

Let $K = (V, \Sigma)$ be a simplicial complex. The n -skeleton of K is the simplicial complex $K^{(n)} = (V^{(n)}, \Sigma^{(n)})$ where

$$V^{(n)} := V \text{ and } \Sigma^{(n)} := \{\sigma \in \Sigma \mid |\sigma| \leq n + 1\}$$

Corollary 11.17. For $n \geq 2$,

$$\Pi_1(|K|, b) \cong \Pi_1(|K^{(n)}|, b)$$

Proof. Using edge loops, some elementary contraction/expansion on edge loops involve only information on 2-simplex in. □

Remark: If $n = 1$, then we only have the information on edges, but we threw away all the faces. Hence, we can NOT use operation 3. Here we imply the difference between proposition 8.13 above and proposition 8.18 below.

Proposition 11.18. Let $K = (V, \Sigma)$ where $V = \{1, 2, 3\}$ and $\Sigma = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$. Then K is simply-connected.

Proof. It's equivalent to show that $\Pi_1(K, 1) = \{e\} = \{[1]\}$. For any edge loop $\alpha = (1 = v_0, v_1, \dots, v_{n-1}, v_n = 1)$ based at 1, by operation 1, we can take $\{v_{i_1}, \dots, v_{i_k}\} \subseteq \{v_1, \dots, v_{n-1}\}$ to be such that $v_{i_1} \neq 1$, $v_{i_k} \neq 1$, $v_{i_j} \neq v_{i_{j+1}}$ for any j , and

$$[\alpha] = [1v_{i_1} \cdots v_{i_k} 1]$$

WLOG, let $v_{i_1} = 2$. Then $[\alpha] = [123 \cdots]$ or $[121 \cdots]$. Hence, if $[\alpha] = [123 \cdots]$, then by operation 3, $[\alpha] = [13 \cdots]$ and if $[\alpha] = [121 \cdots]$, then by operation 2, $[\alpha] = [1 \cdots]$. Repeat this procedure, and then we can get $[\alpha] = [1]$. Then we're done. $\square$

Proposition 11.19. *For any $n \geq 2$,*

$$\Pi_1(\mathbb{S}^n) \cong \{e\}$$

Proof. Consider the simplicial complex $K = (V, \Sigma)$ given by the boundary of the simplicial complex generated by $V = \{1, \dots, n+1\}$. Then one has

$$\mathbb{S}^n = |K|$$

Hence, by theorem 8.14 (*) and corollary 8.17 (**),

$$\Pi_1(\mathbb{S}^n) = \Pi_1(|K|) \stackrel{(*)}{\cong} \Pi_1(K) \stackrel{(**)}{\cong} \Pi_1(|K^{(2)}|) \quad (\#)$$

For any edge loop $\alpha = (1 = v_0, v_1, \dots, v_{n-1}, v_n = 1)$ based at 1, after using operation 1 and 2, we can get $\alpha \sim (1 = v_{i_0}, v_{i_1}, \dots, v_{i_{k-1}}, v_{i_k} = 1)$. Then for any $j \in \{1, \dots, k-1\}$, $\{v_{i_{j-1}}, v_{i_j}, v_{i_{j+1}}\}$ forms a face of $K^{(2)}$. Hence, after using operation 3, we can finally get $[\alpha] = [1] \in \Pi_1(|K^{(2)}|, 1)$. Therefore,

$$\Pi_1(|K^{(2)}|, 1) \cong \{[1]\} \cong \{e\}.$$

$\square$

Theorem 11.20. *Browwer's Fixed Point Theorem*

For any $n \in \mathbb{N}_+$, let $\mathbb{D}^n$ be the n -dim'l disk, i.e.:

$$\mathbb{D}^n := \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}$$

For any continuous map $f : \mathbb{D}^n \rightarrow \mathbb{D}^n$, there exists a fixed point, i.e.: there exists $x_0 \in \mathbb{D}^n$, s.t.:

$$f(x_0) = x_0$$

Proof. Suppose on the contrary that there is a cont map $f : \mathbb{D}^n \rightarrow \mathbb{D}^n$ s.t.: for any $x \in \mathbb{D}^n$, $f(x) \neq x$. Then we can construct a ray pointing from $f(x)$ to x uniquely, denoted as l_x . Hence, there exists a point p_x , s.t.:

$$\{p_x\} = l_x \cap \partial \mathbb{D}^n = l_x \cap \mathbb{S}^{n-1}$$

Then we can construct a map $\bar{f} : \mathbb{D}^n \rightarrow \mathbb{S}^{n-1}$ by

$$\bar{f}(x) := p_x$$

- **Claim:** $\bar{f}$ is a retraction from $\mathbb{D}^n$ onto $\mathbb{S}^{n-1}$.

Proof. Let $i : \mathbb{S}^{n-1} \rightarrow \mathbb{D}^n$ be the inclusion map.

- $\bar{f}$ is continuous: Since f is cont and $f(x) \neq x$ for all $x \in \mathbb{D}^n$, then the change in the direction of the rays is cont and hence $\bar{f}$ is cont.
- $\bar{f} \circ i = id_{\mathbb{S}^{n-1}}$: By def of $\bar{f}$, for any $x \in \mathbb{S}^{n-1}$

$$(\bar{f} \circ i)(x) = \bar{f}(i(x)) = \bar{f}(x) = x = id_{\mathbb{S}^{n-1}}(x) \Rightarrow \bar{f} \circ i = id_{\mathbb{S}^{n-1}}$$

- $i \circ \bar{f} \simeq id_{\mathbb{D}^n}$: Consider $H : \mathbb{D}^n \times I \rightarrow \mathbb{S}^{n-1}$ given by

$$H(x, t) := (1 - t)p_x + tx$$

Then one has $H(x, 0) = p_x = (i \circ \bar{f})(x)$, $H(x, 1) = x = id_{\mathbb{D}^n}(x)$. Hence, H gives a homotopy between $i \circ \bar{f}$ and $id_{\mathbb{D}^n}$.

By claim, $\Pi_1(\mathbb{D}^2) \cong \Pi_1(\mathbb{S}^1)$. But $\Pi_1(\mathbb{D}^2) \cong \{0\}$, $\Pi_1(\mathbb{S}^1) \cong \mathbb{Z}$, contradiction since $\{0\}$ is trivial and $\mathbb{Z}$ is NOT. □

□

Theorem 11.21. Every non-constant polynomial $p(x) \in \mathbb{C}[x]$ has a root in $\mathbb{C}$.

Proof. Suppose on the contrary that there exists a non-constant polynomial $p(x) \in \mathbb{C}[x]$ having NO root in $\mathbb{C}$. Then for any $x \in \mathbb{C}$, $p(x) \neq 0$, i.e.: $p : \mathbb{C} \rightarrow \mathbb{C} \setminus \{0\}$ is a map given by $p(x) := a_n x^n + \cdots + a_1 x + a_0$. For $r > 0$, consider

$$\begin{array}{ccc} C_r & \xrightarrow{i} & \mathbb{C} \\ & \searrow p|_{C_r} & \downarrow p \\ & & \mathbb{C} \setminus \{0\} \end{array}$$

where $C_r := \{re^{i\theta} \mid 0 \leq \theta < 2\pi\}$ denotes the circle in complex plane of radius r , i.e.: $p|_{C_r} = p \circ i$. Then we have a diagram of group homomorphisms:

$$\begin{array}{ccc} \mathbb{Z} \cong \Pi_1(C_r) & \xrightarrow{i_*} & \Pi_1(\mathbb{C}) \cong \{0\} \\ & \searrow (p|_{C_r})_* & \downarrow p_* \\ & & \Pi_1(\mathbb{C} \setminus \{0\}) \cong \Pi_1(S^1) \cong \mathbb{Z} \end{array}$$

Hence, for any $m \in \mathbb{Z} \cong \Pi_1(C_r)$,

$$(p|_{C_r})_*(m) = (p_* \circ i_*)(m) = p_*(i_*(m)) \stackrel{\Pi_1(\mathbb{C}) \cong \{0\}}{=} p_*(0) = e_{\Pi_1(\mathbb{C} \setminus \{0\})} \quad (\#)$$

- **Claim 1:** For large r , we have

$$p|_{C_r} \simeq q : C_r \rightarrow \mathbb{C} \setminus \{0\} \text{ with } q(z) := \frac{p(r)}{r^n} z^n$$

Proof. Define $H : C_r \times I \rightarrow \mathbb{C} \setminus \{0\}$ by

$$H(z, t) := (1 - t)p(z) + q(z)$$

- **Check** that H is well-defined, i.e.: $H(z, t) \neq 0$ for all $z \in C_r, t \in I$: Suppose on the contrary that $H(z, t) = 0$ for some $z \in C_r, t \in I$. Then by the def of H and q , one has $(1 - t)(a_n z^n + \dots + a_1 z + a_0) = -t(\frac{p(r)}{r^n} \cdot z^n)$ and hence

$$\begin{aligned} a_n z^n + \dots + a_1 z + a_0 &= t \left(a_n z^n + \dots + a_1 z + a_0 - \frac{p(r)}{r^n} \cdot z^n \right) \\ &= t \left(a_n z^n + \dots + a_1 z + a_0 - a_n z^n - \frac{a_{n-1}}{r} z^n - \dots - \frac{a_1}{r^{n-1}} z^n - \frac{a_0}{r^n} z^n \right) \\ &= t \left(a_{n-1} z^{n-1} + \dots + a_1 z + a_0 - \left(\frac{a_{n-1}}{r} + \dots + \frac{a_1}{r^{n-1}} + \frac{a_0}{r^n} \right) z^n \right) \end{aligned}$$

For large r , the coefficient of z^n on the right hand side is nearly 0 but the coefficient of z^n on the left hand side is $a_n \not\approx 0$, contradiction.

- It's easy to check that H gives a homotopy between $p|_{C_r}$ and q .

□

Therefore, by *claim 1*, for large r , one has

$$(p|_{C_r})_* = q_* : \Pi_1(C_r) \cong \mathbb{Z} \rightarrow \Pi_1(\mathbb{C} \setminus \{0\}) \cong \mathbb{Z}$$

- **Claim 2:** $q_*(m) = n \cdot m$

Proof. Take $1 = [l] \in \Pi_1(C_r) \cong \mathbb{Z}$ with $l(t) = re^{2\pi it}$ and $q_*(1) := [q \circ l](1)$. Since

$$(q \circ l)(t) = q(re^{2\pi it}) := \frac{p(r)}{r^n} \cdot (re^{2\pi it})^n = p(r) \cdot e^{2\pi int}$$

then $q_*(1) = n$.

□

Contradict (#).

□

11.3 Fundamental Group of Graph

Theorem 11.22. *Let $\Gamma = (V, E)$ be a connected graph with $|V|, |E| < \infty$. Then*

$$\Pi_1(|\Gamma|) \cong \langle \{e \mid e \in E \setminus E_T, \forall \text{ maximal tree } T \text{ of } \Gamma\} \rangle$$

i.e.: $\Pi_1(|\Gamma|)$ is isomorphic to free group generated by $e \in E \setminus E_T$ for all maximal tree T of Γ .

Corollary 11.23. *Let $\Gamma = (V, E)$ be a connected graph with $|V|, |E| < \infty$. Then*

$$\Pi_1(|\Gamma|) \cong \langle a_1, \dots, a_n \rangle$$

where $n = |E| - |V| + 1$.

Fact: Planar graph: $\chi(|\Gamma|) = 2$ where

$$\chi := |V| - |E| + |F|$$

(for Γ , $|F| = 0$).

Chapter 12

Seifert-Vankampen Theorem

Theorem 12.1. *Let $K = K_1 \cup K_2$ be topological spaces with*

- K_1, K_2 *path-connected open in K*
- $K_1 \cap K_2$ *path-connected*

Take $b \in K_1 \cap K_2$. Suppose

$$\Pi_1(K_1, b) \cong \langle X_1 \mid R_1 \rangle$$

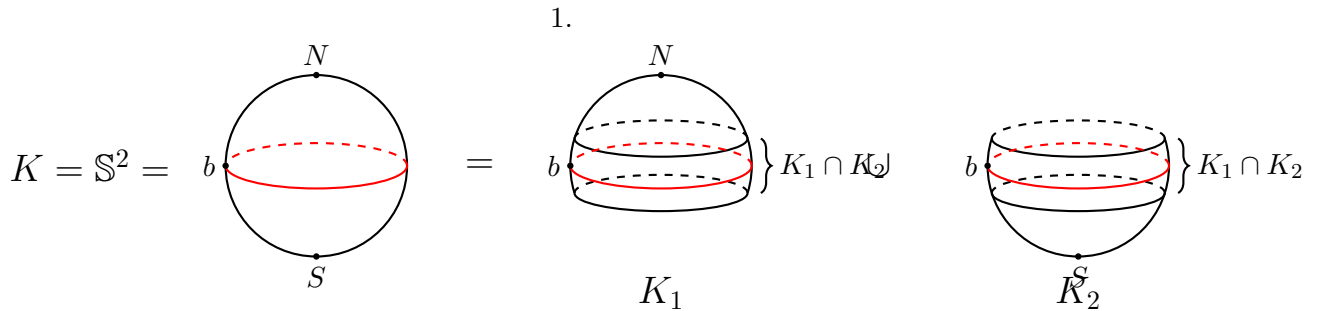
$$\Pi_1(K_2, b) \cong \langle X_2 \mid R_2 \rangle$$

with $i_1 : K_1 \cap K_2 \rightarrow K_1$ and $i_2 : K_1 \cap K_2 \rightarrow K_2$ inclusion maps. Then

$$\begin{aligned} \Pi_1(K, b) &\cong \langle X_1, X_2 \mid R_1, R_2, (i_1)_*(\alpha) = (i_2)_*(\alpha) \forall \alpha \in \Pi_1(K_1 \cap K_2, b) \rangle \\ &\cong \Pi_1(K_1, b) *_{\Pi_1(K_1 \cap K_2, b)} \Pi_1(K_2, b) \end{aligned}$$

here $$ is called the Amalgamation Product.*

Example:



$$\begin{aligned}
 \text{Then } K_1 \simeq \{N\} &\implies \pi_1(K_1, b) \cong \langle e \rangle = \langle \alpha \mid \alpha \rangle \\
 K_2 \simeq \{S\} &\implies \pi_1(K_2, b) \cong \langle e \rangle = \langle \beta \mid \beta \rangle \\
 K_1 \cap K_2 \simeq \mathbb{S}^1 &\implies \pi_1(K_1 \cap K_2, b) \cong \mathbb{Z} = \langle \gamma \mid \emptyset \rangle
 \end{aligned}$$

$$\text{Hence, } (i_1)_* : \pi_1(K_1 \cap K_2, b) \longrightarrow \pi_1(K_1, b) \quad \text{and} \quad (i_2)_* : \pi_1(K_1 \cap K_2, b) \longrightarrow \pi_1(K_2, b)$$

$$\gamma \longmapsto e$$

$$\gamma \longmapsto e$$

Then, apply Seifert-Vankampen Th'm,

$$\begin{aligned}
 \pi_1(K, b) &= \langle \alpha, \beta \mid \alpha, \beta, (i_1)_*(\gamma^n) = (i_2)_*(\gamma^n) \quad \forall n \in \mathbb{Z} \rangle \\
 &= \langle \alpha, \beta \mid \alpha, \beta, e = e \rangle = \langle e \rangle
 \end{aligned}$$

$$2. \quad K = \text{Diagram of two circles sharing a point } b \quad \cup \quad \begin{matrix} \text{Diagram of circle } K_1 \text{ with base point } b \\ \text{Diagram of circle } K_2 \text{ with base point } b \end{matrix}$$

$$\text{Then we have } K_1 \simeq S^1 \implies \pi_1(K_1, b) \cong \langle \alpha \mid \emptyset \rangle$$

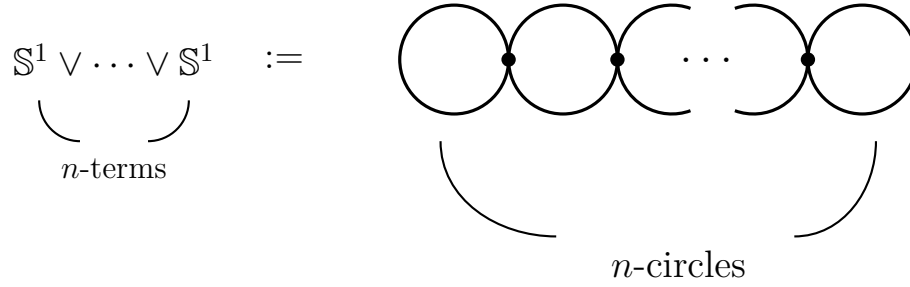
$$K_2 \simeq S^1 \implies \pi_1(K_2, b) \cong \langle \beta \mid \emptyset \rangle$$

$$K_1 \cap K_2 \simeq \{b\} \implies \pi_1(K_1 \cap K_2, b) \cong \langle e \rangle$$

Hence, $(i_1)_* : \alpha \mapsto e$ and $(i_2)_* : \beta \mapsto e$. Then by S-V Thm,

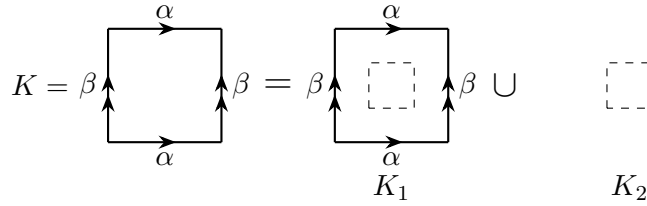
$$\pi_1(K, b) \cong \langle \alpha, \beta \mid e = e \rangle \cong \langle \alpha, \beta \mid \emptyset \rangle \cong \mathbb{Z} * \mathbb{Z}.$$

Then by induction, if Γ is shown as figure 9.1, then we have:


 Figure 12.1: Wedge sum of n circles

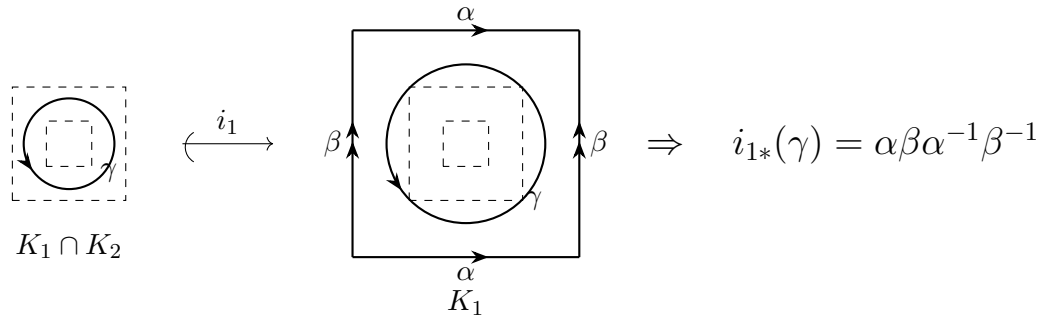
$$\pi_1(\Gamma, b) \cong \underbrace{\mathbb{Z} * \cdots * \mathbb{Z}}_{n\text{-tuples}}$$

3. $K = \mathbb{T}^2$



where $K_1 \cap K_2 \simeq S^1$

$$\Rightarrow \pi_1(K_1) \cong \langle \alpha, \beta \rangle, \quad \pi_1(K_2) \cong \langle e \rangle, \quad \pi_1(K_1 \cap K_2) \cong \langle \gamma \rangle.$$



It's easy to get $i_{2*}(\gamma) = e$. Then applying SVK,

$$\begin{aligned} \Pi_1(\mathbb{T}^2) &\cong \langle \alpha, \beta \mid i_{1*}(\gamma) = i_{2*}(\gamma) \rangle \\ &= \langle \alpha, \beta \mid \alpha \beta \alpha^{-1} \beta^{-1} = e \rangle \cong \mathbb{Z} \oplus \mathbb{Z} \end{aligned}$$

4.

$$\Pi_1(\mathbb{T}^2 \# \mathbb{T}^2) \cong \Pi_1(\mathbb{T}^2) *_{\Pi_1(S^1)} \Pi_1(\mathbb{T}^2) \cong \langle \alpha_1, \beta_1, \alpha_2, \beta_2 \mid [\alpha_1, \beta_1][\alpha_2, \beta_2] = e \rangle$$

5.

$$\Pi_1(S_g) := \Pi_1(\#_{i=1}^g \mathbb{T}^2) \cong \langle \alpha_1, \beta_1, \dots, \alpha_g, \beta_g \mid \Pi_{i=1}^g [\alpha_i, \beta_i] = e \rangle$$

Proposition 12.2. *Let X, Y be two topological spaces with basepoints $x_0 \in X, y_0 \in Y$. Then*

$$\Pi_1(X \times Y, (x_0, y_0)) \cong \Pi_1(X, x_0) \times \Pi_1(Y, y_0)$$

Proposition 12.3. *Suppose X, Y are of dimension > 2 . Then*

$$\Pi_1(X \# Y) \cong \Pi_1(X) * \Pi_1(Y)$$

Proof. Since $\Pi_1(\mathbb{S}^{n(\geq 2)}) \cong \langle e \rangle$, then

$$\Pi_1(X \# Y) \cong \Pi_1(X) *_{\Pi_1(\mathbb{S}^{n(\geq 2)})} \Pi_1(Y) \cong \Pi_1(X) *_{\{e\}} \Pi_1(Y) \cong \Pi_1(X) * \Pi_1(Y)$$

□

Proposition 12.4. *Every closed (compact without boundary) orientable surface is given by S_g for some $g \geq 0$.*

Definition 12.5. Amalgamated Product

Let $\{H_{ij}\}_{(i,j)}$ be a symmetric collection of groups where we have a collection of homomorphisms

$$\{\varphi_{ij} : H_{ij} \rightarrow G_i\}$$

The *amalgamated product* of $\{G_i\}$ over $\{H_{ij}\}$ is defined as

$$*_{H_{ij}} G_i := *G_i / \langle \varphi_{ij}(\alpha) \varphi_{ji}(\alpha)^{-1} \mid \alpha \in H_{ij} \rangle$$

Here $\langle \varphi_{ij}(\alpha) \varphi_{ji}(\alpha)^{-1} \mid \alpha \in H_{ij} \rangle$ is normal closure of set given by $\varphi_{ij}(\alpha) = \varphi_{ji}(\alpha)$, $\alpha \in H_{ij}$, for any $(i, j) \in J \times J$ in $*G_i$.

In presentation,

$$*_{H_{ij}} G_i = \langle \bigsqcup S_i \mid (\bigcup R_i) \cup \{\varphi_{ij}(\alpha) \varphi_{ji}(\alpha)^{-1} \mid \alpha \in H_{ij}\} \rangle$$

Example:

$$F_2 *_Z F_2 \cong F_3$$

This is by taking isomorphic copy of $\mathbb{Z}$ in F_2 , i.e.:

$$\langle a, b, c, d \mid a = b \rangle \cong F_3$$

Definition 12.6. HNN-Extension

Suppose $H, F \leq G$ are two isomorphic subgroups of G given by φ isomorphism. Let $\lambda \in F$. Then

$$\langle G, \lambda \mid \lambda^{-1}\alpha\lambda = \varphi(\alpha), \alpha \in H \rangle = G * \langle \lambda \rangle / \langle \lambda^{-1}\alpha\lambda\varphi(\alpha)^{-1} \mid \alpha \in H \rangle$$

Example: Bamlog-Solitan Group:

$$BS(n, m) = \langle \alpha, \lambda \mid \lambda^{-1}\alpha^n\lambda = \alpha^m \rangle \Rightarrow BS(1, -1) = \Pi_1(Klein\ Bottle)$$

Theorem 12.7. *Van-Kampen Theorem*

Suppose $X = \bigcup X_\alpha$ collection of open path-connected sets X_α with basepoint $x_0 \in X_\alpha \forall \alpha$. If $X_\alpha \cap X_\beta$ are path-connected, then there is a unique surjective homomorphism

$$\Phi : * \Pi_1(X_\alpha) \rightarrow \Pi_1(X)$$

If in addition $X_\alpha \cap X_\beta \cap X_\gamma$ are path-connected, then $\ker \Phi$ is normal subgroup N generated by

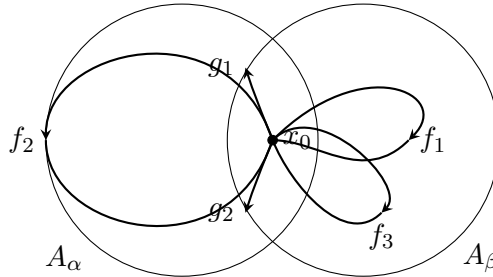
$$\langle i_{\alpha\beta*}(\omega)i_{\beta\alpha*}(\omega)^{-1} \rangle = N$$

i.e.: $\Pi_1(X) \cong * \Pi_1(X_\alpha) / N$

Proof. First we show Φ is surjective. Given a loop $f : I \rightarrow X$ at the basepoint x_0 , we may choose as in the proof of Proposition 1.5 a partition $0 = s_0 < s_1 < \dots < s_m = 1$ so that each subinterval $[s_{i-1}, s_i]$ is mapped by f to a single A_α , which we denote A_i . Let f_i be the path $f|_{[s_{i-1}, s_i]}$, so f is the composition $f_1 \cdots f_m$. Since $A_i \cap A_{i+1}$ is path-connected, we may choose a path g_i in $A_i \cap A_{i+1}$ from x_0 to the point $f(s_i) \in A_i \cap A_{i+1}$. Inserting $\bar{g}_i \cdot g_i$ into f at s_i for each i then yields a loop

$$(f_1 \cdot \bar{g}_1) \cdot (g_1 \cdot f_2 \cdot \bar{g}_2) \cdot (g_2 \cdot f_3 \cdot \bar{g}_3) \cdots (g_{m-1} \cdot f_m)$$

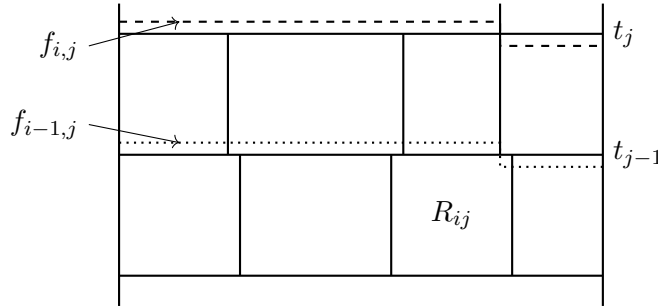
homotopic to f . This loop is a composition of loops each lying in a single A_i , so $[f]$ is in the image of Φ , and Φ is surjective.



The harder part of the proof is to show that the kernel of Φ is N . Suppose we are given a reduced word $w = w_1 \cdots w_k$ in $*_{\alpha} \pi_1(A_\alpha)$ such that $\Phi(w)$ is trivial. If we can show that w has trivial image in $*_{\alpha} \pi_1(A_\alpha) / N$, then this will imply that the kernel of Φ is contained in N . The opposite inclusion has already been noted, so this will finish the proof.

Corresponding to the word $w = w_1 \cdots w_k$ there is a product $f = f_1 \cdots f_k$ of loops f_i lying in A_α 's. Since $\Phi(w)$ is trivial, there is a map $F : I \times I \rightarrow X$ defining a homotopy from f to the constant loop. There exist partitions $0 = s_0 < s_1 < \cdots < s_m = 1$ and $0 = t_0 < t_1 < \cdots < t_n = 1$ such that each rectangle $R_{ij} = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ is mapped by F into a single A_α , which we label A_{ij} . These partitions may be obtained by covering $I \times I$ by finitely many rectangles $[a, b] \times [c, d]$ each mapping to a single A_α , using a compactness argument, then partitioning $I \times I$ by the union of all the horizontal and vertical lines containing edges of these rectangles. We may assume the s_i -partition subdivides the partition decomposing f as $f_1 \cdots f_k$. Since F maps a neighborhood of R_{ij} to A_{ij} , we may perturb the vertical sides of the rectangles R_{ij} so that each point of $I \times I$ lies in at most three R_{ij} 's. We can do this perturbation just on the rectangles R_{ij} with $j > 1$, leaving the rectangles R_{i1} in the bottom row unchanged.

Let f_{ij} be the loop obtained by restricting F to the path in $I \times I$ which goes in linear segments from $(0, t_j)$ to the upper right corner of R_{ij} , then to the lower right corner of R_{ij} , then to $(1, t_{j-1})$.



The restriction of F to the rectangle R_{ij} gives a homotopy from $f_{i-1,j}$ to $f_{i,j}$, and in this way the homotopy F is broken up into a sequence of smaller homotopies:

$$f \simeq f_{01} \simeq f_{11} \simeq f_{21} \simeq \cdots \simeq f_{m1} \simeq f_{02} \simeq f_{12} \simeq f_{22} \simeq \cdots \simeq f_{mn}$$

Some of these homotopies, for example $f \simeq f_{01}$ and $f_{m1} \simeq f_{02}$, are just reparametrizations since the left and right edges of $I \times I$ map to the basepoint x_0 . The final path f_{mn} is the constant loop.

Let us call the corners of the R_{ij} 's *vertices*. For each vertex v with $F(v) \neq x_0$, choose a path g_v from x_0 to $F(v)$ in the intersection of the two or three A_{ij} 's corresponding to the R_{ij} 's containing v . If we insert into f_{ij} the appropriate paths $\bar{g}_v g_v$ at successive vertices, as in the first paragraph of the proof, then we obtain a product $f_{ij}^1 f_{ij}^2 \cdots$ homotopic to f_{ij} , with each f_{ij}^k a loop in a single A_α , namely the A_{pq} corresponding to one of the at most two R_{pq} 's containing the relevant horizontal or vertical segment. Making a choice of such an A_α for each f_{ij}^k gives a way of writing $[f_{ij}] \in \pi_1(X)$ as $\Phi(w_{ij})$ for a word $w_{ij} = w_{ij}^1 w_{ij}^2 \cdots$ in ${}_{*\alpha}\pi_1(A_\alpha)$, with $w_{ij}^k = [f_{ij}^k]$. Making a different choice A_β instead of A_α changes the corresponding w_{ij}^k from $w_{ij}^k = i_{\alpha\beta}(\omega) \in \pi_1(A_\alpha)$ to $w_{ij}^k = i_{\beta\alpha}(\omega) \in \pi_1(A_\beta)$. Hence w_{ij} changes from a product of the form $\omega_1 i_{\alpha\beta}(\omega) \omega_2$ to $\omega_1 i_{\beta\alpha}(\omega) \omega_2$. Modulo the normal subgroup N , these two w_{ij} 's are equivalent since

$$\omega_1 i_{\alpha\beta}(\omega) \omega_2 = [\omega_1 i_{\alpha\beta}(\omega) i_{\beta\alpha}(\omega)^{-1} \omega_1^{-1}] \omega_1 i_{\beta\alpha}(\omega) \omega_2$$

and the bracketed expression is in N .

The homotopy $f_{i-1,j} \simeq f_{ij}$ across R_{ij} changes the subword of $w_{i-1,j}$ corresponding to the left and bottom edges of R_{ij} to the subword of w_{ij} corresponding to the upper and right edges of R_{ij} . If we choose the terms in these subwords to lie in $\pi_1(A_{ij})$, then the homotopy across R_{ij} shows that these two subwords define the same element of $\pi_1(A_{ij})$, hence $w_{i-1,j} = w_{ij}$ in $*_{\alpha}\pi_1(A_{\alpha})$. Thus we have shown that the image of w_{ij} in $*_{\alpha}\pi_1(A_{\alpha})/N$ is independent of i and j .

We may assume that the homotopy $F(s, t)$ is independent of t in the bottom row of rectangles R_{i1} . Then since the s_i -partition subdivides the partition defining the factorization $f = f_1 \cdots f_k$, the bottom edge of each R_{i1} is contained in the domain of an f_{ℓ} and we may choose A_{i1} to be the A_{α} for which $[f_{\ell}] \in \pi_1(A_{\alpha})$. The effect of this is that the word w_{01} reduces to $w = w_1 \cdots w_k$ in $*_{\alpha}\pi_1(A_{\alpha})$. The final word w_{mn} is the identity element of $*_{\alpha}\pi_1(A_{\alpha})$ since f_{mn} is the constant loop. Applying the conclusion of the preceding paragraph, we deduce that w has trivial image in $*_{\alpha}\pi_1(A_{\alpha})/N$, finishing the proof. $\square$

Example:

- *Wedge Sum:* $\bigvee_{\alpha} X_{\alpha}$ at basepoint $x_{\alpha} \in X_{\alpha}$ is $\bigsqcup_{\alpha} X_{\alpha} / \{x_{\alpha} \sim x'_{\alpha}\}$. Assume that $\{X_{\alpha}\}$ path-connected, then $\bigvee_{\alpha} X_{\alpha}$ is path-connected. Take $U_{\alpha} \ni x_{\alpha}$ to be an open neighborhood deformation retract to x_{α} . Then for any α , $A_{\alpha} := X_{\alpha} \vee (\bigvee_{\beta \neq \alpha} U_{\beta})$ is an open neighborhood of x_{α} in $\bigvee_{\alpha} X_{\alpha}$, deformation retract to x_{α} . Since $A_{\alpha} \cap A_{\beta} = \bigvee_{\gamma} U_{\gamma}$ deformation retract to point, then $\Pi_1(A_{\alpha} \cap A_{\beta}) = \{e\}$. Hence,

$$\Pi_1\left(\bigvee_{\alpha} X_{\alpha}\right) \cong *_{\alpha}\Pi_1(A_{\alpha}) \cong *_{\alpha}\Pi_1(X_{\alpha})$$

- *Finite Graph:* Let X be a path-connected space defined by connected finite graph with V vertices, E edges. Set $g := 1 - V + E$. Then

$$\Pi_1(X) \cong F_g$$

Proof. Let T be a maximal tree of X . Recall that T is contractible and that T contains all the vertices of X . Adding a new edge e_{α} to T , then we can get $T \cup \{e_{\alpha}\} \simeq \mathbb{S}^1$. Continuing adding edge in $X \setminus T$, then we are done. $\square$

- *Suspension of 3 points:* Let X be the suspension of 3 points, i.e.:

$$X := S\{x, y, z\} := \{x, y, z\} \times I / (\{x, y, z\} \times \{0\} \cup \{x, y, z\} \times \{1\})$$

Take X_{ω} for $\omega \in \{x, y, z\}$ open cover as $X \setminus \{\omega\}$. Then $X_{\omega} \cap X_{\omega'}$ is contractible. Hence,

$$\Pi_1(X) \cong \Pi_1(X_{\omega}) * \Pi_1(X_{\omega'}) \cong \mathbb{Z} * \mathbb{Z}$$

Chapter 13

Knot Theory

13.1 Torus Knot

Definition 13.1. Torus Knot

$T_{m,n} : \mathbb{S}^1 \rightarrow \mathbb{S}^1 \times \mathbb{S}^1 \subseteq \mathbb{R}^3$ given by $T_{m,n}(z) := (z^m, z^n)$ is called a (m, n) -torus knot, i.e.: m -times in the longitude and n -times in the meridian.

Proposition 13.2. $T_{m,n}$ is embedding if and only if $\text{g.c.d.}(m, n) = 1$.

Proof. • ($\Leftarrow$): Let $\theta, \theta' \in [0, 1)$, s.t.: $\theta \geq \theta'$ $(e^{2\pi i m \theta}, e^{2\pi i n \theta}) = (e^{2\pi i m \theta'}, e^{2\pi i n \theta'})$. Hence, $m(\theta - \theta'), n(\theta - \theta') \in \mathbb{Z}$. Since $\text{gcd}(m, n) = 1$, then $\theta - \theta' \in \mathbb{Z} \Rightarrow \theta = \theta'$. Therefore, $T_{m,n}$ is injective. Obviously $T_{m,n}$ is cont. Then by theorem 4.31, $T_{m,n}$ is embedding.

• ($\Rightarrow$): Suppose on the contrary that $d = \text{gcd}(m, n) > 1$. Then $m/d, n/d \in \mathbb{Z}$. Hence, $(e^{2\pi i m \theta}, e^{2\pi i n \theta}) = (e^{2\pi i m \theta/d}, e^{2\pi i n \theta/d}) \Rightarrow T_{m,n}$ is not injective, contradiction. □

Theorem 13.3.

$$\Pi_1(T_{m,n}) := \Pi_1(\mathbb{R}^3 \setminus T_{m,n}) \cong \langle a, b \mid a^m = b^n \rangle$$

Proof. Note that $\mathbb{R}^3 \setminus T_{m,n} \simeq \mathbb{S}^3 \setminus T_{m,n}$.

• *Step 1:*

$$\begin{aligned} \mathbb{S}^3 &= \partial \mathbb{D}^4 \simeq \partial(\mathbb{D}^2 \times \mathbb{D}^2) = \overline{\mathbb{D}^2} \times \overline{\mathbb{D}^2} - \text{int}(\mathbb{D}^2 \times \mathbb{D}^2) = \overline{\mathbb{D}^2} \times \overline{\mathbb{D}^2} - (\mathbb{D}^2)^\circ \times (\mathbb{D}^2)^\circ \\ &= (\overline{\mathbb{D}^2} - (\mathbb{D}^2)^\circ) \times \mathbb{D}^2 \cup \mathbb{D}^2 \times (\overline{\mathbb{D}^2} - (\mathbb{D}^2)^\circ) = \partial \mathbb{D}^2 \times \mathbb{D}^2 \cup \mathbb{D}^2 \times \partial \mathbb{D}^2 \\ &= \mathbb{S}^1 \times \mathbb{D}^2 \cup \mathbb{D}^2 \times \mathbb{S}^1 \end{aligned}$$

• *Step 2:* Denote $T = \mathbb{S}^1 \times \mathbb{D}^2$, $T' = \mathbb{D}^2 \times \mathbb{S}^1$ and $T'' = T \cap T' = \mathbb{S}^1 \times \mathbb{S}^1$. Note that the $T_{m,n}$ is embedded in T'' and that $T'' \setminus T_{m,n}$ is homotopic to an annulus. Let $x_0 \in T'' \setminus T_{m,n}$ be a basepoint. Both T and T' are homotopic to $\mathbb{S}^1$. Hence, one has

$$\Pi_1(T) \cong \langle a \rangle, \Pi_1(T') \cong \langle b \rangle, \Pi_1(T'' \setminus T_{m,n}) \cong \langle c \rangle$$

with $i_{1*}(c) = a^m$ and $i_{2*}(c) = b^n$.

- *Step 3:* Then applying SVK, one can get

$$\begin{aligned}\Pi_1(T_{m,n}) &:= \Pi_1(\mathbb{R}^3 \setminus T_{m,n}) \cong \Pi_1(\mathbb{S}^3 \setminus T_{m,n}) \cong \Pi_1(T) *_{\Pi_1(T'' \setminus T_{m,n})} \Pi_1(T') \\ &= \langle a, b \mid i_{1*}(c) = i_{2*}(c) \rangle = \langle a, b \mid a^m = b^n \rangle\end{aligned}$$

□

13.2 Kont

Definition 13.4. Knot

A knot is an embedding of $\mathbb{S}^1$ in $\mathbb{S}^3$ or $\mathbb{R}^3$.

Definition 13.5. Equivalent Knots

Two knots K_1, K_2 are called *equivalent* if there is a continuous map $F : \mathbb{S}^3 \times [0, 1] \rightarrow \mathbb{S}^3$, s.t.:

$$F(x, 0) = id_{\mathbb{S}^3}, F(x, 1) \circ K_1 = K_2, \text{ and } F(x, t) \text{ is homomorphism } \forall t$$

Remark: It's easy to show that this definition gives an equivalence relation on space of knots.

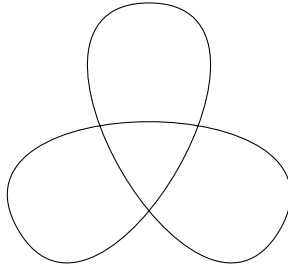
Definition 13.6. Tame Knot

Knot K is called tame if its projection in $\mathbb{R}^2$ have finitely many *crossings*.

Example:

- Unknot (= circle)
- Trefoil knot ($T_{2,3}$)

$$\theta \mapsto \begin{pmatrix} 2 + \cos(3\theta) \cos(2\theta) \\ 2 + \cos(3\theta) \sin(2\theta) \\ -\sin(3\theta) \end{pmatrix}$$

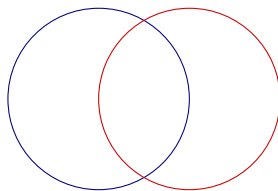


Definition 13.7. Link

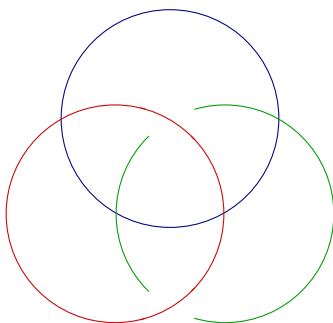
A link is an embedding of a disjoint union of finitely many $\mathbb{S}^1$ to $\mathbb{S}^2$ or $\mathbb{R}^3$.

Example:

- Hopf link

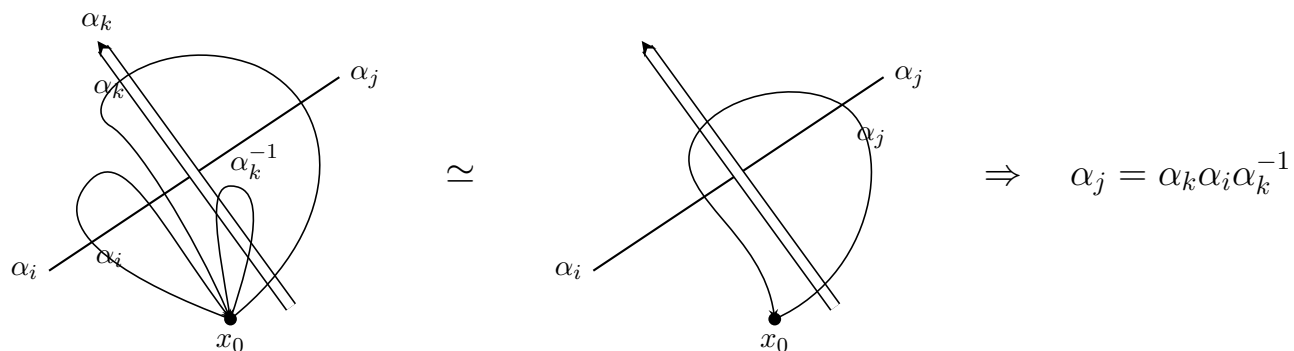


- Bonomean rings



Remark: When m, n are NOT coprime integers, one has shown that $T_{m,n}$ becomes torus link, with the number of component given by $\gcd(m, n)$.

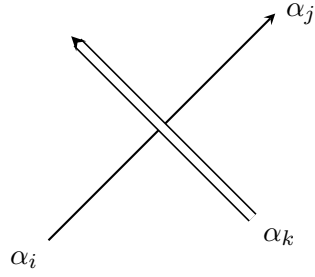
13.3 Wirtinger Presentation of Knot Group



The above figure shows how the formula for negative crossing comes into being. Similarly, we can derive the formula for positive crossing.

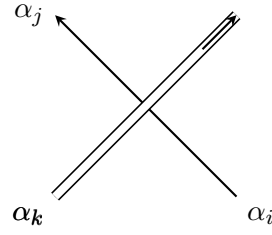
For l arcs in knot diagram with n -crossings, the group is given by presentation

$$\langle \alpha_1, \dots, \alpha_l \mid r_1, \dots, r_n \rangle \cong F_l / N_n$$



negative crossing

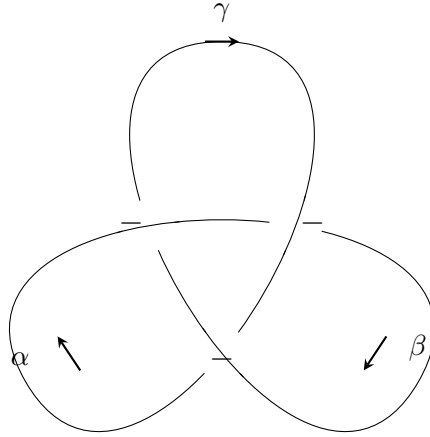
$$\alpha_j = \alpha_k \alpha_i \alpha_k^{-1}$$



positive crossing

$$\alpha_j = \alpha_k^{-1} \alpha_i \alpha_k$$

Example: Trefoil knot ($T_{2,3}$):



We have 3 negative crossings, so we can get three relations:

$$\alpha = \beta \gamma \beta^{-1}, \beta = \gamma \alpha \gamma^{-1}, \gamma = \alpha \beta \alpha^{-1}$$

Therefore, we get

$$\Pi_1(T_{2,3}) = \langle \alpha, \beta, \gamma \mid \alpha = \beta \gamma \beta^{-1}, \beta = \gamma \alpha \gamma^{-1}, \gamma = \alpha \beta \alpha^{-1} \rangle = \langle \alpha, \beta, \gamma \mid \beta = \gamma \alpha \gamma^{-1}, \gamma = \alpha \beta \alpha^{-1} \rangle$$

It's easy to show that the Wirtinger's presentation of $T_{2,3}$ is same as the form in theorem 10.3, i.e.:

$$\langle \alpha, \beta, \gamma \mid \beta = \gamma \alpha \gamma^{-1}, \gamma = \alpha \beta \alpha^{-1} \rangle \cong \langle a, b \mid a^2 = b^3 \rangle$$

13.4 Seifert-van Kampen Theorem applied to knot group

For a knot K , we firstly fatten K into a tubular neighborhood $nbhd(K)$. Note that $\mathbb{R}^3 \setminus K$ can be deformation retract onto $\mathbb{R}^3 \setminus nbhd(K)$. Since K is compact, then we can take $\mathbb{R}^3 \setminus nbhd(K) = U_1 \cup U_2$ where $U_1 = \{(x, y, z) \mid z < \epsilon_1\} \setminus nbhd(K)$ and $U_2 = \{(x, y, z) \mid z > \epsilon_2\} \setminus nbhd(K)$ for some $\epsilon_1, \epsilon_2 \in \mathbb{R}$. Choose basepoint in $U_1 \cap U_2$. Then applying SVK,

$$\Pi_1(\mathbb{R}^3 \setminus K) \cong \Pi_1(U_1) *_{\Pi_1(U_1 \cap U_2)} \Pi_1(U_2)$$

The remaining steps require a case-by-case analysis.

Chapter 14

Manifolds

14.1 Topological Manifolds

14.1.1 Definition and Examples

Definition 14.1. • Given $n \in \mathbb{N} \cup \{0\}$, we define the *upper half-space* in $\mathbb{R}^n$ as

$$\mathbb{H}^n := \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$$

Moreover, we define the *boundary* of $\mathbb{H}^n$ as

$$\partial\mathbb{H}^n := \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n = 0\}$$

Remark: It's easy to show that the map $\begin{smallmatrix} \mathbb{R}^{n-1} & \rightarrow & \partial\mathbb{H}^n \\ x & \mapsto & (x, 0) \end{smallmatrix}$ is a homeomorphism.

Let X be a topological space.

- An *n-dimensional chart* for X around a point $x \in X$ is a homeomorphism $\Phi : U \rightarrow V \subseteq \mathbb{R}^n$ where
 - U is an open neighborhood of x
 - V satisfies one of the following two axioms:
 1. V is an open subset of $\mathbb{R}^n$
 2. an open subset of the upper half-space $\mathbb{H}^n$ and $\Phi(x) \in \partial\mathbb{H}^n$

In the former case we say that Φ is a chart of type 1, while in the latter case we say that Φ is a chart of type 2.

- We say that X is an *n-dimensional manifold* if X is second-countable and Hausdorff, and if for any $x \in X$, there is an *n-dimensional chart* $\Phi : U \rightarrow V$ around x .
- An *n-dimensional atlas* for X is a collection of *n-dimensional charts* $\{\Phi_i : U_i \rightarrow V_i\}_{i \in I}$ satisfying that $\{U_i\}_{i \in I}$ covers X .

Let X be a topological manifold.

- We say that a point x on X is a *boundary point* if x doesn't admit a chart of type 1. Furthermore, we denote by ∂X the set of all boundary points of X . We refer to ∂X as the *boundary of X* .
- We say that X is *closed* if X is compact with $\partial X = \emptyset$.

Proposition 14.2. *Countable Atlas-Second Countable Proposition*

Let X be a topological space.

1. If X admits a countable atlas, then X is second-countable.
2. If X is compact and if X admits an atlas, then X is second-countable.

Proof. 1. Suppose that X admits a countable atlas $\{\Phi_i : U_i \rightarrow V_i\}_{i \in I}$. Since $V_i \subseteq \mathbb{R}^n$ are subsets of $\mathbb{R}^n$, then V_i 's are second-countable. Since Φ_i 's are homeomorphisms, then U_i 's are second-countable and hence so is X .

2. By the compactness of X , the atlas can be reduced into the finite case and then by 1, we're done. □

Example:

- *Examples:*
 - Consider $\mathbb{S}^n$. Note that for any $P \in \mathbb{S}^n$, $\Phi_P : \mathbb{S}^n \setminus \{-P\} \rightarrow \mathbb{R}^n$ is a chart of type 1. Hence, $\mathbb{S}^n$ is a closed n -dimensional topological manifold.
 - Every open subset U of $\mathbb{R}^n$ is an n -dimensional topological manifold with boundary $\emptyset$.
 - Let M be an open subset of $\mathbb{H}^n$.
 - * For any $x = (x_1, \dots, x_n) \in M$ with $x_n > 0$, there exists U open in both $\mathbb{H}^n$ and $\mathbb{R}^n$, s.t.: $x \in U \subseteq \mathbb{H}^n$. Then the identity map on U gives a chart of type 1.
 - * For any $x = (x_1, \dots, x_n) \in M$ with $x_n = 0$, the identity map is a chart of type 2.
- Hence, M is a topological manifold with boundary $\partial M = M \cap \partial \mathbb{H}^n$.
- n -ball $\overline{B}^n$ is an n -dimensional topological manifold:
 - * By the open-ness of B^n , every $x \in B^n$ admits an open neighborhood U in B^n and hence the identity map on U gives a chart of type 1.
 - * Consider $V := \{(x_1, \dots, x_n) \in \overline{B}^n \mid x_n > 0\}$. The map

$$(x_1, \dots, x_n) \mapsto (x_1, \dots, x_{n-1}, \sqrt{1 - x_1^2 - \dots - x_{n-1}^2} - x_n)$$

is a chart of type 2 around any $x = (x_1, \dots, x_n) \in \mathbb{S}^{n-1} = \overline{B}^n \setminus B^n$ with $x_n > 0$. Similarly for $x = (x_1, \dots, x_n) \in \mathbb{S}^{n-1}$ with $x_n < 0$.

- X is a 0-dimensional $\Leftrightarrow X$ is countable and is equipped with $\mathcal{T}_{discrete}$.
- The empty set $\emptyset$ is a topological manifold of any dimension.
- *Non-examples:*
 - $X = \mathbb{R}$ equipped with $\mathcal{T}_{discrete}$ is **NOT** second-countable and hence X is **NOT** a topological manifold.
 - The line with two zeros is **NOT** Hausdorff and hence it is **NOT** a topological manifold.
 - In the later section, we will see that there exists a connected Hausdorff space which admits a 1-dimensional atlas, but which is not second-countable. In particular this topological space is not a topological manifold

Proposition 14.3. *Chart-to-Balls Proposition*

Let X be an n -dimensional topological manifold, let $P \in X$ be a point in X and let N be a neighborhood of P .

1. If $P \in X \setminus \partial X$, then $\exists$ a chart $\Phi : U \rightarrow B^n$ with $\Phi(P) = 0$ and $U \subseteq N$.
2. If $P \in \partial X$, then $\exists$ a chart $\Phi : U \rightarrow B_{\geq 0}^n$ with $\Phi(P) = 0$ and $U \subseteq N$.

Proof. 1. By the def of ∂X , $\exists$ a chart $\Psi : V \rightarrow W$ around P of type 1. WLOG, we can take V to be such that $V \subseteq N^\circ \subseteq N$. By the def of charts of type 1, W is open and hence there exists $r > 0$, s.t.:

$$B_r^n(\Psi(P)) \subseteq W$$

Denote $B := B_r^n(\Psi(P))$. Then $\Phi := \Psi|_{\Psi^{-1}(B)} : \Psi^{-1}(B) \rightarrow B$ is the desired chart.

2. Similarly as 1.

□

Proposition 14.4. *Topological Manifold Product Proposition*

Let X be a m -dimensional topological manifold and let Y be an n -dimensional topological manifold.

1. $X \times Y$ is an $(m + n)$ -dimensional topological manifold.
2. (a) Each point in $(X \times Y) \setminus ((\partial X \times Y) \cup (X \times \partial Y))$ admits a chart of type 1.
 (b) Each point in $(\partial X \times Y) \cup (X \times \partial Y)$ admits a chart of type 2.
 (c) $\partial(X \times Y) \subseteq (\partial X \times Y) \cup (X \times \partial Y)$.
3. If X and Y are closed topological manifolds, then so is $X \times Y$.

Proof. • Since X and Y are Hausdorff and second-countable, then so is $X \times Y$.

- Let $(x, y) \in X \times Y$.
 - **Case 1:** $x \in X \setminus \partial X, y \in Y \setminus \partial Y$. Then we have two charts $\Phi_x : U_x \rightarrow V_x, \Phi_y : U_y \rightarrow V_y$ around x respectively y of type 1. Then consider the "product chart" $\Phi_x \times \Phi_y : U_x \times U_y \rightarrow V_x \times V_y$ given by

$$(\Phi_x \times \Phi_y)(u_1, u_2) := (\Phi_x(u_1), \Phi_y(u_2))$$

which is a chart around (x, y) of type 1.

- **Case 2:** $x \in X \setminus \partial X, y \in \partial Y$ or $x \in \partial X, y \in Y \setminus \partial Y$. Similarly the "product chart" is a chart around (x, y) of type 2.
- **Case 3:** $x \in \partial X, y \in \partial Y$. Note that the image of the "product chart"

$$\Phi_x \times \Phi_y : U_x \times U_y \rightarrow V_x \times V_y \subseteq \mathbb{R}^{m+n-2} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}$$

is **NOT** $\mathbb{H}^{m+n} := \mathbb{R}^{m+n-1} \times \mathbb{R}_{\geq 0}$, so we need an extra step: consider the map $\Psi : \mathbb{R}^{m+n-2} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{H}^{m+n} \mathbb{R}^{m+n-2} \times \mathbb{R} \times \mathbb{R}_{\geq 0}$ given by

$$\Psi(x_1, \dots, x_{m+n-2}, r \cos \varphi, r \sin \varphi) := (x_1, \dots, x_{m+n-2}, r^2 \cos 2\varphi, r^2 \sin 2\varphi)$$

It's easy to verify that Ψ is a homeomorphism and hence

$$\Psi \circ (\Phi_x \times \Phi_y)$$

is a chart around (x, y) of type 2.

- We've proven that (a) and (b) hold in 1 and (c) holds immediately by the def of the boundary of topological manifolds.
- X, Y closed $\Leftrightarrow \partial X = \partial Y = \emptyset \xrightarrow{2.(c)} \partial(X \times Y) = \emptyset \Leftrightarrow X \times Y$ closed.

□

Chapter 15

Additional Chapter

15.1 $\mathbb{S}^1 \times \mathbb{S}^1 \not\cong \mathbb{S}^2$

15.1.1 Method 1: Euler Characteristic

15.1.2 Method 2: Simple Closed Curve

Definition 15.1. Simple Closed Curve (SCC)

A SCC in a topological space X is a set $C \subseteq X$, s.t.: there exists a homeomorphism $h : \mathbb{S}^1 \rightarrow C$.

Theorem 15.2. *Jordan Curve Theorem*

If C is a SCC in $\mathbb{R}^2$, then $\mathbb{R}^2 - C$ has exactly two path-connected components.

Remark: Since $\mathbb{R}^2 \cup \{\infty\} \cong \mathbb{S}^2$, then we have the following corollary.

Corollary 15.3. *Sphere version of Jordan Curve Theorem*

If C is a SCC in $\mathbb{S}^2$, then $\mathbb{S}^2 - C$ has exactly two path-connected components.

Proposition 15.4. *Let $f : X \xrightarrow{\cong} Y$ be a homeomorphism. $f(C)$ is a SCC if C is a SCC.*

Proof. Let $h : \mathbb{S}^1 \xrightarrow{\cong} C(\subseteq X)$ be a homeomorphism by the def of SCC. Then it's easy to show that $f \circ h$ is a homeomorphism between $\mathbb{S}^1 \cong f(C)$. $\square$

Proposition 15.5. *For any $p \in \mathbb{S}^1$, define*

$$\gamma_p := \mathbb{S}^1 \times \{p\}.$$

Then we have:

$$\mathbb{T}^2 - \gamma_p \cong \mathbb{S}^1 \times (0, 1).$$

Proposition 15.6.

$$\mathbb{S}^1 \times \mathbb{S}^1 \not\cong \mathbb{S}^2$$

Proof. Suppose on the contrary that $\mathbb{S}^1 \times \mathbb{S}^1 \cong \mathbb{S}^2$. Then $\mathbb{T}^2 \cong \mathbb{S}^2$. Then there is a homeomorphism $f : \mathbb{S}^2 \xrightarrow{\cong} \mathbb{T}^2$.

By proposition 1.4 and 1.5, $f^{-1}(\gamma_0)$ is NOT a SCC in $\mathbb{S}^2$. Hence, $f^{-1}(\gamma_0) = \{q\}$ for some $q \in \partial\mathbb{S}^2$, contradicting the infectivity of f . $\square$

15.2 Computing $\Pi_1(X)$

15.2.1 Dimensionality Reduction

Definition 15.7. Let X be a topological space, $A \subseteq X$ and let $i : A \rightarrow X$ be an inclusion.

- We say A is a *homotopy retract (deformation retract)* of X , if there exists a continuous map $r : X \rightarrow A$, s.t.:

$$\begin{aligned} r \circ i : A &\xrightarrow{i} X \xrightarrow{r} A = id_A \\ i \circ r : X &\xrightarrow{r} A \xrightarrow{i} X \simeq id_X \end{aligned}$$

Moreover, we call r here is a retraction. Furthermore, for all $a \in A$,

$$r(a) = a$$

- If $r : X \rightarrow A$ is a retraction, then there exists a homotopy $H : X \times I \rightarrow X$, s.t.: $H(x, 0) = x$, $H(x, 1) = r(x)$ and $H(a, t) = a$ for any $a \in A$, $t \in I$.

Proposition 15.8. If A is a deformation retract of X , then

$$A \simeq X$$

Example:

- For any point $p \in \mathbb{S}^n$, $\{p\}$ is a deformation retract, i.e.:

$$\Pi_1(\mathbb{S}^n) \cong \Pi_1(\{p\}) = \{e\}$$

which follows that $\mathbb{S}^n$ is simply-connected.

- $\mathbb{S}^1$ is a deformation retract of $X := \mathbb{R}^2 \setminus \{0\}$.

Proof. Consider $r : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{S}^1$ given by

$$r(\mathbf{x}) := \frac{\mathbf{x}}{\|\mathbf{x}\|}$$

It's easy to check that r is continuous. Then construct $H : X \times I \rightarrow X$ by

$$H(\mathbf{x}, t) := (1 - t)\mathbf{x} + t \cdot \frac{\mathbf{x}}{\|\mathbf{x}\|}$$

It remains to check that $H(\mathbf{x}, t) \neq 0$ for all $\mathbf{x} \in X, t \in I$ for showing the well-defined-ness of H . Then one can find

$$\begin{aligned} H(\mathbf{x}, 0) &= \mathbf{x} = id_X(\mathbf{x}), H(\mathbf{x}, 1) = \frac{\mathbf{x}}{\|\mathbf{x}\|} = r(\mathbf{x}) \quad \forall \mathbf{x} \in X \\ \forall \mathbf{x} \in X, \|\mathbf{x}\| &= 1 \Rightarrow H(\mathbf{x}, t) = (1-t)\mathbf{x} + t\mathbf{x} = \mathbf{x} \quad \forall t \in I \end{aligned}$$

Therefore, X can deformation retract to $\mathbb{S}^1$. □

Proposition 15.9. *Let X be a topological space, $A \subseteq X$, $x_0 \in X$ and let $i : A \rightarrow X$ be an inclusion map. If A is a retract of X , then i can induce a group homomorphism $i_* : \Pi_1(A, x_0) \rightarrow \Pi_1(X, x_0)$ is injective.*

Proof.

$$(r \circ i)_* = r_* \circ i_* = (id_A)_* = id_{\Pi_1(A, x_0)} \quad (\#)$$

For any $[l], [l'] \in \Pi_1(A, x_0)$, if $i_*[l] = i_*[l']$, then

$$[l] \stackrel{(\#)}{=} r_*(i_*[l]) = r_*(i_*[l']) \stackrel{(\#)}{=} [l']$$

Therefore, i_* is injective. □

15.2.2 Dimensionality Raise

Definition 15.10. Let X be a topological space.

- *Cone*

$$CX := (X \times I)/(X \times \{1\})$$

- *Suspension*

$$SX := (X \times I)/(X \times \{1\} \cup X \times \{0\})$$

Example:

$$S(\mathbb{S}^n) \cong \mathbb{S}^{n+1}$$

Proposition 15.11. *For all topological space X , CX is contractible.*

Proof. Consider $H : CX \times I \rightarrow CX$ given by

$$H([x, s], t) := [x, st] \quad \forall$$

Then

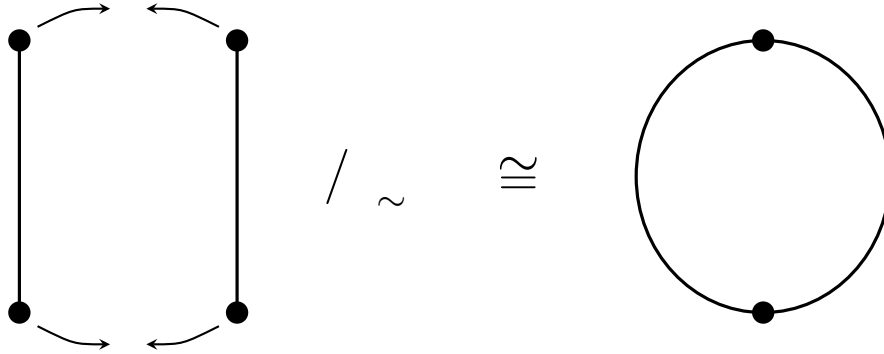
$$\begin{aligned} H([x, s], 0) &= [x, 0] \\ H([x, s], 1) &= [x, s] = id_{CX}([x, s]) \end{aligned}$$

Hence, H gives a homotopy between id_{CX} and $c_{\{*\}}$ where $c_{\{*\}}$ denotes the constant map on the top of the cone, i.e.: CX is contractible. $\square$

Example:

$$S(\mathbb{S}^0 := \{-1, 1\}) = \mathbb{S}^1$$

Proof. $S(\mathbb{S}^0) := \{-1, 1\} \times I / (\{-1, 1\} \times \{1\} \cup \{-1, 1\} \times \{0\})$ $\square$



15.2.3 Exercise

Exercise 15.1. $X \subseteq \mathbb{R}^3$, X be union of n lines through the origin, compute $\Pi_1(\mathbb{R}^3 \setminus X)$.

Intuition:

1. Since $\mathbb{R}^3 \setminus \{0\}$ can deformation retract onto $\mathbb{S}^2$, then if we consider $Y := \mathbb{S}^2 \cap X \subseteq \mathbb{S}^2$, then $\mathbb{R}^3 \setminus X$ can deformation retract onto $\mathbb{S}^2 \setminus Y$.
2. By stereographic projection, one has

$$\mathbb{S}^2 \setminus \{*\} \cong \mathbb{R}^2$$

Solution:

- **Step 1:** Consider $H : \mathbb{R}^3 \setminus X \rightarrow \mathbb{S}^2 \setminus Y$ given by

$$H(\mathbf{x}) := \frac{\mathbf{x}}{\|\mathbf{x}\|}$$

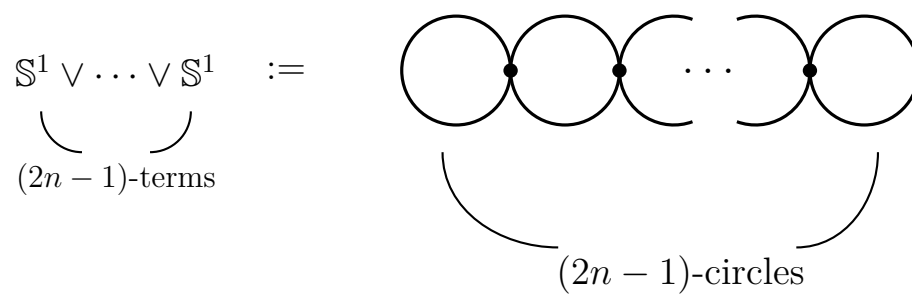
which is a homotopy between $\mathbb{R}^3 \setminus X$ and $\mathbb{S}^2 \setminus Y$, which follows that

$$\Pi_1(\mathbb{R}^3 \setminus X) \cong \Pi_1(\mathbb{S}^2 \setminus Y)$$

- **Step 2:** Pick a point $y_0 \in Y$, denote $Z := Y \setminus \{y_0\}$ and use stereographic projection, one can get

$$\mathbb{S}^2 \setminus Y = (\mathbb{S}^2 \setminus \{y_0\}) \setminus (Y \setminus Z) \cong \mathbb{R}^2 \setminus Z \Rightarrow \Pi_1(\mathbb{S}^2 \setminus Y) \cong \Pi_1(\mathbb{R}^2 \setminus Z)$$

- **Step 3:** $\mathbb{R}^2 \setminus Z = \mathbb{R}^2 \setminus \bigcup_{i=1}^{2n-1} \{*\}_i$ can deformation retract onto $\bigvee_{i=1}^{2n-1} \mathbb{S}^1$:

Figure 15.1: Wedge sum of $2n - 1$ circles

Therefore,

$$\Pi_1(\mathbb{R}^3 \setminus X) \cong \Pi_1\left(\bigvee_{i=1}^{2n-1} \mathbb{S}^1\right) (= F_{2n-1})$$